

# DEVELOPMENTAL PSYCHOLOGY (PSY505)

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## DEFINITION AND NATURE OF DEVELOPMENTAL PSYCHOLOGY

Developmental Psychology is one of the sub-fields of Psychology. It is an **ontogenetic study of human organism from conception to death**. Developmental Psychology seeks understanding and controls the basic processes and dynamics underlying human behavior at the various stages of life. Its investigations encompass the growth and maturation of the individual organism, its cognitive and emotional powers, as well as its personality structure.

Developmental Psychology, as a science of growth, deals with all the processes contributing to becoming an infant, a child, an adolescent, and a mature adult.

According to Hurlock, "Developmental Psychology is the branch of psychology that studies intraindividual changes and interindividual changes within these intraindividual changes. Its task is not only description but also explication of age-related changes in behavior in terms of antecedent consequent relationship."

Some developmental psychologists study developmental changes covering the life span from conception to death. Others cover only a segment of the life span, childhood or old age.

Developmental Psychology is a scientific discipline that attempts: (i) to devise methods for studying organisms as they evolve over time (ii) to collect facts about individuals of different ages, backgrounds and personalities and (iii) to construct a theoretical frame work that can account for the observed behaviors as well as for the changes occurring throughout the life cycle.

### Aspects of Development

There are four aspects of development which are closely intertwined. Each aspect of development affects the other.

- Physical development
- Intellectual development
- Personality development
- Social development

We will now discuss each of these in brief.

1. **Physical development.** Physical development consists of changes in the body, brain, sensory capacities and motor skills. They exert major influences on both intellect and personality. For example, much of an infant's knowledge comes from the senses and from motor activity. A child who has a hearing loss is at risk of delayed language development. In late adulthood, physical changes in the brain as in Alzheimer's disease which has been estimated to affect about 10 percent of people over the age of 65 (Evans et al. 1989) can result in intellectual and personality deterioration.

2. **Intellectual development.** Changes in mental abilities such as learning, memory, reasoning, thinking and language are aspects of intellectual development. These changes are closely related to both motor and emotional development. A baby's growing memory, for example, is related to separation anxiety, the fear that the mother will not return once she has gone away. If children could not remember the past and anticipate the future, they could not worry about the mother's absence.

3. **Memory** also affects babies' physical actions. For example, a one year old boy who remembers being scolded for knocking down his sister's block tower may refrain from doing it again.

4. **Personality and Social development.** Personality and social development affect with the cognitive aspects and the physical aspects of functioning. For example, anxiety about taking a test can impair performance; and social support from friends helps people cope with the negative effects of stress on their physical and mental health.

### Principles of Development

Development follows certain principles. These principles are:

- (i) **Development is similar for all:** All children follow a similar pattern of development, with one stage leading to next.

For instance, the baby stands before he walks, or draws a circle before a square.

- (ii) **Development Proceeds at Different Rates:** Even though all individuals follow much the same pattern of development, the rate of development varies from individual to individual. Because rate of development differs, all children of the same age do not reach the same point of physical or mental development. Nor do all individuals decline physically or mentally at the same rate. In the same individual, different physical and mental traits develop at different ages. Different rates of decline have likewise been observed for different physical and mental traits.

- (iii) **Development is continuous:** From the moment of conception to the time of death, changes are taking place within the individual, sometimes slowly. As a result, what happens at one stage of development varies over and influences the following stages. Unhealthy attitudes developed in childhood, for example, have been found to be at the root of much of the unhappiness and poor adjustment during middle and old age.

- (iv) **Development Proceeds from General to Specific Responses:** In mental as well as motor responses, general activity always precedes specific activity. For example, the baby waves his arms in general movements before he is capable of a specific response as reaching. Studies of speech have revealed that the young child learns general words before he learns to call each toy by its name.

- (v) **All Individuals are Different:** Although all individuals follow a definite and predictable pattern of development each individual has his own distinct style of doing so. Some develop in a smooth, gradual, step by-step fashion, while others move in spurts. Some show wide swings, while others show only slight ones. Individual differences are due partly to differences in hereditary endowment and partly to environmental influences. There are fewer differences in physical structure than in intellectual differences. Difference in special aptitudes seems to be the most mark of all.

- (vi) **Direction of Physical Development:** Physical development follows two laws:

- (a) **Cephalocaudal Principle.** This means that improvements in structure as well as in control of different areas of the body come first in the head region, then in the trunk, and last, in the leg region. Not only do the structures in the head region develop sooner than those in the leg region, but not only control comes first in the upper areas of the body and last in the lower areas.

- (b) **Proximodistal Principle.** According to this principle, development proceeds from near to far-outward, from the central axis of the body toward the extremities. Head and trunk develop before the limbs. Arms develop before the fingers. Functionally, the baby can move his hands as unit before he can control the movement of his fingers.

- (vii) **Each Stage has characteristic Traits:** Our life span is divided into a number of stages, namely, parental stage, infancy, childhood, adolescence, adulthood and old age. Each of these stages is characterized by certain problems of adjustment.

- (viii) **Development comes from Maturation and learning:** Physical traits are developed in two ways, partly from intrinsic maturing of these traits and partly from exercise and experience of the individual. Development through intrinsic maturity is known as Maturation and development through one's experience and exercise is termed Learning.

- (ix) **Development follows a Familiar and Predictable Pattern:** There is a particular pattern of development for each species, animal or human. Development follows a regular genetic sequence during the prenatal period, in progressive stages. Similarly, the postnatal period also witnesses an

organized development, although facts develop more quickly than memory for abstract or theoretical thinking. Cases of incompatibility in the rate of development of different physical and mental traits, lead to problems in adjustment. If the intellectual development exceeds or outpaces the physical, emotional or social development the child will be out of step, with his contemporaries or even other children.

(x) **The global of all development is self-realization:** In general, the overall goal or objective of individual development is **self-realization** which is defined as the **motive to achieve one's potential**. It is difficult to define what maturity is. A person who has attained physical maturity may still want in intellectual maturity with a still longer way to go to reach social or emotional maturity. Further, the level of maturity for one person may not be applicable to the other. Each person is endowed with certain growth potentials which can be developed to maximum level for that individual, given a favorable environment.

Arnold Gessel has identified the following **important principles of development**.

1. **The Cephalo-caudal Principle.** In general, structures in and around the **head area develop first**, and structures in and around the tail area develop last.
2. **The Proximodistal Principle.** In general, the **brain and nervous system and the internal organs develop earlier than the extremities** and the physiological systems associated with them.
3. **The Differentiation Principle.** In general, the development of a new organ or subsystem begins with its growth as a relatively undifferentiated mass which nevertheless is identifiable as a separate part of the organism, and then, once this mass has grown, it differentiates into finer and more interrelated subparts.
4. **The Bilateral Principle.** **Humans have bilaterally organized bodies, with many parts appearing in pairs, one on each side.** During development, each number of a given pair of parts appears at about the same time and develops at about the same rate as the corresponding member of the pair.

### **Definition of Growth and Development**

The terms growth and development are often synonymously used in daily language but not so in Psychology. Before we begin our discussion, therefore, we would carefully note the meanings of the terms "growth" and "development". The term **"growth" refers more to the physical growth-growth from the fertilized egg at time of conception to the fully grown body of the adult, that is, growth means increases in the size of the various parts of the body.** The term **"development" refers to the progressive changes that take place with time in the behavior of the organism and which lead to maturity.** Through separate, growth and development are simultaneous processes. They are also inter depended in the same that stunting of growth may often, though not always, result in deficits of development and vice-versa.

### **Distinction between Growth and Development**

The two main pints of difference between growth and development are as follows:

- (a) **Growth denotes the structural and physical changes** within the body of the individual right from the moment of conception to the adult period. Development refers to growth and the scope of physical and mental progress a person is capable of achieving.
- (b) **Growth denotes quantitative changes**, it shows an increase in the size and structure of the body and organs. **Development refers to changes which are qualitative and directional.** The changes are improvement and move forward rather than backward.

Along with the concept of growth and development, we must also understand the meanings of two other concepts which play an important role in human development. The first one is maturation and the second one is learning.

### **Periods of the Span**

Human development progresses through a series of various stages. The entire life span of human begins can be divided into eight periods. The divisions of the life span in these eight periods are somewhat arbitrary especially in adulthood. We will discuss these eight stages or periods in brief.

1. **Pre-natal Period.** It ranges from birth to conception. During this stage, the following important developments take place:

- (i) Basic body structure and organs form
- (ii) Physical growth is most rapid in this period of the life span.
- (iii) Vulnerability to environmental influences is great.

2. **Infancy and Toddler hood.** This stage ranges from birth to the age of three. The following important developments characterize this stage of development:

- (i) The new-born is dependent and competent
- (ii) All sense operate at birth
- (iii) Physical growth and development of motor skills are rapid.
- (iv) Ability to learn and remember is present, even in early weeks of life.
- (v) Attachments to parents and other form toward end of first year.
- (vi) Self-awareness develops in second year.
- (vii) Comprehension and speech develop rapidly
- (viii) Interest in other children increases.

3. **Early Childhood.** This period of development ranges from 3 to 6 years of life. The following important developments take place during this period:

- (i) Family is still the focus of life, although other children become more important.
- (ii) Strength and fine and gross motor skills improve.
- (iii) Independence, self-control, and self-care, increase.
- (iv) Play, creativity and imagination become more elaborate.
- (v) Cognitive development leads to many 'Illogical' ideas about the world.
- (vi) Behavior is largely eccentric, but understanding of other people's perspective grows.

4. **Middle Childhood.** This period ranges from 5 years to 12 years. The following important changes take place during this period:

- (i) Peers assume central importance
- (ii) Children begin to think logically, although largely concretely.
- (iii) Egocentrism diminishes.
- (iv) Memory and language skills increase
- (v) Cognitive gains improve the ability to benefit from formal schooling.
- (vi) Self-concept develops, affecting self-esteem
- (vii) Physical growth slows.
- (viii) Strength and athletic skills improve

5. **Adolescence.** The period extends from the age of 12 to puberty to around 20 years. The following major developments take place during this period.

- (i) Physical changes are rapid and profound
- (ii) Reproductive maturity is attained
- (iii) Search for identity becomes central
- (iv) Ability to think abstractly and use scientific reasoning develops.
- (v) Adolescent egocentrism persists in some behaviors.
- (vi) Peer groups help to develop and test self-concept
- (vii) Relationships with parents are generally good.

6. **Young adulthood.** This is period that ranges from 20 to 40 years. The following major developments take place during this period.

- (i) Decisions are made about intimate relationships
- (ii) Most people marry; most become parents
- (iii) Physical health peaks, then declines slightly
- (iv) Career choices are made

- (v) Sense of identity continues to develop
- (vi) Intellectual abilities assume new complexity.

**7. Middle age.** This period ranges from the age of 40 years to 65 years. The following important developments take place during this period.

- (i) Search for meaning in life assumes central importance.
- (ii) Some deterioration of physical health, stamina, and prowess takes place.
- (iii) Women experience menopause
- (iv) Wisdom and practical problem-solving skills are high; ability to solve novel problems declines.
- (v) Double responsibilities of caring for children and elderly parents may cause stress
- (vi) Time orientation changes to "time left to live".
- (vii) Launching of children typically empty nest.
- (viii) Typically, women become more assertive, men more expressive.
- (ix) For some, career success and earning powers peak; for others "burn-out" occurs.
- (x) For a minority, there is a mid-life "crisis".

**8. Late Adulthood.** This is period ranging from 65 years of age and over.

The following important changes take place during this period:

- (i) Most people hearty and active, although health and physical abilities decline somewhat.
- (ii) Most people are mentally alert. Although intelligence and memory deteriorate somewhat, most people find ways compensate.
- (iii) Slowing of reaction time affects many aspects of functioning.
- (iv) Need to cope with losses in many areas (loss of one's own faculties, loss of loved ones).
- (v) Retirement from work force creates more leisure time by many reduce economic circumstances.
- (vi) Need arises to find purposes in life to face impending death.

### Research Methods in the Study of Human Development

There are, broadly speaking, two methods of research in the study of human development. There are as follows:

#### I. None-experimental methods

1. Case study method
2. Naturalistic observation
3. Laboratory observation
4. Interview
5. Correlational studies

#### II. Experimental methods

### Non-Experimental Methods

**1. Case Study Method.** Case studies are studies of a single case or individual. Much of the data for psychoanalytical theories comes from case studies- careful notes and interpretations of what patients have said under psychoanalysis.

Our earliest sources of information about infants' development are case studies- careful notes and interpretation of what patients have said under psychoanalysis.

Our earliest sources of information about infants' development are case studies called baby biographies, journals in which parents recorded children's day by-day development. The first that we know about was begun in 1601; one of the most famous was Charles Darwin's about his son; and perhaps the most influential were those of Jean Piaget, whose theories about how children learn drew extensively on observations of his own three children.

An important more recent case study is the poignant story of "Genie". Case studies offer useful, in-depth information, giving a rich description of an individual. But these studies have shortcomings.

- (i) Case study does not tell us about the exact cause-effect relationship.
- (ii) Furthermore case studies do not explain behavior, and if they try to Case study can also be affected by "observer bias". That is, the recorder may emphasize some aspects of a person's

development and minimize others. While case studies may tell a great deal about individuals, then, it is questionable how well the information applies to people in general.

**2. Naturalistic Observation.** In naturalistic observation, researchers observe and record people's behavior in real-life settings. (Like pre-schools or nursing homes). They do not manipulate the environment or later behavior. To gain normative information, researchers observe people; record data about their development at various ages; and derive average ages, or norms, for growth and for the appearance of various skills and behaviors.

One type of naturalistic observation is time sampling, a technique used to determine how often a particular behavior (like aggression, babbling or crying) occurs during a given period of time. The method of naturalistic observation is of great use, especially in those cases where normal human aspects of children's behavior cannot be studied in laboratory settings. They have to be studied as and when the behavior occurs. In such cases, naturalistic observation methods are the best method.

Observation studies have a number of drawbacks.

- (i) Observation studies cannot explain or determine cause and effect.
- (ii) The very presence of the observer can alter the behavior being observed.

**3. Laboratory Observation.** Here the researchers observe and record behavior in settings that have been designed to place all the subjects in the same basic situation and to yield information on the subjects' behavior in this experimental situation.

Laboratory observation has the greatest advantage of telling us about the exact cause and effect relationship. It is one of the most scientific and reliable type of observation. However its greatest disadvantages are as follows:

- (a) Problem of generalizability. It is difficult to generalize with high degree of accuracy, the finding obtained in laboratory settings to everyday life settings.
- (b) Occurrence of demand characteristics and experimenter bias.
- (c) Many important aspects of human behavior cannot be studied in laboratory setup due to ethical and moral reasons.

**4. Interview.** In an interview, researchers ask questions about people's attitudes or opinions or some other aspect of their lives. By interviewing a large number of people, investigators get a picture of what people say they believe or do or have done.

A problem with interview is that the memory and accuracy of the interviews may be faulty. Some subjects forget when and how certain events actually took place. Others distort their replies to make them more acceptable to the questioners or to themselves. Finally, the wording of a question can influence how people answer it.

**5. Correlational Studies.** Correlational means relationship between two variables. One method of studying how two variables are related is to determine the relationship between them. Let us explain this with the help of an example.

Suppose that we want to determine the relationship between two factors; for example, between a parent's intelligence and a child's language ability. By carefully measuring both the factors (which are called variables, because they vary among members of a group or can be varied for purposes of an experiment) we might find that higher a parent's scores on intelligence test, the larger a child's vocabulary is at a given age. If so, we have found a positive correlation between the parent's intelligence and child's vocabulary.

Correlational studies show the direction and magnitude of a relationship between variables. That is, they can tell us whether two variables are related positively (that is, both increase together and decrease together) or negatively (as one increases, the other decreases), and to what degree. Correlation are reported as numbers ranging from - 1.00 (perfect negative, or inverse, or direct, relationship) to +1.0 (a perfect positive, or direct, relationship). The higher the number whether + or -, the stronger the relationship (either positive or negative). A correlation of zero shows that there is no relationship between the two variables. Knowing the correlation (or

relationship) between variables. Knowing the correlation (or relationship (or relationship) between variables allows researchers to make predictions about development.

However, Correlational studies do not give any information about cause and effect. A strong positive correlation does not tell us they, say, a parent's intelligence level causes the size of a child's vocabulary. A child's large vocabulary might have resulted from a third factor; a shared favorable environment that helped the parents to do well on intelligence test and encouraged the child to develop a large vocabulary.

### Types of Experiments

There are three types of experiments:

- (a) Those conducted in the laboratory.
- (b) Those conducted in the "field", setting that is part of the subject's everyday life, and
- (c) Those that may use of naturally occurring events.

**Laboratory Experiments:** In a laboratory experiment the subject is brought to a specific place and experiences conditions under the experimenter's control. The researcher records the subject's reaction to these conditions, possibly contrasting it with the same person's behavior under different condition or with the behavior of people who experience a different set of conditions. In the first type of laboratory experiment parents and children, for example, might be brought into a room so that researchers can measure the strength of parent-child attachment by seeing what happens when the mother leaves the child, when the father leaves the child, or when a stranger leaves the child. In the second type of laboratory experiment, some children might see a person acting aggressively while other children do not; then both groups of children would be measured on the degree to which they act aggressively themselves. Laboratory experiments permit the greatest control over the situation and are the easiest studies to replicate (that is, they are the easiest for other researchers to carry out in exactly the same way). But because of the artificiality of the situation, the subjects may not always act as they would in real life.

**Field Experiments:** In a field experiment, experimenters make a change in a familiar setting, like school or home. The experiment in which the parents adopted new reading routines was a field experiment.

**Natural Experiments:** A natural experiment compares people who have been accidentally divided into separate groups by circumstances of life—one group who were exposed to some naturally occurring event and another group who were not. Natural experiments are not true experiments, because they do not try to manipulate behavior; but they provide a way of studying events that cannot be created artificially. For example, it would be unethical to separate identical twins at birth just to do an interesting experiment, but if we discover identical twins who did happen to be separated at birth and raised in different circumstances, we can compare the effects of different environments on people with the same heredity.

### Methods of Data Collection

Information about development is most commonly gathered by cross-sectional or longitudinal studies. In some cases sequential designs are used.

**Cross-Sectional Study:** In a cross-sectional study, people of different ages are assessed on one occasion. This kind of study provides information about differences in development among different age groups, rather than changes with age in the same person (which longitudinal studies show).

In one cross-sectional study, people in six different age groups from age 6 to old age took a battery of cognitive tests. Age differences in performance appeared: middle-aged subjects scored highest and young children and older people scored lower (Papalia, 1972). Cross-sectional studies look at differences among groups of people. The advantages of the cross-sectional

method include speed and economy; it is faster and cheaper than the longitudinal method. In addition, it does not lose subjects, who drop out. Among its drawback is its masking of differences among individuals, since it looks at group averages. Its major disadvantage is that it cannot eliminate cohort, or generational influences on subjects born at different times. Cross-sectional studies are sometimes misinterpreted as yielding information about developmental changes in groups or individuals; but such information is often misleading and may contradict longitudinal research. For example, it would be incorrect to conclude from the cross-sectional study that intellectual functioning declines in later years. This may be so, but longitudinal data would be needed to determine whether there were actual age changes. All that the cross-sectional method can show is that there were age differences in performance.

**Longitudinal Studies:** In a longitudinal study, researchers measure the same people more than once to see changes in development over time. The researchers may measure one characteristic, such as vocabulary size, IQ, height or aggressiveness.

Or they may look at several aspects of development, to find interrelationships among factors. Since the same people are measured more than once, this design gives a picture of the process of development. One classic longitudinal study of gifted children, started by Terman followed young school children with high IQs into old age and found that their intellectual, scholastic and vocational superiority held up over time. Also, looking at differences within the group itself, researchers identified factors other than intelligence that seem to foster success in life (P. Sears & Barbee, 1978; Terman & Oden, 1959). The great strength of longitudinal studies is their sensitivity to individual patterns of change, since data about individuals can be tracked. Also, they avoid cohort effects within a study—although longitudinal studies done on one particular cohort may not apply to a different cohort. (A study done on people born in 1930 may not apply to subjects born in 1980). Longitudinal studies, however, are time-consuming and expensive. Another shortcoming is probable bias in the sample; people who volunteer tend to be of higher-than-average socio-economic status and intelligence, and those who stay with the project over time tend to be more competent than those who drop out. Also, the results can be affected by repeated testing; people tend to do better in later tests because of a "practice effect".

**Sequential Studies:** The cross-sectional study is one of several strategies assigned to overcome the drawbacks of longitudinal and cross sectional studies. This method combines the other two: people in a cross-sectional sample are tested more than once, and the results are analyzed to determine the differences that show up over time for the different groups of subjects. Some important research on intellectual functioning in adulthood employs sequential techniques. These techniques seem to provide a more realistic assessment than either the cross-sectional method (which tends to overestimate the drop in intellectual functioning in later years) or the longitudinal method (which tends to underestimate it).

## DIFFICULTIES IN STUDY OF DEVELOPMENT PSYCHOLOGY

A research who studies human development has to face a number of difficulties. Hurlock has identified following difficulties or obstacles in the study of human development.

**1. Representative sample** of subjects is one of the most common drawback that a researcher of developmental psychology faces. It is generally difficult to get a representative sample of subject at different age levels. The greatest difficulty arises in selecting children who are newborns or infants because of strong parental objections. Similarly obtaining some of elderly adult belonging to old age is also extremely difficult.

**2. Establishing Rapport with Subjects.** The second obstacle scientists' encounter in studying development is establishing rapport with subjects at different age levels. There is no guarantee that scientists will be able to elicit the information they are seeking from any group unless they are able to establish rapport with their subjects. Therefore, there is no guarantee that the data they obtain is as accurate or as comprehensive as it might have been had a better relationship existed between subjects and experimenters.

The reason for this is that obtaining information from subjects of any age is extremely difficult because most people resent having a stranger pry into their personal affairs.

**3. Methodological obstacles.** The third obstacle scientists' encounter in studying development during the life span is securing a satisfactory method. This is because no one method can be used satisfactorily for studying people at all ages or for investigating all areas of development. Some of the methods that must be restored to, for lack of better ones, are of dubious scientific value.

Because of the wide range of subjects and the variety of different areas of development that must be studies to give a composite picture, assorted methods have had to be used. Some have been borrowed from medicine, from the physical sciences, and from related social sciences, especially anthropology and sociology. Some have made use of laboratory settings, and others of the naturalistic settings of the home, school, community, or work environment. Some are regarded as reliable, while others, especially the retrospective and introspective techniques, are of questionable value.

**4. Accuracy of Data obtained.** The fourth obstacle scientists' encounter in studying development during the life span is ensuring that the data obtained from the studies will be accurate. Inaccuracies may result when a biased sampling of subjects gives a false picture of the normal developmental pattern at a particular age. This can happen, for example, when institutionalized elderly people are used for the study and the subjects try to present as favorable a picture of them as they can and, either consciously or unconsciously, distort their introspective or retrospective report. It can also occur when the only method available for studying a certain area of development is less than satisfactory.

In the measurement of intelligence it is still questionable if the results are accurate for the first two years of life. There is every controversy about the accuracy on intelligence tests for older age levels. Observational techniques for the study of behaviour during the preschool years are questioned for accuracy because of the tendency of observers to draw inferences from their study of children behaviour and speech. Studying well-being life satisfaction or happiness is very difficult because only subjective measures can be used. The accuracy of such measures is open to question.

**5. Ethical Aspects of Research.** The fifth obstacles scientists encounter in studying development during the life span involves the ethical aspects of research. Today there is a growing trend to take this into account, and it has been a stumbling block to certain kinds of studies, which, in the past, were made without consideration of their fairness to the subjects studied. Within the trend nowadays towards considering the rights of subjects, emphasis is being placed on asking their consent to participate in experiments, or for the very young, the consent of their parents or guardians.

Another aspect of ethical consideration is with regard to confidentiality of results obtained. How much confidentiality or privacy has to be kept is also a highly debatable issue.

### Perspective of Human Development

The way people explain development depends on how they view the fundamental nature of human beings. Different psychologists have developed different theories and explanations about why people behave as they do. Some important theories are as follows:

- I) psychoanalytic (a) Freud's psychosexual theory  
 II) Organismic Piaget's cognitive-stage theory

**I. Psychoanalytic perspective** was developed by **Dr. Sigmund Freud** who viewed human beings as **controlled and guided by unconscious forces**. Freud is well known for his theory of psychosexual development.

**(a) Theory of Psychosexual Development**

Freud believed that **personality is developed during the first few years of life when children learn to deal with conflicts between their biological and sexual urges and the demands of society**. According to this theory, during each stage of development, psychic energy gets localized on certain erogenous zones - i.e. certain areas of the body which, when stimulated, leads to pleasure. There are **three such zones** that become **important at each stage of development**:

(i) Mouth, (ii) Anus and (iii) Genitals

**Fixation, an arrest in development, may occur if children receive too little or too much gratification** at any of these stages, **they may become emotionally fixated or stuck**, at that stage and may **need help in order to move beyond it**. For example, a baby who is weaned too early or is allowed to suck too much may become an excessively distrustful or dependent adult. (However, Freud was vague about what constituted "too early" or "too much".)

**The various stages of psychosexual development are as follows:**

**Prenatal Stage:**

- Oral stage** which ranges from birth to **12-18 months**. During this stage, the baby's chief sources of pleasure are **mouth-oriented activities** like sucking and eating.
- Anal stage** extends from **12-18 months to 3 years**. The child derives sensual gratification from **wild drawing and expelling feces**. Zones of gratification is the anal region.
- Phallic stage** extends from **3 years to 6 years**. During this stage, sexual energies in the children are activated. The **male child develops sexual attraction towards the mother which is called as Oedipus complex**. The **female child develops sexual attraction towards the father which is known as Electra complex**. During this stage of a female child develops penis envy and male child develops castration anxiety. All this affects his later personality development.

**Latency Stage:** This **stage extends from 6 years to puberty**. During this stage no psychosexual development takes place. The child mixes among his own sex peers.

**Genital Stage:** This **stage begins from puberty and lasts through adulthood**. During this stage, the child develops mature sexuality and indulges in heterosexual adjustment.

Id., ego and superego. Freud saw the human personality as made up of **three elements** which he called the **Id**, the **ego**, and the **superego**.

Now-borns are governed by the **id**, and **unconscious source of motives and desires** that is present at birth and seeks immediate gratification under the **differentiate themselves from the outside world**. All is there for their gratification, and only when it is delayed (as when they have to wait for food) do they develop an ego and begin to tell themselves apart from their surroundings).

Id has all the qualities of a spoilt child. It wants, what it wants when it wants. It cannot delay gratification. It is illogical, irrational and has no regard for time, place and person. The **ego represents reason, or common sense**. According to Freud, the ego develops during the first year of life and operates under the reality principle as it strives to find acceptable and realistic way to obtain gratification. Freud calls ego as the executive of personality. It is rational, logical and has to serve three masters; id, superego and the reality principle. Freud calls ego as battlefield where three forces are constantly fighting among themselves. These are id, superego and reality principle.

The **superego**, which develops by about age 5 to 6, represents the values that parents and other adults **communicate to the child**. Largely though the child's identification with the parent of the same sex, the superego incorporates socially approved duties and prohibitions "shoulds" and "should nots" into the child's own value system. **Superego is also rigid and seeks perfection**. Whenever Id and superego takes over the ego, pathology results.

## PRENATAL DEVELOPMENT

### Characteristic of Prenatal Period

1. Although this period is relatively short it has six important characteristics according to Hurlock. The **heredity endowment**, which serves as the **foundation for later development is fixed**, once and for all, at this time. While favorable or unfavorable conditions both before and after birth may and probably will affect to some extent the physical and psychological traits that make up this hereditary endowment. The changes will be quantitative not qualitative.
2. Favorable conditions in the mother's baby can foster the development of hereditary potentials while unfavorable conditions can stunt their development, even the point of distorting the pattern of future development.
3. The **sex of the newly created individual is fixed at the time of conception** and conditions within the mother's body will not affect it. as is true of the hereditary endowment. Except when surgery is used in sex transformation operations, the sex of the individual, determined at the time of conception, will not change; such operations are rare and only partially successful.
4. Proportionally **greater growth and development take place during the prenatal period than at any other time** throughout the individual's entire life. **During the nine months before birth**, the individual grows from a microscopically small cell to an infant who measures approximate twenty inches in length and **weights on the average, 7 pounds**. It has been estimated that weight during this time increases eleven million times. Development is likewise phenomenally rapid. From a cell that is round in shape, all the bodily features, both external and internal, of the human being develop at this time. At birth, the newly born infant can be recognized as human even though many of the external features are proportionally different from those of an older child, an adolescent, or an adult.
5. The prenatal period is a time of many hazards, both physical and psychological. While it cannot be claimed that is the most hazardous period in the entire life span - many believe that infancy is more hazardous - it certainly is a time when environment or psychological hazards can have a marked effect on the pattern of later development or may even bring development to an end.
6. The prenatal period is the time when significant people form attitudes towards newly created individuals. These attitudes will have a marked influence on the way these individuals are treated especially during their early, formative years. If the attitudes are heavily emotionally weighted, they can and often do play havoc with the mother's homeostasis and, by so doing, upset the conditions in the mothers' body that are essential to the normal development of the newly created individual.

### Body System of the New-Born Infant

A young infant has its own body systems. However, before birth the fetus blood circulation respiration, nourishment, elimination and temperature regulation are all accomplished through its connections with the mother's body.

Some important body system of young infants are as follows:

- (a) **Circulatory system:** Before birth, mother and baby have independent circulatory system and separate heartbeats, but the fetus's **blood is cleansed through the umbilical cord**, which carries blood to and from the placenta. At birth, the body's own system must take over. The neonate's heartbeats are still fast and irregular, and blood pressure does not stabilize until the tenth day.
- (b) **Respiratory system:** The **fetus gets oxygen through the umbilical cord**, which also carries away carbon dioxide. The new-born who needs much more oxygen must now get it independently. Most infants start to breathe as soon as they emerge into the air. A **baby who is not breathing within 2 minutes after birth is in trouble**; if breathing **has not begun in 5 minutes** or so, some degree of **brain injury** from anoxia (lack of oxygen) may result. Infants' lungs have only one-tenth as many air sacs as adults thus infants are susceptible to respiratory problems.
- (c) **Gastrointestinal system:** The **fetus relies on the umbilical cord to bring food and carry body wastes away**. The new-born has a strong **sucking reflex to take in milk and has gastrointestinal secretions of digest it**. Meconium (sticky, greenish-black waste matter formed in the fetal intestinal tract) is excreted during the first two days or so after birth. When the neonate's bowels and bladder are full, the sphincter muscles open automatically. Many months will pass before the baby can control these muscles.

Three or four days after birth, about half of all babies- and a larger proportion of babies born prematurely develop physiologic jaundice. Their skin and eyeballs look yellow. This kind of jaundice is caused by the immaturity of the liver, usually it is not serious and has not long-term effects. In some cases, it is treated by putting the baby under fluorescent lights.

(d) **Temperature Regulation:** The layer of fat that develops during the last months of fetal life enables healthy full-term infants to keep their body temperature constant despite changes in air temperature. Newborn babies also maintain body temperature by increasing their activity in response to a drop in air temperature.

### Fertilization for Conception

The process, by which sperm and ovum fuse to form a single cell, is most likely to occur about 14 days after the beginning of a woman's menstrual period. The new single cell formed by the two gametes or sex cells the ovum and the sperm is called a zygote. Once conceived this zygote duplicates itself again and again by cell division.

At birth, a human female has about 400,000 immature ova in her two ovaries each ovum in its own small sac, or follicle. The ovum is the largest cell in the human body. From the time a female matures sexually until menopause, ovulation occurs about once each menstrual cycle (for most women, every 28 days): a mature follicle in one ovary ruptures and expels its ovum. The ovum is swept along through the fallopian tube by tiny hair cells called cilia, toward the uterus, or womb. It is in the fallopian tube that, if the ovum meets a sperm cell, fertilization normally occurs. The sperm shaped like a tadpole but only 1/600 inch from head to tail - is one of the smallest cells in the body. Sperm are produced in the testicles (testes), or reproductive glands of a mature male at the rate of several hundred million a day and are ejaculated in the semen at sexual climax.

The sperm enter the vagina and begin to swim up through the opening of the cervix, the neck of the uterus. From the uterus, they head into the fallopian tube, but only a few sperm actually get that far. That protective layer around the ovum needs to be worn down so that one, and only one, sperm cell will actually penetrate the membrane. About 20 million sperm cells must enter a woman's body at one time to make fertilization likely, but only one of them can fertilize an ovum to conceive a new human being. From the time of ejaculation sperm maintain their ability to fertilize an ovum for up to 48 hours: and ova can be fertilized for about 24 hours after release from the ovary. Thus there is a "window" of about 48 hours during each menstrual cycle when sexual intercourse can result in fertilization. Sperm reach a woman's reproductive tract up to 24 hours before or after the ovum is released and are capable of fertilizing that ovum. If fertilization does not occur, the ovum and the sperm cell in the woman's body die. The sperm are absorbed by the woman's white blood cells, and the ovum passes into the uterus and exits through the vagina. If sperm and ovum do meet, they conceive a new life and endow it with a rich genetic legacy.

### Heredity and Environment Influences

Psychologists and other social scientists have attempted to find the influence of heredity and environment on the individual. What determines an individual's behavior, develop his/her personality. It is heredity or environment or interaction of both.

People have asked this question for years, and the answer differs for different traits. Some physical characteristics like eye color and blood type are clearly inherited. But more complex traits having to do with health, intelligence and personality are subject to interplay of the both hereditary and environmental forces.

How much of a given trait is influenced by heredity and how much by environment is difficult to answer.

Some traits governed by genes (like eye color) do not seem to be affected by the environment or by a person's own behavior. But many traits are subject to variation, within the limits set by the genes. Genetic influences are not necessarily all-powerful, then; they may set up a range of possible reactions among people living in a particular range of environmental conditions. Also, genes are not expressed directly as behavior, thus they may be expressed differently in different environments. How our genetic inheritance shows itself depends to a considerable extent on our specific environment.

Heredity and environment have different influences and affect the different traits and behavior differently.

## METHODS OF PRENATAL ASSESSMENT

Following are the methods of prenatal assessment.

**Amniocentesis:** In amniocentesis, a sample of the fluid in the amniotic sac is withdrawn and analyzed to detect the presence of various genetic defects. This fluid in which the fetus floats in the uterus contains fetal cells. The procedure can be done in the fifteenth or sixteenth week of pregnancy; it takes about 2 weeks to get the results. Amniocentesis can also reveal the sex of the fetus which may be crucial in the case of a sex-linked disorder like hemophilia. Amniocentesis is generally recommended for pregnant women if they are at least 35 years old, if they and their parents are both known carriers of Tay-Sachs disease or sickle cell disease, or if they or their partners have a family history of such conditions as Down syndrome; spina bifida, Rh disease, or muscular dystrophy. One analysis of 3000 women who had the procedure indicated that it was "safe, highly reliable and extremely accurate" (Golbus et al, 1979, p-157). But another study of 4600 women found a slightly higher risk of miscarriage in women who had the procedure (Tabor et. al, 1986).

**Chorionic Villus Sampling:** Chorionic Villus sampling (CVS) consists of taking tissue from the end of one or more villi-hair like projections of the membrane around the embryo which are made up of foetal cells. These cells are then tested for the presence of various conditions. This procedure can be performed earlier than amniocentesis (in the first trimester) and it yields results sooner (within about a week). CVS has a higher risk of procedural failure and loss of the foetus than amniocentesis (Rhoads et al., 1989); and these other problems are raising some doubts about its use (E. Rosenthal; 1991).

**Maternal blood tests:** Blood taken from the mother between the fourteenth and twentieth week of pregnancy can be tested for the amount of alpha fetoprotein (AFP) it contains. This maternal blood test is appropriate for women at risk of bearing children with defects in the formation of the brain or spinal cord (like anencephaly or spina bifida), which may be detected by high AFP levels may suggest Down syndrome (DiMaio, Baumgarten, Greenstein, Saal & Mahoney 1987). To confirm or refute the presence of suspected conditions, ultra-sound or amniocentesis, or both, may be performed.

Blood tests can also identify carriers of sickle-cell disease (a blood disorder seen mostly in black people). Tay-Sachs disease the thalassemia (a blood disorder that affects people of mediterranean origin). And they can reveal the sex of the foetus which can be of help with sex-linked disorders (Loetal., 1989).

**Ultra Sound:** Ultra sound is used to measure a baby's head size, to judge gestation age, to detect multiple pregnancies to evaluate uterine abnormalities, to detect major structural abnormalities in the foetus and to determine whether a foetus has died as well as to guide other procedures like amniocentesis. Although some obstetricians administer ultrasound routinely it has been in use only for very short time, and so its long-term effects are yet unknown (Kleinman, Cooke, Machlin & Kessel, 1983). Because animal studies have suggested possible harmful effects, the National Institute of Health (1984b) recommended ultrasound only for a specific medical reason, (e) Umbilical cord assessment - By threading a needle into tiny blood vessels of the umbilical cord under the guidance of ultrasound doctors can take samples of a fetus's blood. They can then get a blood count, examine liver function and assess various other body functions. This procedure can test for infection, anemia, certain metabolic disorders and immunodeficiency's, and heart failure and it seems to offer promise for identifying still other conditions. The telephonic is associated with such occasional problems as bleeding from the umbilical cord, early labour and most serious infectious (Chervenak, Isacson & Mahoney, 1986; Kotala, 1988).

### Neonatal Period

The neonatal period refers to the first few weeks of life, during which, the new born or neonate makes radical adjustment to the demands of the extra uterine environment. Though the length of this period has been defined variedly by different authors, we can assume that it is a period extending up to the first four weeks after birth in the life of an individual. Following are some of the important physical characteristics of the neonate.

#### Physical characteristics of neonate

1. Average neonate is 20 inches long, 7 ½ pounds in weight and 18-22 inches long.
2. During first few days, neonates lose as much as 10 percent of their body weight, primarily because of a loss of fluids. They begin to gain weight again at about the fifth day and are generally backed to birth-weight by the tenth or fourteenth day.
3. At first the skin of the neonate is thin, sensitive and very ruddy in appearance some have little white bumps around the nose and on the cheeks: darker complexioned babies often have bluish colorations on the back or the buttocks. Some are born with a full head of hair as well as a fine down on the ears, the lower back, and the shoulders.
4. The neonate's body has a frail look: it does not yet have the thick fatty tissue that gives baby the usual soft roundness. It keeps its legs bent in at the knee with its feet flexed outward, which tends to give the limbs an awkward look.
5. The eyes of the newborn seem small. The lids are puffy: the eyes have a dull color because pigmentation has not yet developed and the gaze has an absent almost unseeing quality. Most babies have blood spots on the eyes for the first few days, due to pressure during delivery. The nose, too, often suffers some temporary distortion from having been pressed down during birth, and the head is molded.

#### Types of Adjustment

Neonate is a period of adoption and adjustment. Some of the adjustments that neonate has to make are as follows:

**Adjustment to temperature changes:** The temperature inside the mother's uterus is about 100°F. After the birth the temperature of the body will be around 68 to 70°F, which will again change later depending upon the temperature of the environment. The neonate, for its survival and healthy adjustment has to adapt to such temperature changes.

**Adjustment to circulation of blood:** When the umbilical cord is clamped and cut. One major branch of the circulatory system, as it existed in the fetal stage, closes down. The main blood vessels to and from the placenta shut off shortly after birth.

As circulation through the umbilical cord stops; circulation through the lungs increases. The neonate has to adjust to such circulation which appears for the first time.

The newborn's heart beats rapidly from 120 to 140 beats per minute. During the first weeks of life the blood is very rich in hemoglobin, the substance that carries oxygen to the tissues.

**Adjustment to breathing:** The fetus is surrounded by a fluid environment within the amniotic sac in the uterus. The supply of oxygen in this situation comes from the placenta through the umbilical cord. This cord is cut after birth and thereafter it becomes necessary for the infant to inhale and exhale air. The birth cry is usually timed, with the beginning of breathing, and helps to inflate the lungs, and set the process of breathing in motion. The breathing may be irregular in the beginning but the yawning, gasping, sneezing and coughing of the baby helps to regulate the supply of air. Gradually the new born develops the ability to harmonize the breathing process to an effortless routine.

**Adjustment to taking Nourishment:** Sucking and swallowing is reflex actions for the new born, it takes time to perfect these reflex. Until such time the new born is able to master these reflexes, he is not able to take adequate feed which makes him lose weight.

As long as he was within the uterus, he continued to receive his nourishment without any effort, through the umbilical cord. It takes time for the neonate to adjust to the postnatal situation, where he has to exert himself through his reflexes to get his nourishment.

**Adjustment of elimination:** While inside the uterus, the waste products were being eliminated through the umbilical cord and the maternal placenta. After birth, the new born has to get him adjusted to the elimination of waste products from the body through the functioning of the excretory organs.

**Adjustment to infection:** Before birth the baby receives a valuable supply of antibodies directly from its mother especially toward the end of pregnancy. These antibodies protect it at birth when it is suddenly confronted with an environment filled with germs, many of which invade its skin, digestive tract, and respiratory tract almost instantly. The protection given by some of the mother's antibodies lasts for about a month in the baby. Other antibodies from the mother, against diseases such as measles, polio and hepatitis, lasts from six to twelve months.

The mother's antibodies protect the baby against most of the common infection diseases. The neonate gradually has to build up his own immunity to infection.

## INFANCY

The word infancy is derived from the Latin word "infants". This period extends from birth to 2 weeks after birth.

Infancy begins at birth and extends up to two weeks (14 days). During this period no new development takes place and there is some reduction in the weight at this stage the adjustment to the totally new environment outside the mother's body must be made. The development again starts only when the infant adjusts itself to the external environment.

### Characteristics of Infancy

The following are the important characteristics of this period:

**Extreme helplessness-** This is a period of extreme helplessness. A young infant is generally dependent upon the caretakers, especially his mother for his basic needs. However during this period newborn gradually becomes independent.

**Radical Adjustment-** takes place during this period. A young infant makes adjustments with respect to Breathing, temperature regulation elimination etc.

**Loss of Weight-** During this period while making adjustments, the young infant loses considerable weight.

**Disorganized behavior-** For the first day or two of postnatal life, all infants show relatively disorganized behavior, such as irregularities in breathing rate, frequent urinations and defecations and wheezing. This is due partly to pressure on the brain during birth, which results in a stunned state and partly to the underdeveloped state of the autonomic nervous system, which controls body homeostasis.

**High infant mortality rate-** Even today, the rate of infant mortality during the first two days of postnatal life is high. The causes of infant mortality are many and varied.

**Developmental tasks-** The developmental task of the infancy includes learning new modes of locomotion, learning to take solid foods, establishing day time toilet control, learning to communicate by gestures and words, as well as learning to relate emotionally to parents, siblings and other persons in the home environment.

### Physical Growth and Development

1. The period of infancy is one where rapid physical development and growth takes place. In infants following important changes take place in physical growth and development.

Besides increase in height and weight, which are obvious signs of growth, there is also less obvious signs of growth in muscles. Muscles increase not only in size but in the precision and control with which they can be used.

2. **Bones harden and skeleton is transformed** from a gelatinous mould into a sturdy frame.
3. **Brain cells grow and become specialized.**
4. **Infants' body weight doubles in first five months and triples by the end of the first year.** While the **length increases by 20 percent in three months and 50 percent in one year.**
5. At-the birth the skeleton is composed largely of cartilage and in an Zray looks somewhat like a puzzle whose pieces are on the verge of being locked into place. Most of the bones are still separated, and the relatively large spaces between them give the infants joints flexibility. Infants bones contain more water and protein like substances than do mature bones, and they have a lower mineral content.
6. **By birth all twenty baby teeth and a few permanent teeth are developing,** although they will not usually begin to make their appearance until around the middle of the first year. The baby teeth paper in a predictable sequence, but the age at which they erupt varies greatly. So does the amount discomfort that accompanies teething. The first teeth usually erupt between four and twelve month of age with the average at seven months. A small number of babies are actually born with one or two teeth, while a few children do not get their first tooth until sixteen months. Neither extreme is cause for concern.
7. **By six months the child has gained considerable control over voluntary movements.** Different parts of the brain develop in accordance with the movements. First are the cell controlling the upper trunk, neck and upper arm, those controlling the legs and head follow.
8. **Motor development occurs in an orderly,** predictable manner according to a sequential pattern. The sequence follows the law of developmental direction of **"cephalocaudal" or head to foot, "Proximodistal"** which is development from the main axis to the periphery. Muscle control for e.g. first appears in arms then in fingers. In motor development in infancy there is one important achievement which is the achievement of upright locomotion or walking.
9. Another **important motor achievement during infancy is the ability of the infant to use his hands.**
10. **Brain weight increases dramatically in the first year of life.** As the brain develops and the cerebral cortex increases in size, more areas of the child's body as well as visual and auditory faculties cone under greater control. As the brain stem ceases to exert much control, there Ilex actions of the new born disappear.
11. The **autonomic nervous system is not well developed in the young infant.** Hence due to this he is unable to maintain homeostasis, which is one of the causes of the high mortality rate.
12. The **respiration rate at first range from 40 to 45 breathing movements per minute.** By the end of the first week of life, it normally drops to approximately 35 per minute and is more stable then it was at first.
13. The infant's sleep is broken by short waking period which occur every two or three hours, with fewer and shorter waking periods during the night than during the day.

### Perception among Infants

Perception **is one of the most important sense which helps the child to interact with world.** We would now briefly discuss the manifestation of different aspects of perception among infants:-

**Sight:** Much is know about the **sight of infants than about their other senses.** **Newborn baby can distinguish between shades of light and dark.**

A few weeks after the birth, babies can indulge in visual co-ordination. It is the ability of the eye muscles to turn the eyes in the direction of an object.

Convergence is also an important step in the refinement of visual experience. Babies do make a rough attempt at convergence in the fist hours of life but do not perfect it until the end of the second month. During the first three months visions is blurry because the muscles cannot yet adjust the lens curvature to focus the light rays reflected by objects sharply on the retains. This adjustment is called accommodation of the lens. It begins in the second month and is fully developed by the third of fourth month (Haynes. White, and Held. 1985) long before infants can recognize faces, and even before the physical processes of vision are fully developed, babies

seem predisposed to look around and examine the world. In a recent study (Mendelson and Haith. 1976) newborn infants were tested for visual scanning activity. Even in darkness; when there was nothing to be seen, they scanned quite actively, which suggests that humans are born with some built in procedures for learning about their world. In light the infants tended to scan areas that contained many contours- lines edges of shape and soon. These areas would, of course, be more likely to offer useful information than would blank, undifferentiated areas such as a bare wall. Even though newborn infants certainly do not form any clear concept or memory of what they "see" by this early scanning, they are already building up a fund of experience that will help them later.

**Perceptual attention:** An infant's perceptive frame of reference changes however, almost from month to month. During the first three months an object must have relatively well-defined characteristics to attract the baby's attention- such characteristics as movement sound sharp color contrast or distinctive contours and patterns. The youngest babies seem interested in very simple patterns older ones in more complex designs. However, there must not be too much novelty. Apparently babies must be able to perceive the new object as in some way like things they have been seen before. If it is totally new and different, they cannot fit it into any frame of reference and soon seem to lose interest.

**Depth perception:** The ability of baby to perceive depth during the first year has been established by a famous experiment using a "visual cliff" (Gibson & Walk. 1960: Walk. 1966). A sheet of heavy plate glass was laid over a surface with a checkerboard pattern. On one side the check board pattern was directly under the glass; on the other side it was several feet below; giving the impression of a cliff just along the brink of the 'cliff a wide platform was laid over the glass. Six-month-old babies were placed on the platform and coaxed it in mothers from the far side of the deep area. They could have crossed it in perfect safety to the glass was strong and securely anchored. The question was would they see the depth and be afraid to cross it or would they be reassured by the feel of the glass under their hands and knees and take no notice of the depth? As it turned out, the visual impression was stronger. Most of the babies could not be tempted into venturing off the platform on the "cliff side. They had no hesitation about crossing on the other side, where there was no visible drop. And when the check board pattern on the deep side was raised to within a few inches of the glass-making the "cliff" look much less dangerous some babies were willing to risk the crossing. Evidently then infants do have a definite awareness of depth.

How much judgment is acquired is not clear. Some authorities maintain that babies have built in depth perception at birth others believe they are taught by the experience of seeing objects at various levels and distances.

### Motor Development among Young Infants

Infants indulge in a wide variety of motor movement. This movement is evident even in the womb.

By about the fourth month, voluntary cortex-directed movements largely take over, Motor control, the ability to move deliberately and accurately, develops rapidly and continuously during first 3 years, as babies begin to use specific body parts consciously. The order in which they acquire this control follows three principles of development; (i) head to toe, (ii) inner to outer and (iii) simple to complex.

Two of the most distinctively human motor capacities are the precision grip, in which thumb and index finger meet at their tips to form a circle.

### Characteristics of Infants Emotions

There are certain characteristic features of infant's emotions which make them different from those of adults.

**Brief:** The young child's emotions last only a few minutes and then end abruptly.

**Intense:** A response to even a trivial situation calls forth an emotional reaction of great intensity.

**Transitory:** The young child shifts rapidly from laughter to tears, from age to smile.

**Appear frequently:** A child displays his emotions frequently than the typical adults.

**Differences in emotional responses:** There is a wide variability in infants emotional responses. For example, New Delhi child may run away from the fear-producing object, the other may hide behind the mother and the third may stand and cry.

**Easily detectable:** Infants emotions can be detected easily.

**Change in strength:** An emotion which is strong at a certain age level reduces in strength as the child grows older and the emotion of anger becomes stronger.

**Change in patterns of emotional expression:** The pattern of emotional expression varies from one age level to the other. For example, a very young child expresses anger by temper tantrums and other physical means. The older child expresses anger verbally.

### Expression and Type of Emotions

Infants express their emotions in a wide variety of ways; they express unhappiness more obviously than happiness. The manners in which individuals express emotions help them to gain control over their environment. The meaning of emotional language changes as babies develop. We will discuss how certain emotions are shown and expressed by infants.

**Crying:** Important needs of the infant are communicated through crying. Babies have four patterns of crying (i) the basic **hunger cry** (a rhythmic cry which is not always associated with hunger), (ii) the **angry cry** (a variation of the rhythmic cry in which excess air is forced through the vocal chords), (iii) the **pain cry** (a sudden onset of loud crying without preliminary moaning, sometimes followed by holding of the breath). Babies in distress cry louder, longer, and more irregularly than hungry babies

**Smiling:** A baby's smile is almost irresistible. Parents usually greet a baby's first smile with great excitement, and adult who see a smiling baby will almost always smile back. The early faint smile that appears soon after birth occurs spontaneously as a result to central nervous system activity. It generally appears without outside stimulation, often when the infant is falling asleep (Sroufe & Waters, 1976)

In their second week, babies often smile after a feeding, when they are drowsy and may be responding to the caregiver's sounds. After this, smiles come more often when babies are alert but inactive. At about 1 month, the smiles become more frequent and more social, directed towards people. Babies smile now when their hands are clapped together or when they hear a familiar voice (Kruutzer & Charles Worth, 1973; Wolf, 1963). During the second month, a visual recognition develops; babies respond more selectively, smiling more at people they know than at those they do not know.

Some infants smile much more than others. Babies who generously reward care taking with smiles and gurgles are likely to form more positive relationships with their caregivers than babies who smile less readily.

**Laughing:** At about the fourth month of life, babies, start to laugh out loud. They chortle at being kissed on the stomach, hearing various sounds, and seeing their parent do unusual things. In additions, some of their laughter may be related to fear. Babies sometimes react to the same stimulus (like an object looking toward them) with both fear and laughter (Sroufe & Wunshc, 1972).

As babies grow older, they laugh more often and at more things.

## PIAGET THEORY OF COGNITIVE DEVELOPMENT

The Swiss psychologist Jean Piaget took a different approach from that of the psychometricians to children's cognitive development - the growth in thought processes that enables them to acquire and use knowledge about the world.

The Piagetian approach describes quantitatively different stages of cognitive development that characterize children from infancy through adolescence.

Piaget's theory of cognitive development conceives of intelligence as developing through four stages.

1. Sensorimotor development
2. Preoperational development
3. Concrete operational development
4. Formal operational development

Since Sensorimotor development takes place in infancy and toddler hood we will discuss it in brief.

Piaget's first cognitive stage is called the Sensorimotor stage. It is a developing when infants learn about themselves and their world through their own developing sensory and motor activity. During the first 2 years babies change from creatures who respond primarily through reflexes and random behaviour into goal-oriented toddlers. They now organize their activities in relation to their environment, co-ordinate information they receive from their senses and progress from trial and error learning to using rudimentary insights in solving simple problems. During the Sensorimotor stages, children develop several important cognitive concepts:

1. One is object permanence: the realization that an object or person continues to exist even when out of sight. Object permanence is the basis of children's awareness that they exist apart from objects and other people. It allows a child whose parent has left the room to feel secure in the knowledge that the parent continues to exist and will return. It is essential to understanding time, space, objects.
2. Another important concept that emerges during this stage is causality, the Piagetian term for the recognition that certain events cause other events. Awareness of causality develops at about 10 months. Sub stages of sensorimotor development; there are six sub stages of sensorimotor development. These are as follows:-

**Sub stage 1:** Use of reflexes (birth to 1 month)

**Sub stage 2:** Primary circular reactions and acquired adaptations (1 to 4 months).

**Sub stage 3:** Secondary circular reactions (4 to 8 months)

**Sub stage 4:** Co-ordination of secondary schemes (8 to 12 months)

**Sub stage 5:** Tertiary circular reaction (12 to 18 months).

**Sub stage 6:** Beginning of thought (18 to 24 months)

We will discuss these in brief.

**Stage-1: Reflex Activity (1st month of life)** In this early stage the child's behaviour is essentially limited to exercising innate skills - that is, reflex actions like sucking, that are the product of heredity. Initially these actions are automatically triggered by external stimulation. By the end of this stage, through visual and other sensory experiences with environment, infants are able to use their innate skills in a somewhat discriminating and proficient (though basically primitive) manner. When really hungry, for example, infants now appear able to distinguish the nipple from other objects and surfaces and to maneuver their head and mouths in order to find it.

**Stage-2: Primary Circular Reactions (1 -4 months).** Infants begin to define some of their reflex actions and to express repetitive behaviour such as opening and closing their fists, hand

sucking, foot kicking etc. The reason Piaget calls such behaviors circulation is that the infants are reaching to the pleasure they have derived from chance actions our unplanned behaviour by seeking to repeat this behavior in a trial and error fashion. Infants also begin to develop primitive anticipations on the basis of past experience: For example they start to make sucking motions on being put into a posture that signals to them that they are about to be fed.

**Stage-3: Secondary Circular Reactions (4 - 10 months).** In the previous stage infants were preoccupied with their own bodily activities, such as the movements of their hands. In this stage they begin to turn their attention toward objects and occurrences in the external environment. Infants begin to perceive some connection between a given action and its consequences in the "outer" world. Through accidental body movements, for example, an infant may cause a toy to make a sound. Finding this even interesting or pleasurable, the infant is motivated to try to reproduce the action or series of movements that resulted to try to reproduce the action or series of movements that resulted in the sound. The infant's behaviour is now becoming somewhat intentional or purposeful, involving a more refined trial and -error process than observed previously.

**Stage-4: Coordinating Secondary Schemes (10-12 months).** The infant begins applying patterns of behaviour learned in the previous stage to new or problematic situations. Piaget noted how his infant son Laurent had already developed the habit of hitting hanging objects and then, in the fifth

month, of actually striking the objects to make them swing. Previously an end in itself, this "schemes of striking" was used by Laurent in state 4 as a means: in an effort to grasp a box of matches held by Piaget. the son first struck at a cushion that Piaget held up as an obstacle. According to Piaget the infant's behaviour has become much more intentional and thus for the first time, intelligent.

**Stage-5: Tertiary Reactions (12 - 18 months).** The child becomes interested in objects in themselves. For example, infants may repeatedly move or drop different items, like a piece of food or a toy, from various positions and heights, and then watch how the objects "responds". From these "experiments" children begin to understand more about the nature of the external environment. Children also become more proficient at imitating simple behaviors of adults - such as pointing a finger to the forehead then they were in previous stages.

**Stage-6: Invention of New means through Mental Combinations. (18-24 months).** This stage heralds the beginning of thought. Children begin to solve practical problems by thinking them out before taking action, rather than simply using hit-or-miss physical means to achieve an end. As Piaget noted in observing his son Laurent, children of this stage who find that a desired object is beyond their reach may use some other objects such as a stick to bring the desired objects closer to them. The mental imagery or internalized symbolism this requires can also be seen children's imitative behaviour. In the past the child could only imitate behaviour in the presence of a model. Now the child can observe the behaviour, make an internal representation, and reproduce the behaviour later on, when the model is absent. For example, Piaget's daughter Jacqueline watched a young boy throw a taper tantrum in his play pen; on the next day she copied that tantrum in her own playpen.

Infant's mental imagery now permits a very different conception of objects. In the earliest stage, when an object or person goes out of sight, the infants act as though it no longer exists. In the course of the sensorimotor period the infant slowly begins to realize that objects have a reality of their own. Separate from his or her body or actions. It is not until stage 6, however, that the concept of object permanence is truly acquired, an object say a spoon of object permanence is truly acquired, an object-say, a spoon - can be shown to the child and then concealed in the adult's hand. The spoon can then be moved from one hiding-place to another as the child watches. At stage 6 infants will search for the spoon in the last place where the adult had been seen enter. They understand that the object exists even when concealed. And they now have the ability to make a mental image of the object, so that they can "follow" it through a series of invisible displacements.

### Development of Language

Language learning or developing largely depends upon maturation of brain processes and physical structures concerned with making of speech sounds. By the end of the second month, infants begin to laugh and coo, making soft how vowel sound in response to others.

At about 5 or 6 months, babbling being. Infants combine vowel and consonant sound in strings of syllables like bababa and dabata. Babbling increases until the infant is between to 12 months of age then decreases after the first real words are produced.

Infants utter their first word around the age of 11 to 12 months. They know and understand more words than they actually utter. That is, comprehension develops earlier and more rapidly than production. Infants frequently respond to questions and instructions like 'Where is hour bottle?' or 'pat the doggy' with gestures and actions before they say any words. Around the age of 18 month's the child has the vocabulary of about 50 words. These generally refer to things that are important to the child. After the age of two years the vocabulary continues to grow at a rapid age. The child learns to string these word into sentences. The first two word sentences usually appear at about 18 months. Language development among infants occurs in stages. Children, the world over, go through certain stage is acquiring knowledge, which are as follows:-

- (a) Crying and babbling
- (b) Followed by one-word sentences
- (c) Followed by longer or two-word sentences
- (d) Telegraphic speech and
- (e) The use of complex grammatical rules.

At every stage and in every process the child appears to test hypothesis about appropriate usage; sometimes, even generalizing rules of construction and overextending the meaning of words, but eventually mastering language.

According to Papalia and Olds (1992) there are four stages of languages development. These are follows:-

1. Prespeech. 2. The first word, 3. The first sentence, 4. Early syntax. During the infancy stage pre speech is most important. It refers to sound that the children make even before they learn cooing and babbling. Babies ca distinguish between sounds long before they can utter anything but a cry. In the first few months of life, they can tell apart similar sounds like bah and pah (Ei as et al.. 1971). This ability to differentiate sound seems to be an inborn capacity that people lose as they her the language spoken around them. Japanese infants, for example, can easily tell ra from la, but Japanese adults have trouble making the distinction (Bates, O Connell, & Shore, 1987). Babies seem to lose this ability at about 9 to 10 moths or age, when they begin to understand meaningful speech, but before they are physically mature enough to produce their own.

Prespeech can be divided into:

- (a) Crying, (b) babbling and (c) imitation of sounds

We would discuss each of these in brief.

a. Crying is the new-born's first and only means of communication. To a stranger a baby's cries may sound alike, but the baby's parents can often tell, for example, the cry for food from the cry of pain. Different pitches, patterns, and intensities signal, hunger, sleepiness, or anger.

At anywhere from 6 weeks to 3 months of age, babies start to laugh and coo when they are happy making squeals gurgles and vowel sounds like ah. A kind of "vocal tennis" begins at about 3 months, when they begin to play with speech sounds producing a variety of sound that seem to match the one's they hear from the people around them (Bates et al., 1987).

b. Babbling repeating consonant-vowel strings like ma-ma-ma-occurs rather suddenly between 7 to 10 months of age, and these strings are often mistaken for a baby's first word. Early babbling is not real languages, since it does not seem to have meaning for the baby, but it become more word-like, leading to early speech.

c. At first, babies accidentally **imitate sound** they hear. Then they imitate themselves making these sounds. At about **9 to 10 months of age they deliberately imitate other sound without understanding them.** Once they have this basic repertoire of sounds, they string them.

### Theories of Language Acquisition

There are **two theories of language** acquisition. These are;

1. Learning theory
2. Nativism

We would discuss these two theories in brief.

**Learning Theory:** According to learning theory, **children learn language in the same way that they learn other kinds of behavior through reinforcement.** Parents reinforce children for making sounds that resemble adult speech, and so children make more of these sounds, generalizing and abstracting as they go along.

Behaviorists believe that **children utter sound at random,** and that those which sound like adult speech are the n reinforced. Social learning theorists maintain that children imitate the sounds they hear adults making and then are reinforced for doing so, thus children English-speaking countries learn English rather than another language. Mutation may explain why children generally outgrow incorrect usage even when their parents do not correct their grammar (R. Brown, Czden. & Beilugi, 1969)

Learning theorist point to the fact that children reared at home, who presumably hear more adult speech and get more attention and more reinforcement than those who grow up in institutions, do babble more (Brodbeck & Irwin, 1946). However, learning theory does not account for children's marvelously imaginative ways of saying things they have never heard,

**Nativism:** According to Nativism, **human beings have an inborn capacity for acquiring language and learn to Talk as naturally as they learn to walk.** Evidence for this viewpoint comes from several facts:

- (a) **Almost all children learn their native language.** No matter how complex, mastering the basics in the same age-related sequence without formal teaching.
- (b) Human beings, the only animals that have a spoken language, are also the only species in which the **brain is larger on one side than the other** and seems to have an **inborn mechanism for language localized on the larger hemisphere** (for most people, the left hemisphere).
- (c) New-born respond to language in sophisticated way. They move their bodies in the rhythm of the adult speech they hear (Condon & Sander, 1974): they can tell their mother's voice from those of strangers (A De Casper & Fifer, 1980); and in the first month of life, they can distinguish very similar sounds. All this suggests that language is inborn.

Most developmentalists today draw on both Nativism and learning theory; they believe that children enter the world with an inborn capacity to acquire a language which is then activated and enhanced by learning through experience.

### Factors Influencing Language Development

Language development is one of the most important aspects of general development of young infant. **The basic speech sounds are acquired through interaction of maturation and learning.** Certain amounts of neuron physiological growth are necessary to produce certain sound and a certain stage in cognitive development is required to use certain words.

There are many factors that influence the development of language among infants and Toddlers. These are

**Maturation:** The **two general variables** on which the development of language depends are maturation and learning. The stages in the development of **speech, vocabulary**, sentence formation and skills as regarding and writing are related so much to certain age levels that these

levels are auspicious times requiring encouragement, stimulation and proper care. Expectation of standards higher than the maturational level in general mental and motor development result in difficulties in speech reading or writing.

**Intelligence:** Another variable that determines language development is learning. Ability and achievement in learning depends on intelligence, physical factors and social influences. As the growth and development of speech on imitating the sounds, comprehending the context situation and establishing relation between the spoken sound and word concept, it also depends on intelligence. That is why the feeble minded are late in speech development. The extent of vocabulary, which is primarily a measure of language development, is so clearly related to the level of intelligence during preschool years that is also used as a measure of intelligence.

**Physical Factors:** The speech development of the young child suffers from prolonged illnesses. The physically disabled child is isolated from other children. Moreover, he is so much protected by the parents that he gets everything he wants without asking for. Thus there is no need of talking. In the case of deaf children, as they do not get any model to imitate they show poor pronunciation and vocabulary.

Meaning it is and other similar disorder also affects language development among infants.

**Social influences:** As languages is a means of communication, it develops in the social context. Hence the environmental conditions at home, school and pay are very closely related to the language achievement. It is seen very clearly from the language achievement by children in the orphanage as compared with that shown by children from good families and well equipped schools. If orphanage children are given a training program for enriching language experience in listening, reading and expression, it stimulates the growth of language and intelligence too. Cultural handicaps in poor families and in the case of neglected children even in good families affect the development of language of language. The environment conditions are more associated with the occupation of fathers. Children from higher professional groups show early speech development extensive vocabulary, and expressiveness.

**Bi-lingualism:** Many children have to learn two languages at a time, their mother-tongue and the foreign or regional language. This creates confusion. Learning two words for one thought, and two sets of grammar, is not an easy task. In this case, his thinking is likely to be confused and he becomes therefore, self-conscious about talking. It interferes with school work and formation of abstract symbols, and consequently results in the retardation of growth of intelligence and creativity.

**Family Constellation:** The size of the family and number of siblings make a difference in the language experience and the language model for the child. The only child is more in contact with adults and shows acceleration in all phases of language development. Twin children learn to talk more slowly because they imitate each other's speech and do not get model from older children and adult to copy. Presence of twin gives social and emotional satisfaction without any efforts. While the singleton has to seek for it through conversation with adults.

**Sex:** In the beginning boys are behind girls in the acquisition of verbal skills. This might be because of their interest in motor activities and play. Superiority of girls is found at very age in tests of vocabulary and grammar. A few recent studies show that boys make up for this initial retardation.

### Ways of Learning

Learning is defined as a relatively permanent change in behavior due to experience or practice. Children learn in different ways. Some of which are as follows:-

- (i) Habituation
- (ii) Classical conditioning
- (iii) Operant conditioning
- (iv) Through more complex learning that combines more than one model
- (v) Limitation

We would discuss each of these in brief.

Habituation is a simple of learning. Experience plays an important role in habituation. Because habituation is associated with normal development, its presence or absence, as well as the speed with which it occurs, can tell us a great deal about a baby's development. Since the capacity for habituation increases during the first 10 weeks of life, it is regarded as a sign of maturation (Rovee-Collier, 1987). Habituation studies show us how well babies can see and hear, how much they can remember, and what their neurological status is. Babies with low Apgar scores and those with brain damage, distress at the time of birth, or Down syndrome show impaired habituation (Lipsitt, 1986) as do neonates whose mothers were heavily medicated during childbirth.

## EARLY CHILDHOOD

Early childhood is also called as preschool years and it is a period which generally ranges from 3-6 years. The childhood is the most crucial year of one's life span.

### Physical Development and Growth

Physical development during early childhood proceeds at a relatively slow rate as compared to the rapid rate of growth in infancy. The early childhood period covers from age 3-6 years.

Early childhood is a period of relatively even growth. However, there are variations indifferent individuals with respect to how they develop physically.

Children of superior intelligence, for example, tend to be taller in early childhood than those of average or below-average intelligence and to shed their temporary teeth sooner. While sex differences in height and weight are not pronounced, ossification of the bones and shedding of the temporary teeth are more advanced, for age, in girls than in boys. Because children from higher socioeconomic groups tend to be better nourished and receive better prenatal and postnatal care, variation in height, weight, and muscular development are in their favor.

Following are important points to note with respect to physical development:-

- a. At age three the average boy is a little over three feet (90 cm) tall and weighs over thirty pound (13.5) by age five had grown to about forty-four inches (110 cm) and forty three ponds (19.3 kg). There are individual variations, but girls tend to be slightly shorter and lighter than boys. Children who are big for their age at the beginning of the preschool years will probably still be big for their age at the end of the period. There is also some correlation between the height of preschool children and the height they will attain as adults, though there are exceptions: A child who is small at four or five may well develop into a lamer than average adult.
- b. Boys and girls develop at about the same pace during the early childhood till they reach puberty.
- c. One of the most significant developmental differences between boys and girls is that boys have more muscle and girls more fatty tissue. In both sexes the infant fatty tissue is gradually replaced, but girls tend to retain it longer than boys.
- d. Body proportions change dramatically in the preschool period. The trunk and legs grow rapidly but cranial growth is not so fast as before. As a result, by age six the legs are about half the length of body, a ration that will remain constant for the rest of the child's life. In other words, the average six-year old already has the body shape of an adult.
- e. The bones ossify at different rates in different parts of the body, following the laws of developmental direction. The muscles become larger, stronger, and behavior, with the result that children look thinner as early childhood progresses, even though they weigh more.
- f. Differences in body build become apparent for the first time in early childhood. Some children have an endomorphic or flabby, fat body build, some have a mesomorphic or sturdy, muscular body build, and some have an ectomorphic or relatively this body build.
- g. Around the age of five years the brain has reached 75% of its adult weight and only a year later, 90%

- h. It is during the early childhood that hyalinization- a sheathing of nerve fibers in the brain composed of a white insulating material called myelin is completed.
- i. During early childhood many important physiological changes occur. Breathing becomes slower and deeper, the heart beats more slowly and steadily and especially in boys; blood pressure increases.
- j. Another important physical development during early childhood is with respect to growth of teeth. During the first four to six months of early childhood, the last four baby teeth - the back molars erupt. During the last half year of early childhood, the baby teeth begin to be replaced by permanent teeth. The first to come out are the front central incisors - the first baby teeth to appear.  
When early childhood ends, the child generally has one or two permanent teeth in front and some gaps where permanent teeth will eventually erupt.

Highlight by Buzybee

## Lesson 5

## COGNITIVE DEVELOPMENT DURING EARLY CHILDHOOD

The word **cognitive** is roughly equivalent to **thought**. Hence cognitive development means development of thought. Intelligence is one important aspect of our thought.

Piaget regards the development of intelligence as an **interaction between an individual's maturation and his or her social and physical environment**. **Cognitive development**, according to Piaget, **takes place by means of two processes**: organization, in which one psychological structure is integrated with another, and adaptation, in which psychological structures are modified in the course of interactions with the environment. **Adaptation** consists of **assimilation and accommodation**. Individuals **assimilate each new experience to what they already know**, and the **same time accommodate or modify what they know to take the new experience into account**.

According to Piaget there are **four stages** of cognitive development these are:

1. **Sensorimotor period** (birth to 2 years)
2. **Pre-operational period** (from two years to seven years)
3. **Period of concrete operation** (from 7 years to 11 years)
4. **Period of formal operations** (from 11 years onwards)

The Early childhood is a period of pre-operational thought.

The pre-operational stages is the significant step beyond the Sensorimotor period because preoperational children can learn not only by sensing and doing but also by thinking symbolically, an no only by acting but also by reflecting on their actions. The major important development of the preoperational stages is the symbolic function.

The **symbolic function is the ability to learn by using symbols**. A symbol is a mental representation to which consciously or unconsciously, a person has attached meaning. It is something that stands for really comes into it between ages 2 and 6. Symbols allow us to think about objects or events without actually having them in front of us. An object can be a symbol, taking on in people's mind the qualities of whatever it stand for.

The most common symbol - and probably the most important one for thought - is the word, at first spoken and then written. Knowing the symbols for things helps us to think about them to incorporate their qualities, to remember them, and to communicate with other people about them. Symbolic thought is, therefore, a great-advance over the sensorimotor stage.

**Three ways in which children show the symbolic function are**

**Deferred imitation**

**Symbolic play**

**Language**

**Deferred imitation:** is imitation of **an observed action after their time has passed**. For example, Raju aged 3, **sees his father shaving**. Later, at his pre-school, he heads for the housekeeping corner and begins to "shave" According to Piaget, Raju saw the shaving, formed and stored a mental symbol (probably a visual image), and later when he could no longer see it reproduced the behavior by calling up the stored symbol.

**In symbolic play:** **children make an object stand for something else**. Shoba, age 4, makes her finger stand for a bottle and **she "feeds" her doll by putting her finger to its mouth**. Her laughter as she does this show that she knows her finger is not really a bottle.

**Language:** The **symbolic function is most impressive in language**. Preoperational children use language to stand for absent things and for events that are not taking place at the time. By using the words apple tree to stand for something that was not there. Shoba made an utterance with symbolic character

The **preoperational period of cognitive development begins at about two years of age** and characterizes a child's mental development until he is about seven years old. The symbolic

representation that **characterizes preoperational thought differs from the sensorimotor intelligence** of infancy in several fundamental ways.

1. **First, representational thought is faster and more flexible.** Unlike Sensorimotor intelligence, which can link actions and perceptions one at a time but cannot achieve a cohesive overview, representational thought can grasp many events at one. For example, a boy may enjoy sailing a toy boat. Representational thought allows him to think about sailing it in the bathtub in a mud puddle, or in a pond. He can consider which of these mothers will allow and where he would be most likely to meet his friends. Representational thought helps him to decide where he wants to sail his boat on this particular occasion.
2. **Second, representational thought is not limited to goals of concrete action.** Children at this stage are capable of reflection and of re-examining their knowledge; they can contemplate as well as act. The young sailor can remember that last time he sailed a boat in a mud puddle it got stuck, and the when he sailed in the pond, he lost it.
3. **Third, representational thought enable children to deal with numbers and such qualities as size.** By thinking with symbols they can extend their scope beyond themselves and the concrete objects they encounter every day. This sailors boat may have one most and one sail, and although he may never have seen a boat with more, his understanding of number enables him to conceive of a boat with three - or, in principle, six or twelve - masts and any number of sails per mast.
4. **Finally, representational thought can be codified and most important, socialized; that is, we can translate our thoughts into form that can be communicated to other individuals.** Whereas sensorimotor intelligence is private and unshakeable, the child at the preoperational stage can sail his boat with a fries, and the two of them can imagine together that one is that captain and the other the mate. The enriched intelligence and imagination of the preoperational stage can make even private play into a representation of shared social experience.

### **Limitations of Preoperational Thought**

In some ways, of course, **preoperational thinking is still rudimentary compared with what children will be able to do when they reach the stage of concrete operations in middle childhood.** For example, preoperational children do not yet clearly differentiate reality from fantasy. Some of the limitations of the preoperational thought are as follows:

**Centralization:** **Preoperational children tend to centrate, they focus on one aspect of a situation and neglect others, often coming to illogical conclusions.** They cannot decentre or think simultaneously about several aspects of a situation.

A classic example is Piaget's most famous experiment. He designed it to test children's development of conservation - the awareness that two things that the equal remain so if their shape altered so long as nothing is added or taken away. He found that children do not fully understand this principle until the stage of concrete operations, normally in middle childhood.

Preoperational children cannot consider height and width at the same time. They centre on one aspect or the other and so cannot understand what is happening. Their logic is flawed because their thinking is tied to what they "see", if one glass looks bigger, they seem to think it must be bigger.

**Irreversibility:** **Preoperational children's logic is also limited by irreversibility; failure to understand that an operation can go tow ways.** One a child can conceptualize restoring the original state of the water by pouring it back into the other glass; he or she will realize that the amount of water in both the glasses is the same. **The proportional child does not realize this.**

**Focus on states rather than on transformation.** Preoperational children think as if they were watching a filmstrip with a series of static frames. They focus on successive states and are not able to understand the meaning of transformation from one state to another. Preoperational children do not grasp the meaning of pouring the water from the original to the new one. That is, they do not understand the implication of transforming B into B - the even though the appearance changes, the amount does not.

**Transductive reasoning:** Logical reasoning is of two basic types; deduction and induction. Deduction goes from the general to the particular. "Eating a lot of candy can make people sick. I ate a lot of candy today, and so I may get sick". Induction goes from the particular to the general: "yesterday I ate a lot of candy and felt sick. Last week I ate a lot of candy and felt sick. The same thing happened to Ram and Anju. Therefore it looks as if eating a lot of candy can make people sick."

Preoperational children, said Piaget, do not think along either of these lines. Instead, they reason by transduction: they move from one particular to another particular without taking the general into account. The kind of reasoning leads to see a casual relationship where none actually exists; "I had bad thoughts about my sister. My sister got sick. Therefore I made my sister sick". Because the bad thoughts and the sister's sickness occurred around the same time, the child assumes illogically that one cause the other.

**Egocentrism:** Egocentrism is an inability to see things from another's points of view. A classic Piagetian experiment known as the mountain task illustrates egocentric thinking. A child would sit on a chair facing a table on which were three large mounds. The experimenter would place a doll on another chair, on the opposite side of the table, and would ask the child to tell or show how the "mountains" look to the doll. Young children could not answer the question; instead, they persistently described the mountains from their perspective. Piaget took this as proof that they could not imagine a different point of view (Piaget & Inhelder 1967).

Egocentrism, to Piaget, is not selfishness but self-centered understanding, and it is fundamental to the limited thinking of young children. Egocentrism is a form of centration. These children are so centered on their own point of view that they cannot take in another's view at the same time. Three year-olds are not as egocentric as newborn babies, who cannot distinguish between the universe and their own bodies; but young children still think that the universe centers on them.

### Speech Disorders

Though there are many different childhood speech disorders, psychological disturbance appears to be implicated in only two of these conditions: delayed and stuttering.

**Delayed speech** - Most children say their first words within a few months after their first birthday. And between eighteen and twenty four months, they usually begin to formulate two or three-word sentences. There are wide individual differences in this schedule, and a few months delay in developing normal speech is rarely thought to have any diagnostic significance. Some normal children begin to speak much later than others. Albert Einstein, for example, did not utter his first words until he was fully three years old. However, a prolonged delay in speaking is usually taken very seriously, as a possible indication of an organic or functional disorder. In some cases, failure to speak is an early sign of autism, deafness, mental retardation, or some other specific form of brain damage. Often, however, there is more manageable, functional cause: lack of encouragement or verbal stimulation from the parents, a trauma such as hospitalization or an\ long separation from the parents, or a discouragement of the child's independence on the part of the parents.

**Stuttering** - Stuttering refers to the interruption of speech fluency through blocked, prolonged, or repeated words, syllables, or sounds. Many people stutter on occasion, and speech hesitation in young children is a very common phenomenon. Consequently, with stuttering as with so many other childhood disorders, it is often difficult to decide what a serious condition is and what is not. Persistent stuttering occurs in approximately 1 percent of the population, males outnumber females four to one. The disorder is not likely to appear either between two and three and-a-half years or between five and seven years. In any case, onset is almost always before age twelve (DMS-III 1980).

**Causes** - As usually, the etiological theories of stuttering are divided between biogenetic or psychogenetic causes. The bioorganic view suggests that stuttering results from organic impairment in the central nervous system. Support for this position again comes from the over representation of males among stutters (three to eight times more prevalent than females. In addition, it is often noted that stuttering is seen uniformly throughout the words, implying that culture makes little differences (Sheehan 1975).

The psychogenetic position maintains that stuttering is learned behaviour. One particular learning theory model was proposed by Bloodstein (1975): the anticipatory struggle hypothesis. This hypothesis proposes that the stutterer behaves as though he or she has "acquired a belief in the difficulty of speech..." For example, the child takes for granted that he or she will fail in speaking. Bloodstein further notes that stuttering occurs primarily in social situations and "become intensified when (he or she feels) socially ill at ease, tense, insecure or uncomfortable..." Since the stutterer can, in certain situations, speak rather fluently, is in the anticipation of failure and the struggle not to fail (anticipatory struggle) that undermines normal speaking. As Polow (1975) remarks, the person does not stutter so much when he or she feels confident. The harder the person tries, the more likely he or she is to fail.

Bloodstein offers several antecedent environment conditions that may cause a child to develop a poor self-concept and a belief in his or her ability to speak. Parents often demand perfection, or they may seem over concerned about their child's speech. All these factors lead to stuttering or other speech difficulties.

**Treatment** - Treatment based on the anticipatory struggle hypothesis involves counseling with the parents to lessen environmental pressure. Other essentials of treatment are 1 desired speech patterns should be reinforced; 2 the child's anxiety about stuttering should be reduced by bringing the problem out in the open; 3. The child should recognize that other persons do not always speak perfectly; 4 the child should not avoid the opportunity to speak Sheehan (1975) similarly suggests that the key element in therapy is "avoidance-reduction". That is, the stuttered should not try to avoid speaking, but should make every effort to speak when it is appropriate to do so. The stutterer should, therefore, talk more, not avoid activities because of the possibility that he or she will stutter.

### School Phobia

A phobia is an intense but unwarranted fear of some object or situation. Somatic symptoms such as nausea, refusal to eat, vomiting and abdominal pains may accompany the fear or anxiety. While children may develop animal phobias and transportation phobias (Kessler 1966), school phobia is the most common.

Specifically, a school phobia consists of anxiety, panic and often abdominal pains that develop when the child is faced with having to go to school. Bakwin and Bakwin (1972) remark that the fear is usually related to a particular teacher, classmate, or anticipation of an exam.

**Causes** - The hypothesized causes of school phobia, are rather simple learning theorist, for instance hold that the phobia has been acquired because it is reinforcing to avoid school. Slated differently the school setting is perceived as aversive; any behaviour that keeps the child from the dreaded situation will be reinforced or strengthened somatic complaints (such as nausea) are also thought to be learned, for they reduce the likelihood that the child will have to go to school. Another etiological thesis states that going to school is feared because the child has not learned to "separate" from his or her mother. The child is pathologically attached to her. and the prospect of separation engenders anxiety and panic.

Children with diagnosis of anxiety disorder are characterized by fear and apprehension associated with varying factors. Children with anxiety disorders are often unable to play, go to school, attend special events, or take part in many of the everyday activities of childhood.

**Form of Anxiety Disorders in Children**

Childhood anxiety may manifest itself in several forms. Among these are

Separation anxiety

Avoidant disorder

Over anxious disorder

**Separation Anxiety:** In this disorder, children show "exaggerated distress at the separation from parent home, or familiar surroundings" (DSM-III). They may ruminate about their parents becoming ill, injured, or killed. In some cases these worries may include fantasies about being kidnapped or banned when they are separated from their parents. Fears are shown by "expressing discomfort about leaving home, engaging in solitary activities and continuing to use the mothering figure as a helper in buying clothes and entering social and recreational activities" (Weckman. 1980). Anxiety increases during transitions such as among going to and from school, changing schools, or moving away from home.

**Avoidant Disorders:** As with most of the anxiety related disorders, in avoidant disorders children also have difficulties making transition. While usually fine at home, youngsters with avoidant disorders shrink from interaction and show embarrassment, timidity, and withdrawal when forced to come in contact with strangers. Timidity is a great roadblock to the building of normal peer relationships and to the experiencing of interpersonal activities necessary to growth and maturity. Avoidant youngsters, though not usually participating in many activities, do seem to want to be accepted by peers and to be competitive, in academic and athletic situations. However, should their initial effort meet with failure, they typically stop trying and quickly withdraw from the anxiety.

**Overanxious Disorder:** While rumination can be part of all anxiety disorders, it is the major symptom in the overanxious disorder. These youths ruminate about things such as examinations, possible future events, and past difficulties. Interested in pleasing others and usually quite conforming, overanxious children also are prone to gain attention by exaggerating their pains or illness and having more than their share of accidents. Their sleep is often disturbed because night time appears to be an especially favorable time to ruminate about the past day's event.

**Treatment of Anxiety Disorders**

Most of the interventions for children with anxiety - related disorders have been derived from the psychological paradigm. They can be subsumed under the main headings of play therapy and behavioral therapy. While play therapy was derived specifically for children, the behavioral methods are a subset of the more general behavioral therapies applied to adults as well.

## TREATMENT OF ANXIETY DISORDERS

Most of the interventions for children with anxiety - related disorders have been derived from the psychological paradigm. They can be subsumed under the main headings of play therapy and behavioral therapy. While play therapy was derived specifically for children, the behavioral methods are a subset of the more general behavioral therapies applied to adults as well.

### Play Therapy

Play therapists take advantage of the fact that children often can express themselves better in play than in talking. The approach can be used to treat less severely disturbed children, but it is especially useful with children who have limited verbal ability.

In psychoanalytical play therapy, the therapist role is to identify problems in psychosexual development and to attempt to understand the symbolic nature of the child's play behaviour.

Through "corrective" play and interpretation, insights and behaviour change may occur as the child is placed back on the normal track of psychic development. By skilful interpretation of the use of play materials, the psychoanalytic therapist can gain an understanding of the child and can direct the efforts towards more developmentally appropriate play object. If a child smears fingers paints for a long time, a therapist may try to help the child to use crayons - a more controlled mode of emotional expression. It is assumed that this more controlled mode of expression will generalize to situation outside, the playroom.

Instead of trying to redirect deviant psychosexual development, therapists exposing nondirective play therapy (Axling, 1964) provide an atmosphere of acceptance that will help the child work out problems with a minimum of direction and guidance.

### Behavior Therapies

Rather than focusing on the relationship between child and therapists, therapists using behavioral techniques emphasize procedures derived from learning theory. In token-reinforcement methods, tokens are given to children when they perform acceptable behaviour. Teachers have long used "tokens" like stars and smiling face stickers to reward positive performance of their students. A token is any object that has acquired value because it can be traded for something else of value. For example, a nickel is a "token" that can be traded for some gum. In token-reinforcement system, tokens are given desirable behaviour and can be used to obtain other valued items. Ten stars may enable a child to get a candy bar. Perhaps they receive one star every time they clean their room. Thus, each star represents one-tenth of a candy bar.

In a typical application of a token reinforcement system, parents, child and therapists all agree on target behaviors, or behaviors that may need to be altered. Target behaviors may include "dos" like home work and cleaning up rooms, or "don'ts" like not poking baby brother or talking with one's mouth. There are various ways to keep track of the children's positive behaviour. In one application children are presented with a pictorial representation of a road that has 10 "token booths" or "gates" with 10 stars or tokens required for passage through a gate. At the very end of the road is a representation of a previously agreed upon reinforcer (toy or a similar desirable object) that the child may obtain only by traversing the "road". Behaviors that will result in the awarding of a token are clearly set forth for the child. At each 10 token gate, a smaller reinforcer (a candy bar or comic book) may be given to the child. These intermediate reinforcers ensure the maintenance of behaviour from the beginning of the program to the final reward which may be as many as 1000 tokens down the road. Using the token reinforcement star road, such behaviors as school advances, lying, irritating a younger sibling, and refusing to do homework may be effectively changed and replaced with more acceptable behaviors.

Some professionals have suggested using progressive relaxation training for children who have high levels of tension or identifiable fears (e.g. O'Leannon, 1981). Procedures may have to be modified for treating children. They need to be rewarded for complying with the instruction and their sessions should be much shorter, probably about 15 minutes in length. However, it appears that therapists may be too creative in

adapting desensitization procedures for children (Hat/enbuehler and Schroeder. 1978) to the extent that there is little evidence for particular repeatable interventions (Rickard and Elkins, 1983).

Attention Deficit with Hyperactivity is extremely widespread. It is most common behavior disorder seen by child Psychologist Psychiatrist and the most common cause of childhood referral to mental health clinic. According to conservative estimates, it affects at least 5 percent of the elementary school population with boys outnumbering girls ten to one.

Inmost salient features are incessant restlessness and an extremely poor attention span, leading in turn to impulsive and disorganized behavior. These handicaps affect a almost every area of child's functioning. The inability to focus attention has a ruinous effect on the academic progress. Hyperactive children have great difficult in understanding instructions and finishing tasks; often they cannot even remember what they set out to do. Consequently, while often intelligent, they have severe learning problems. They are also extremely disruptive in the class-room, interrupting, darting her and there, and making incessant demands for attention.

Hyperactive children also show poor social adjustment. They knock over the other children's blocks, disrupt games, get into fight, refuse to play fair and throw temper tantrums when they do not get their ways. They do this not out of aggressiveness but due to low frustration tolerance.

Attention deficit with hyperactivity are usually variable. The child may seem greatly improved one week and then much worse the next week. The disorder also varies situational. In many cases, hyperactive children function adequately on a one-to-one basis but fall apart in group situations; hence they may do well at home but not at school. Hyperactive children seem to be easily distractible, and readily exited. One moment they may be frustrated and tearful because they cannot find their pencil, while at the next moment they will be laughing loudly because one of the other children made a strange noise. They are described by their parents as frequent criers, very active and erratic eaters also sleepers (Laufer & Shetty. 1980).

Although attempts have been made, no paradigm has generated an acceptable theory for the origins of hyperactivity. Some investigators with biological perspectives have implicated defects in cortical arousal (Rosenthal & Allen. 1978: Hastings & Barkley. 1978). environment toxins such as lead poisoning or food additives (Feingold. 1975), and minimal brain dysfunction (Mc Glannon, 1975) in causing hyperactivity. The wide variety of possible causes suggests that there may be more than one way to become hyperactive.

The proponents of psychological and biological paradigms have classed over the correct ways to treat hyperactivity. For example, drugs that usually stimulate the brain and energize behaviour, like amphetamines and caffeine, paradoxically calm hyperactive children (Goodman & Oilman. 1982; Safer & Allen, 1973). Others have pointed out that the prolonged use of antianxiety drugs like Dexedrine and Ritalina may result in a number of serous side effects such as liver disorders and suppressed growth (Laufer & Shetty, 1980). However, those opposed to drug treatment for hyperactivity have suggested the use of psychosocial^ based programmed learning procedures and, in some instances, cognitive behavioral techniques that combine positive reinforcement with verbal self instructions (e.g., Kendall & Finch, 1978; Meichenbaum & Goodman, 1971).

Long-term follow-up of hyperactive children shows that they fall further and further behind their peers academically. The resulting loss of self-esteem compounds their personal problems and they may become isolated or delinquent in their later years. (Minde et. al., 1971; Pelham, 1978;)

### **Middle or Late Childhood**

The term middle and late childhood is generally used to cover almost the same period. The middle childhood generally covers a period ranging from 6 years to 9 years. Whereas late childhood covers a period 9 years to puberty. However many textbooks consider the period from 6 years to puberty as belonging to middle childhood or late childhood. Physical development during middle childhood or late childhood is generally uniform. This is a period of slow and relatively uniform growth until the changes of puberty begins. Physical growth follows a predictable pattern. The physical growth during this period is not spectacular.

Following are some of the important aspects or characteristics features of this period of development.

1. Some children do experience a small burst of growth between all ages of six and eight. As children grow, their trunk tends to become slimmer, their chest broader, and their arms and legs longer and thinner. At the same time the size of the head gradually approaches 'the adult proportion of one-seventh to one-eighth of total height.
2. The annual increase in height is 2 to 3 inches. The average eleven year-old girl is 58 inches tall, and the average boy of the same age is 57.5 inches tall. Both boys and girls gain an average of 7 pounds. (Hi) Permanent teeth appear, making the bottom part of the face look "heavier". Towards the end of middle childhood youngsters may look as though parts of their body are out of proportion and indeed some parts do grow faster than others, causing even the most normal children to feel self-conscious or worried about their bodies.
3. Physical development is considerably influenced by the nutritional intake.
4. Physical, social and cognitive development is considerably influenced by nutrition. This has demonstrated by the result of a wide variety of research studies.
5. Physical development also influence academic achievement, social skills, self-concept and may other aspects of personality.
6. Although physical rates of growth and development are rather slow during the middle years of childhood, marked advancement are made in motor skills abilities. But the age of 6, most children are able to participate in numerous activities that require large-muscle movement and refined coordination.
7. One important aspect of physical development is health. We would now discuss health during middle and late childhood years.

#### Health problems during middle and late childhood years:

1. The average child has fewer illnesses during middle childhood than either before or afterward, and most of the illness is accounted for by the common cold. However, when children start school they go to more places and meet more people than before and thus are exposed to more new diseases.
2. One widespread health problem for which there are as yet no vaccine is dental cavities, or tooth decay. In one study fully 93 percent of the children in a poor city neighborhood were found to be suffering from some degree of tooth decay (Myers et. al. 1968) Levels for middle-class children are not so high, but tooth decay is undoubtedly a major health problem among all school-age children in the United States and India. Researchers have now established that caries is caused, at least for the most part, by a specific bacterium cell *Streptococcus mutans*.
3. Obesity (also called as fatness) has become a major health issue among children. It is most common among 6 to 11 years olds.
4. In a six-year study of nearly 2600 mostly white and middle class-children under age 12 who were enrolled in a pre-paid health maintenance plan, about 14 percent of 8 to 11 year olds were diagnosed as obese (Gortmaker, Dietz, Sobol, & Wehler, 1987, Stettin et. al., 1984). What makes children fat? Research findings are most often correlational, meaning that we cannot draw cause - and effect conclusions. However, there seems to be a strong basis for believing that over weight often results from an inherited predisposition, aggravated by behaviour involving too little exercise and too much food. Some people seem to be genetically predisposed toward obesity. Fat children do not usually "outgrow" being fat; they tend to become fat adults (Kohata, 1986) and obesity in adulthood puts them at risk of health problems like high blood pressure, diabetes, and orthopedic problems.
5. Minor medical conditions such as upper respiratory infections virus, or eczema are common. Migraine and near -sightedness are found to be less common. Injury is a very common medical problem. Sore throat, bedwetting and ear infections are also common among children of this period.

6. Vision is another common health problem.
7. Children under 6 years of age tend to be farsighted because their eyes have not matured and are shaped differently from those of adults. After that age, the eyes are not only mature but can focus better. In a minority of children, however, vision does not develop normally. Ten percent of 6 year old have defective near vision and 7 percent defective distance vision; the latter number jumps to 17 percent by 11 years of age.
8. Dental problems are common to middle childhood as the primary teeth begin to fall out at about the age of 6. Toothache, tooth decay and related problems are very common in middle childhood.
9. Injury is one of the leading causes of disability and death among middle childhood.
10. Boys average more accidents than girls, probably because they take more physical risks (Ginsburg & Miller, 1982). Injuries increase from ages 5 to 14. Possibly because children become involved in more physical activities and are supervised less (Schor, 1987). The most common cause of serious injury and death in young children is being hit by a moving vehicle.
11. Most childhood accidents occur in (or are inflicted by) automobiles, or occur in the home; but between 10 and 20 percent take place in and around schools. Elementary school children are most likely to be injured from playgrounds.
12. Secondary school students are most often injured in sports. Some of these injuries could probably be avoided if players were grouped by size. Skill and maturation level rather than by age.

### Development Tasks of Middle hood

Following are the important developmental skill of middle hood:

1. Learning physical skills necessary for ordinary games.
2. Building wholesome attitudes toward our self as a growing organism.
3. Learning to get along with age-mates.
4. Learning an appropriate masculine or feminine social role.
5. Developing fundamental skills in reading, writing and calculating.
6. Developing concepts necessary for everyday living.
7. Developing conscience, morality and a scale of values.
8. Achieving personal independence
9. Developing attitudes toward social groups and institution

### Self-awareness, Self-concept and Self-esteem

Social psychologists, sociologists, anthropologists and developmental psychologists have done considerable work on self-awareness, self concept and self-esteem among children.

**Self-awareness:** Children between the ages of eight and twelve start to define "What makes me, me". In this process they become more aware of their own emotions and exhibit them in new ways for e.g. Fears that in the past were expressed primarily through actions such as crying or running may now be channeled into anxiety.

At the same time that children begin to understand what makes them who they are, they also develop an understanding of what makes others who they are. Through the self-awareness that arises from interactions with other people children become more sensitive to their own attitude about others they may seek out those who are similar to themselves and shun those they may seek out those who are similar to themselves and shun those they perceive as different and, likewise, they become more aware of the forms of social behaviour that win popularity and they may adopt some of the traits that seem to win approval for others.

**Self-concept:** Self-concept is one's conception of the self. It includes self-understanding and self-control or self-regulation. Self-concept tells us what and who we are. Self-concept includes two things: One is a "real self" (i.e. who they/we are) and the "ideal self" (who they/we would like to be). In the growth of self-concept, the following processes are more important.

**Self-awareness:** The self grows slowly. It begins with self-awareness, the gradual realization (beginning in infancy) that we are beings separate from other people and things, which the ability to reflect on ourselves and our actions.

**Self-recognition:** Growing self-awareness around 18 months of age lead to self-recognition. It occurs when toddlers see themselves in the mirror.

**Self-definition:** Self definition involves identifying the inner and outer characteristics of the self. At about age 3, children think of themselves in terms of externals. What they look like, where they live, what they do. Not until about age 6 to 7 do children begin to define themselves in psychological terms. Self-concept as a social phenomenon.

Most theoreticians and researchers see the self-concepts as a social phenomenon. "The meeting ground of the individual and society" (Markus & Nurius, 1984 p. 147). Middle childhood seems to be the appointed time for that meeting. Children peer into the looking glass of their society and blend an image that they see reflected there with the picture they already have of themselves.

Children are now able to do more than they could earlier. They also have more responsibilities; homework, chores, rules at home and at school, and perhaps some are of younger brothers or sisters. Children begin to regulate their behaviour not only to get what they need and want (as they did earlier) but also meet other people's needs and wants.

Children must complete several important tasks in the development of the self-concept (Markus & Nurius, 1984).

- Expand their self-understanding - to reflect the people's perceptions, needs and expectations. They have to learn what it means to be a friend or a team-mate.
- Learn more about how society works - about complex relationships, roles and rules. A child comes to realize, for example, that his or her own mother had a mother, and that the same person can be since at one moment and mean at another.
- Develop behavioral standards - standards that are both personally satisfying and accepted in society. This is sometimes hard, since children belong to two societies - that of the peer group and that of adults - which sometimes have conflicting standards.
- Manage their own behaviour - as children take responsibility for their own actions, they must believe that they can behave according to both personal and social standard, and they must develop the skills and strategies to do it.
- Self-Esteem - Children's self-esteem is essentially their judgment of their own abilities, influence and popularity.
- According to Sullivan children's self judgments, their self-esteem or lack of it comes to resemble evaluations of them made by significant others in their lives. Their degrees of self-esteem will attempt, whether in academic task, sports, or friendships. Low self-esteem tends to make children less original and more imitative, whereas high esteem brings out initiative and independent judgment. In comparing children of high and low self-esteem one researcher Coppersmith, discovered that those with high self-esteem were able to devote more time to others and to external activities because. On the basis of this research. Coppersmith concluded that people based their self-image on four criteria.

- **Significance** (the extent to which they feel loved and approved of by people important to them).
- **Competence** (in performing tasks they consider important)
- **Virtue** (attainment of moral and ethical standards) and
- **Power** (the extent to which they influence their own lives and the lives of others).

Although people may draw favorable pictures of themselves if they rate high on some of these dimensions and low on others, they are more likely to rate themselves high if they rate high on all four criteria. Students who were found to be high on self-esteem were more popular and did better in school than those with low self-esteem who were more likely to be loners, bed wetter or poor students. Birth order and family also influences one's self-esteem first born or only children. Especially those with warm parents and dominant mothers were likely to have high self-esteem.

The parents of the boys with the self-esteem tended to have authoritative parenting style. They loved and accepted their sons and made strong demands for academic performance and good behavior. They showed respect and allowed individual expression. They defined and enforce limits, relying more on reinforcements than on punishment. Further more, they themselves had high self-esteem and led active, rewarding lives.

Gender role development means learning the basic skills or roles that is related to one's gender in one's culture. By the time children are school age. They have already learned the concepts of male and female and have begun to display attitudes and behavior appropriate to their own sex. Once children leave the relatively closed environment of their families to enter the larger world of school, they begin to broaden their knowledge of gender role, adopting the stereotypes of masculinity and femininity that are dominant in their society. Despite all the efforts in recent years to equalize the power and prestige of the male and female role, changing deeply held attitudes is a slow process. Our own traditions still teach, for example, that boys should be tough and aggressive, girls polite and submissive; that athletic skills is a male trait, and cooking skill a female trait.

There are many theories of gender role development; but the most important theory is that of social learning and modeling. According to this theory children learn sex roles by observing and continue to display them because they are reinforced by social approval and a need for cognitive consistency. Children learn these roles by observing not only among parents but also among peers, teachers and other adults as well.

### Parents Role on Gender Development

Both the parents (the mother as well as the father) influence the gender role development.

In gender role development the evidence point to fathers as having the more important influence, not only in fostering a male self-concept in boys, but feminists role but have little influence on the masculinity of their sons. In a survey of research on the influence of the father in the gender-role development of both boys and girls, **David Lynn (1974) concludes that the father's role is paramount.** Most studies, he says "support the position that the father, in his instrumental function of launching his children into society, is more concerned than the mother with enhancing his boys masculinity and his girls femininity.

Important characteristics of the father who is successful in developing appropriate gender role identification seem to be not only the highly masculine traits of decisiveness and dominance, but also nurturance and warmth. For example, highly masculine boys are more likely to have fathers who are firm and decisive in setting limits and dispensing both rewards and punishments. Such fathers play an active role in the discipline of their sons (Hetherington, 1965) however discipline by fathers fosters a

strong masculine identity in their sons only when it is accompanied by fairness, support and warmth.

Between the ages of 8 and 11, children are able to perceive sex roles in much the same fashion of adults. When presented with a series of test items related to what they would like to do when they grow up, most children will reject those items related to activities of the opposite sex. Children who have working mothers are less likely to indulge in sex-type activities than those whose mothers are not employed. Fling and Manestevitz (1972) found that parents of the same sex of the child have a tendency to encourage more sex typing than parents of the opposite sex.

Highlight by Buzybee

## PIAGET THEORY OF COGNITIVE DEVELOPMENT

According to Piaget the cognitive development during middle childhood is that at the stage of cognitive operations. The period of concrete operations range from 7 years to 11 years (the period of middle childhood). This is the period during which the children being to think and reason logically about objects in their environment and mentally perform sections that previously had to be carried out in actuality. In the stage of cognitive operations there important cognitive developments take place:

1. Conservation or law conservation
2. Operational thinking
3. Adherence to logic and reason

### Law of Conservation

Law of conservation is a principle that states that object properties will remain the same despite changes made in their shape or physical arrangement. In other words conservation is the ability to recognize that two equal quantities of matter remain equal in substance, weight or volume so long as nothing is added or taken away.

Children develop different types of conservation at difference times. At age 6 to 7, they typically are able to conservative substance, at 9 to 10 weight, and at 11 to 12, volume. the underlying principles is identical for all three kinds of conservation but children are unable to transfer what they have learned about one type of conservation to a different type. In concrete stage of operations, however, a child can successfully reasons that objects remain the same even though out side appearances have been changed. The principle of reversibility is also developed which enable the child to understand that objects can have their original conditions restored after changes have been made on physical shape. By understanding the principle of reversibility, a child can reason that the flattened ball of clay can be remolded into original ball and the liquid can be poured back on to the first container to restore its original level. The child knows that the substance had been conserved despite the transformation that is not something that the child needs to check by repeating the operation.

### Operational Thinking

Children in Piaget's third stages are capable of operational thinking they can use symbols to carry out operations that is, mental activities, as opposed to the physical activities, that were the basis for most of their earlier thinking.

For the first time, logic becomes possible. Even though preoperational children can make mental representations of object and events that are not immediately present, their learning is still closely related to physical experience. Concrete operational children are much better than preoperational children at classifying, working with numbers, dealing with concepts of time and space and distinguishing reality from fantasy. Since they are much less egocentric by now, children in the stage of concrete operations can decentre-they can take all aspects of a situation into account rather than focusing on only one aspect, as they did in the preoperational stage. They realize that most physical operational are reversible. Their increases ability to understand other people's viewpoints lets them communicate more effectively and be more flexible in their moral thinking.

### Adherence to Logic and Reason

Mental operations during Piaget's concrete stage of development enable the child to understand the properties of groups and to adhere to the following rules of logic and reasons.

- **Closure** - Any two mental structures (also called elements or operations) can combine to form a third structure for e.g.  $3+4=7$
- **Reversibility** - For any structure there exists an opposite operation which can cancel it. For e.g.  $3+4=7$  but subtracting 4 from 7 = 3.
- **Associativity** - When three structures are combined it does not matter what the order of combination is, for e.g.  $3+4+7$  is same as  $4+7+3$ .
- **Identity** - An awareness that structures can exist that will leave others unchanged for e.g.  $4+0=4$ .

### Development of Memory in Middle Childhood

In the middle years of childhood we become more capable of recalling information from long-term memory. Moreover, a child at this age is capable of making a conscious or unconscious effort to retain short term memories long enough for them to be transferred into the long term reserve. Ten year old children have much better recall than children of four because they can shift information into long term memory and have more efficient devices for retrieval.

Immediate memory increases rapidly in middle childhood. We can see this by asking children to recall a series of digits in the reverse of the order in which they heard them (to recite "8-37-5-1-6" if they have heard "6-1-5-7-3-8"). At ages 5 to 6 children can typically remember only two digits; by adolescence they can remember six. Younger children's relatively poor immediate memory may help to explain why they have trouble solving certain kinds of problems (such as conservation). They may not be able to hold all the relevant pieces of information in memory. Two processes that seem to play an important role in remembering and learning are Metamemory and metacognition. Metamemory, which is one part or aspect of metacognition, is conscious or intuitive knowledge about memory. It seems to affect both how well we remember and the techniques we employ in memorization process (Wellman, 1978). Metamemory includes such kinds of information as our knowledge about how good we are at memorizing what kinds of information we remember best, and what techniques we might employ to remember different kind of information. Children develop their knowledge of memory and their ability to understand the elements of memory and their interaction grow. Although metamemory is considered an important part of cognition, its development in children is not fully understood. We do know that metamemory is one portion a large cognitive process called metacognition, which is the conscious knowledge we have about factors and beliefs involved in the learning processes.

Metacognition is the faculty that provides information about such matter as how fast we learn whether our parents are good at math, what a particular learning task requires of us, and what strategy we should employ to learn the material involved in that task. All of these parts of metacognition are known as metacognitive knowledge. However, metacognition are known as metacognitive knowledge. However, metacognition also includes metacognitive experiences, which are the conscious experiences that either accompany or are related to learning and knowing.

During middle childhood four concepts which play an important role in our perceptual cognitive development are as follows:-

- (a) Form concept (b) Number concept (c) Time concept and (d) Death concept.

We would discuss each of these concepts in brief.

### Form Concept

The ability of children to increase their understanding of form and size concepts and how they relate to the environment is especially rapid during this period. Older children learn from concepts much earlier as compared to younger children. A large number of research studies confirm this fact.

### Number Concept

By the 8th year, most children can add, subtract and deal with multiplication, division and simple fraction. The majority of children can also begin to grasp the durability and abstractness of number concepts, something that had eluded them is the past.

Related to the child's growing understanding of numbers are memory concepts. While the 5 year old understanding of money was restricted to the identification of paisa and rupees, the middle aged child is able to realize the value and worth of coins as well as being able to understand complex money combination. At this age child also develops the understanding of "indeterminate" money concepts such as realization of meaning of "few" rupees, "several" coins or "some" money.

### Time Concepts

Children during time period develop not only clock time but also understand the concept of historical time. They also can group and understand the relationship between past, present and future.

### Death Concept

One of the more difficult concepts for the child to learn is death. Not only does death represent a strange and puzzling phenomenon for young minds to understand but it also an extremely difficult concept for adults to explain.

As compared to earlier years, by the age of 7 and 8 years a more accurate concept of death is being nurtured. At 7 years interest is frequently directed to its possible course such as disease, overacting violence or old age. Some 8 year old may even begin to accept the notion of everyone in the world including them will eventually die. This becomes more firmly established between 9 and 15 a time when advanced cognitive abilities enable individuals to develop mature concepts and attitudes about death and its consequences.

The development of morality is the process whereby individuals learn to consciously adopt standards of right and wrong.

During early childhood, a sense of morality is born when younger realize that certain behaviors parents are classified as "good" and are some times rewarded by their parents. On the other hand some actions are considered to be "bad" and frequently accompanied by punishment. As children become older, morality begins to encompass a complete set of ideas, values and beliefs.

There psychologists have developed elaborate theories of moral development during middle childhood, these are:-

1. Jean Piaget
2. Lawrence Kohlberg
3. Selman

We would discuss each of these theories of moral development in brief.

### Jean Piaget

Piaget contents that all morality consist of a system of rules which are handed down from adults to children. Through training, practice and developing consciousness, children learn to nature respect for these standards of conduct.

In order to obtain an understanding of how moral concepts develop in children, Piaget constructed pairs of stories and asked children to describe which of the two were "naughtier". For e.g. one pair of stories were concerned with two children who accidentally cut holes in their dresses each in a different manner.

The first child wanted to help and surprise her mother by cutting out a pattern of material for her, in the process, she cuts a big hole in her own dress. The second girl

took a pair of scissors from her house to play with while her mother was away one day. Not knowing how to use them, she cut a small hole in her dress.

In a second set of stories, a little boy named Raju is called to dinner. As he opens the dining door, the door slams into a tray holding 15 cups, which was totally obstructed from his view when he entered. As a result, all the cups are broken. In the other story, a little boy named Sunil wants to get some jam from the cupboard while his mother is away from the house. Climbing on the chair, he reaches for the jam, but in the process knocks over a cup and breaks it.

In a follow up conversation, Piaget found that young children felt that the first story in each set was "Naughtier" simply because of the larger hole that was made in the dress and the number of cups that were broken. The child's intention of pleasing and surprising her mother or unintentionally knocking over the cups that were out of sight had no influence on the subjects; rather, they were affected by the total property value of each mistake. Older children, however, considered the second story to be worse because, as Piaget explains, they began to judge right and wrong behavior on the basis of motives involved.

The development of such moral judgment begins during the early years in a stage which Piaget refers to as moral realism. In this stage (generally below 8 to 9 years of age), a child will perceive rules from parents without being totally aware of or understanding their reason. Rather, rules are viewed as being sacred and untouchable. Also the child believes that the purpose of punishment is atonement for one's sins. It is not until the state of ethics and mutual respect is reached that the child becomes aware of the meaning of rules and reasons for them. At this point (beginning after 9 to 10 years of age) rules are gradually viewed by the child as a product of mutual consent and respect and they are understood in relation to the principles they uphold. Children also come to realize the seriousness of wrong doing if the punishment fits the act itself. Justice is based on an "eye for an eye" "tooth for a tooth" ideas, whereby the pain felt by the transgressor must be proportional to the pain inflicted upon others. Older children believe that punishment should serve the function of putting things into proper perspective thus for e.g. upon hearing the story of a boy who broke his little brother's toy, older youngsters recommend that the boy should be deprived of his own toys for a week. There emerges a social cooperative logic in favor of the older child's moral reasoning.

According to Piaget, children's conception of morality develops in two major stages which coincide approximately with the preoperational and operational stages. People go through these moral stages at varying times, but the sequence is always the same.

The first stage, morality of constrained (also called heteronomous morality) is characterized by rigid, simplistic judgments. Young children see every thing in black and white not gray. Because they are egocentric they cannot conceive of more than one way of looking at a moral question. They believe that rules are unalterable, that behavior is either right or wrong, and that any offence- no matter how minor -deserve punishment.

The second stage, morality of co-operation (also called autonomous morality), is characterized by more flexibility. As children mature and interact more with other children and when adult, they think less, egocentrically. They have ever-increasing contact with a wide range of view points, many of which contradict what they have learnt at home. Children conclude that there is not one unchangeable absolute moral standard, but that rules are made by people and can be changed by people, including themselves. They look for the intent behind the act, and they believe that punishment should fit the "crime". They are on the way of formulating their own moral codes.

### Lawrence Kohlberg

Kohlberg is acknowledged as the major figure in the study of moral development. Kohlberg's theory considers six stages categorized within three major levels. The three

major levels are **Level-I Pre-conventional morality** (age 4-10) **Level - II conventional morality** (10 to 13 years) **Level-III Post Conventional morality** (13 or no until young adulthood or never)

The first two levels are important during middle childhood. We would now discuss the first two levels.

**Level-I Re-conventional Morality:** At this level children have little conception concerning what socially acceptable moral means of behavior may be. but through two sub stages, they being to display signs of initial moral behavior.

**Stage - I Obedience and punishment orientation:** Children obey the rules of others to avoid punishment. They ignore the motives of an act and focus on its physical form (such as the size of a lie) or its consequence (for example, the amount of physical damage).

**Stage-II Natively egoistic orientation:** In stage two they reason that by taking the right action, they usually earn some tangible sort of reward. Kohlberg feels that a sense of reciprocity is in operation here, that is children will do the right thing not only to satisfy their own needs but also to satisfy the needs of others. If the latter, is the case, they reason that some sort of "return favor" will be in order ("you scratch my back and I'll scratch yours").

**Level-II Conventional Morality (Age 10 to 13):** Children now want to please other people. They still; observe the standards of others, but they have internalized these standards to some extent. Now they want to be considered "good" by those persons whose opinions are important to them. They are now able to take the role of authority figures well enough to decide whether an action is good by their standards. There are two stages of reasoning at this level which are as follows:-

**Stage-III Good boy-nice girl orientation.:** During this stage there is a considerable degree of conformity. Children realized that they must live up to rules in order to win praise or approval from others. There is also identification during this phase, usually with emotionally important persons.

**Stage-IV Authority and social order-maintaining orientation:** At this stage of moral development an individual is concerned with doing their duty, showing respect for higher authority, and maintaining the social order. They consider an act always wrong, regardless of motive or circumstances, if it violates a rule and harm others.

The morality during this stage is also called as "law and order" morality. Here the focus is one obeying the rules for their own sake. Justice is seen the reciprocity between each individual and the social system. Social order is very important in making judgments at this stage.

**Level-III Post conventional level (Age 13 onwards):** During this period child's morality reaches a mature state.

**Stage-V Morality of contract of individual rights, and of democratically accepted law:** People think in rational terms, valuing the will of the majority and the welfare of society. They generally see these values best supported by adherence to the law. While they recognize that there are times when human need and the law conflict, they believe that it is better for society in the long run if they obey the law.

**Stage-VI Morality of universal ethical principles:** People do what they as individuals think right, regardless, of legal restrictions or the opinions of others. They act in accordance with internalized standards, knowing that they would condemn themselves if they did not.

### Selman

Selman has proposed a theory of moral developmental which links moral development to a role-taking. Role-taking can be defined as putting oneself. Selman (1973) describes the development of role-taking in five stages (0 to 4) which are summarized below.

At stage 0 (from about age 4 to 6) children think and judge egocentrically, and so they cannot assume other people roles or points of view. At stage 1 (from about age 6 to 8), children realize that other people may interpret a situation differently. Reciprocal awareness makes stage - 2. From age 8 to 10. In stage 3, from about age 10 to age 12, a child can step outside a relationship and view it from a third point of view - for example that of an objective outsider like a judge.

Stage 4 arrives, usually, during adolescence, when a person realizes that mutual role-taking does not always resolve disputes. There might be rival values that simply cannot be communicated away.

### Middle Childhood and Family Life

During the middle childhood years children spend relatively much less time with their families than they did during infancy. Following points are worth noting with respect to middle childhood and family life.

- 1) During middle childhood there is a great change in parent child relationship. Children are no longer too much dependent on their parents as they were in the previous years.
- 2) Middle childhood is a transitional stage of co-regulation, in which parents and child share power, "parents continue to exercise generally supervisory control, while children begin to exercise moment-to-moment self regulation" (Maccoby, 1984 p. 191). Co-regulation reflects the child's developing self-concept. As children of this age begin to coordinate their own wishes with societal demands, they are more likely to anticipate how much their parents or other people will react to what they do or to accept a reminder from their parents that others will think better of them if they behave differently.
- 3) Child's personality during middle childhood is greatly influenced by the family form which he comes from. A few decades back we used to have joint family and Nuclear family, but today due to changes in values and culture we have single family especially unwedded mothers or never married mothers. Children coming from divorced families or children coming from families where both the parents are working.
- 4) Relationship between family members is also important in the development of children.
- 5) Children's work in school and their attitudes toward school are greatly influenced by their relationships with family members. Wholesome happy family relations lead to motivation to achieve while unwholesome, unhappy relationship cause emotional tension which usually have a detrimental effect on a child's abilities to concentrate and to learn. Family relationships also effect social adjustments outside the home. When family relationships are favorable, children's social adjustments to people outside the home are better than when family relationships are stressful.
- 6) Similarly, role playing in the home sets the pattern for role playing outside the home. The reason for this is that the roles children learn to play in the home and the kind of relationships they have with their sibling form the basis for their relationships with peers outside the home. This in turn influences children's patterns of behavior toward their peers.
- 7) Children's aspirations and achievements in different areas of their lives are greatly influenced by their parent's attitude. First born and only children usually come under more pressure to achieve than late born children and are given more aid and encouragement to achieve the goals their parents encourage them to set for themselves.
- 8) Whether children will be creative or conformist in their behavior is greatly influenced by their home training. Democratic child training methods encourage creativity, while authoritarian methods tend to foster conformity.

- 9) Children coming from divorced families do have considerable problems. Divorce affects middle childhood period. Children suffer when their parents split up. The children as much as or more than the parents, may feel pain, confusion, anger, hate, bitter disappointment, a sense of failure, and self doubt. For many this family disruption is the central event of childhood, with ramifications that follow them into adult life. No matter how unhappy marriage has been, its break up usually comes as a shock for children. The children of divorcing parents often feel afraid of the future, guilty about their own (usually imaginative) role in causing the divorce, hurt at the rejection they feel from the parent that moves out and angry at both parents. They may become depressed, hostile disruptive, irritable, lonely, sad, accident prone or even suicidal. They may suffer from fatigue, insomnia, skin disorders, and loss of appetite or instability to concentrate and they may lose interest in school work and in social life. Children of different ages react to divorce in different ways.
- 10) The absence of parents, for whatsoever reason creates a special family situation and may have important consequences for a child's social adjustments. If the home has been broken by death or marital discord, the child may as we have seen, feel isolated, rejected or abandoned. On the other hand if there has been a relatively amicable divorce the child will probably maintain contact with both parents, and any-sense of dislocation will be less severe.
- 11) How is a child's personality development affected by growing up without parents of both sexes? Some studies have shown that boys without fathers at home have trouble establishing a secure masculine identity of their own and relating to women in later life, they seem more likely to become delinquents.

### Aggression

Aggression is another common emotion seen during middle childhood. However the rudimentary forms of aggression that is common in earlier years disappear. During 7th year the physical aggression decreases as compared to what seen in 5th or 6th year. During middle childhood aggression takes the form of arguing, alibis, name-calling, or making disagreeable remarks.

Sex differences in aggression are clearly seen. Boys are more aggressive as compared to girls. Parental influence, considerably influence the development of aggression during this period. Most psychologists agree that children exhibit hostile reactions when they are unwanted or given little or no warmth and affection from their parents or are subjected to extreme destructiveness. Non aggressive children generally come from families where children were brought up in warm and affectionate families.

### Fear

During this period fear and anxiety is also seen. Social stress is a primary source of fear during this period. Not knowing how to act in a new social situation, many of these children are easily frightened by exposure to unfamiliar people and places. Shyness and silent withdrawal are common responses. This may not be serious if it occurs only occasionally, but a child may need support and encouragement from the parents in such situations. Without such help, shyness may develop to the point where the child is seriously hampered in social relationships, developing a fear of the unfamiliar that can remain for many years, fears may also be learned through conditioning and reinforcement. Children may acquire fears through exposure to frightening films or television shows.

Some of children's most terrifying fears originate in their own imaginations. From year to year preschoolers develop a more and more powerful capacity to conjure up vivid scenes and situations. They use imagination to entertain ideas that they would not enact in reality and to vent their aggression privately. They also use it to invent fearsome perils. At bedtime a shadow on the ceiling may become a ravaging beast; a dark corner

may conceal nameless horrors. It is important for parents to take these fears seriously because they are very real to the children even if their cases are not.

The nature of children's fears changes as they grow older in one study (Bauer, 1976), the researcher encouraged children in Kindergarten, second grade, and sixth grade to talk about their fears and to draw a picture of them. The Kindergarteners were most afraid of monsters and ghosts, not because they might be physically injured, but because they were ugly frightening to look at. The fear that they might be physically injured had not yet occurred to most of the Kindergarteners. The sixth graders, on the other hand, more realistically, were afraid of physical danger and of bodily harm to themselves or their parents. Thus as children get older, they are less likely to have fears with imaginary themes, but more likely to express realistic fears of danger or bodily injury.

Highlight by Buzybee

## ADOLESCENCE

### Definition and Characteristics of Adolescence

The term adolescence comes from the Latin word *adolescere*, meaning "to grow" "to maturity". As it is used today, the term adolescence has a broader meaning. Adolescence eludes mental, emotional, and social maturity as well as physical maturity.

G. Stanley Hall is usually credited with formalizing the concept of adolescence. Hall is noted for early research into growth and development during this time of life and his ideas about the nature of adolescence continue to influence thinking today.

Adolescence in Hall's views came to represent "storm and stress" period that reflected the unsettling turbulence of growth in modern societies.

According to Bigner (1983) Adolescent can be defined as a stage in the life cycle between 13 and 18 years of age characterized by increasing independence from adult controls, rapidly occurring physical and psychological changes, exploration of social issues and concerns increased focus on activities with a peer group and establishment of a basic self identity.

Piaget has given a definition of Adolescence from a psychological view point. According to him "Psychologically, adolescence is the age when the individual becomes integrated into the society of adults, the age when the child no longer feels that he is below the level of his elders but equal, at least in rights".

The period beginning at about 13/14 years and extending up to 20/21 years in an individual life is known as adolescence. Adolescence begins at about 13 or 14 years and extends up to 20 or 21 years. The period from 13 years to 17 years is called early adolescence and the period from 17 to 21 years is called later adolescence. The major developmental task during adolescence is the preparation for adult life. In early adolescence the emphasis is on learning to be independent of adult guidance and control. In the later adolescence the emphasis is on learning specific skills needed for adult life. The beginning of the adolescence (about 13 to 14 years) is called Puberty. Puberty means the beginning of sexual maturity. Due to sexual maturity various changes in the physiological functions and the structure of body take place. The body proportions suddenly change and the individual tends to look like a grown up man. He acquires primary and secondary sex characteristics. There is intense attraction for the person of opposite sex.

The following are some of the major characteristics of adolescence: (a) Fast Growth and Structural Changes in the body - The individual desires to be free and independent of all authority. Due to the sudden changes in bodily and other features (like change in voice, changes in body proportions, secondary sex characteristics, etc.) he feels inadequate at the beginning. He desires to be like adult and therefore, imitates the behavior of adults.

**A Traditional period** - Adolescence is a transitional period. It is a period of change from childhood to Adulthood. During this period, as in any other transitional period, status roles etc are vague and there is confusion. The adolescent is a period where individual is neither a child nor an adult.

**Development of New Attitude towards one self and others** - Adolescent develops new attitudes towards himself and towards others. Parents also change their expectations from the adolescent; he is expected to behave more responsibly and adopt adult ways of behavior. Adolescent expresses his dislike towards the established rules and regulation in the society, and strongly defends his own views and opinions. Some boys show leadership traits in this period.

**Rising Awareness of One's Rights and Duties** - Adolescent becomes aware of his rights and duties. He also becomes conscious about social approval. Due to sexual maturity, many physical changes take place in the body and this leads to changes in the personality of the individual. The ideas, thinking and personality traits admired before seem to the adolescent as childish and undesirable. A new set of values and norms emerges and he is also confused. He establishes new relationships with others; finds new friends and may break old friendship. He always eager to test his abilities.

**Admiring Qualities of Peers of Opposite Sex** - Certain traits of peer of opposite sex are admired by the adolescents. Boys are admired by girls for such traits as frankness, protectiveness, pleasing personality, neatness and good sense of humor. Boys admire such qualities of girls as good appearance, intelligence,

friendly manners, kindness, etc. Boys are later in sexual maturity by about a year than the girl of the same age. Therefore most girls are attracted towards boys who are older than them.

**Acquiring Skills and Building up Attitude** - During early adolescence various skills and special abilities he developed. In the later adolescence the individual is more interested in the vocational courses which will be useful for his adult life. He also takes interest in reading novels, stories and other literature concerned with romance, scientific invention, adventures, biographies etc. Reasoning and abstract thinking also develops considerably in adolescence.

**Changes in the Expression of Emotions** - Emotional expressions of adolescence are greatly influenced by maturity and training. He does not express his emotions directly like a young child but has great control over them. Emotions become less general and more specific in expressions. With adult guidance he is able to control babyish anger, jealousy, fear etc. He learn to use those expressions which are more likely to be socially approved e.g. when he is angry he will not express it directly by attracting other persons but will make use of critical remarks which will express his anger indirectly.

In case of an adolescent, the heightened emotionally, result from the physical and glandular changes. Sometimes, these changes make it difficult for him to make necessary adjustments to new patterns of behavior and new social expectation. As a result the adolescent may suffer from anxiety and social pressure. Hence, the stage is also called the period of storm and stress. Thus a number of changes to take place in the individual's personality during the adolescence period.

**Adolescent is a period of stress and storm** - Many adolescence come into conflict with norms. They are divided between independence and conformity. Their need to establish their indemnity often makes them confront others.

**Adolescence is a Time of unrealism** - Adolescence have a tendency to look at life through rose-tinted glasses. They are themselves and others as they would like them to be rather than they are. This is especially true of adolescent aspirations. These unrealistic aspirations, not only for themselves but also for their families and friends, are, in part, responsible for the heightened emotionality characteristic of early adolescence.

**Adolescence is a period of individual Emancipation** - Western culture emphasizes the teenage years as appropriate for establishing one's independence as a mature person. The experience of adolescence represents a struggle toward advancement of the person from behavior patterns of childhood. The adolescent becomes an emancipating child who attempts to achieve a new self-definition in many ways.

**It is a time of experimentation, idealism conflict, and uncertainty** - Western culture has created the unique characteristics and expectation of adolescence. There are certain issues central to this period of development. These are as follows:

Tension between self and society (attempt to define who one is and where one stands in relation to others). A high value placed on movement and action of the self, and others and the culture. And tendency to band with others youthful subcultures that maintain a deliberate social distance from established social orders. There is a focus on gaining freedom to explore and examine a wide variety of attitudes, values and behavior as well as life style.

Adolescence is focused psychological on certain preoccupation these include:

1. Identity crystallization
2. Autonomy
3. Alternating stimulation with boredom
4. Begin sociable with one's age group
5. Experiencing physical maturation
6. Achieving a variety of environmental experiences.

A wide variety of physical changes take place in adolescence. These physical changes are dramatic. Some of these changes are as follows:

- There is an adolescence growth spurt
- Beginning of menstruation for girls
- Presence of sperms in males
- Maturation of the reproductive organs and
- Development of secondary sex characteristics

**Physical Changes of Adolescence** - Physical changes in adolescent is often accompanied by many psychological changes. Development of primary and secondary sex characteristics and changes in anatomy is a cause of concern, anxiety and conflict among adolescents.

**Self Consciousness** - An adolescent is very much self-conscious about his/her looks. They spend a great deal of time in front of mirror, at this age boys want to be tall, broad-shouldered, athletic; girls want to be pretty, slim but shapely, with nice hair and skin (Tobbin-Chahards, Boxe, and Petersen, 1983). Anything that boys think that they look feminine or girls think that they look masculine makes them miserable. Teenagers of both sexes worry about their weight, their complexion, and their facial features.

Girls tend to be unhappier about their looks than boys of the same age, no doubt because our culture places greater emphasis on woman's physical attributes. When adolescents are asked what they like least about their bodies, they say "nothing", while girls mostly complain about their legs and hips (Tobin Richards et al. 1983). Changing body image - Sexual changes in adolescent bring about changes in perception about their "body image" Stolz and Stolz (1944) have found in a longitudinal study, that approximately one-third of the adolescent males studied and almost one-half of the females expressed concern or worry over some aspect of physical growth.

**Depression and Physical Appearance** - Adolescent girls are more prone to depression than boys, mainly because of worries about their appearance. They feel "ugly", consider themselves too fat, too short, or too tall; or hate their hair or their complexion. Before puberty, rates for depression are the same in boys and girls; but at about age 12 girls start to have higher rates, and by age 15 girls' rates are twice as high as boys. Sex role and body build - One's body build also affects New Delhi's views on sex role preference. Muscular youth reported stronger muscular interest and teachers had a tendency to perceive these individuals as enjoying in more masculine type of behaviour than those who were either fat or thin.

**Menarche and Psychological Development** - Girls who have begun to menstruate seem more conscious of their "femaleness" than girls of the same age who have not yet reached menarche. They are more interested in boy-girl relation and in adorning their bodies, and when they draw female figures, they show explicit breast. They also seem more mature in certain personality characteristics (Grief and ulman 1982). The better prepared a girl is for menarche; the more positive her feeling and less her distress.

Early and late matures and its impact on psychological development-Joves and Byley (1950) found that early matures, more than late matures, tend to be more adept in overall social adjustment. It was found that early maturing boys were more popular and had a greater capacity for leadership than late maturity, who frequently encountered high school adjustment problem. The effect of early and late matures have been discussed separately in the next section.

**Acne** - Because the sebaceous glands are quite active during these years, skin blemishes become quite common adolescent becomes more anxious and concerned. The youngster may have vague feeling that he is seriously diseased or that the acne is a deserved infliction related to sinful deeds or thoughts.

**Awkwardness** - As the adolescent growth spurt is experienced, some youths may go through a rather awkward and unconditioned stage of development. As legs and arms grow rapidly, adolescents at times may misjudge the length of their stride or their reach. They may result in their humping into furniture or people or their being unable to exhibit delicacy of manners in certain social setting. Their awkwardness, however, may be attributed more to self-consciousness and social inexperience than to any real lack of physical coordination. Voice break - Voice naturally deepens as adolescent's nature. For boys, the change in pitch is greater and is accompanied by a period during which control of the voice is lost. To their great dismay and frequent embarrassment, many males may find their voice cracking at the most inappropriate times. While voice change is completely normal phenomenon, it can be a problem if it creates social uneasiness.

**Weight and Height** - When the adolescent "fat-period" is experienced a considerable amount of self-concern and over anxiety can result. Some youngsters, feeling overly conspicuous because of their weight,

may choose to combat the situation through several channels, such as turning to dieting, withdrawing from social activities or finding social approval by becoming "clowns" or "mascots".

Hurlock (1980) lists several psychological changes in early adolescence that are associated with or are result of the physical changes produced by puberty.

**A Desire for Isolation** which teenagers demonstrate at time by withdrawing from interaction with peers and family and devoting much time day dreaming, being alone, and craving privacy.

**Heightened Emotionality** which is demonstrated by emotional out burst of anger, and crying easily. Excessive modesty which is shown through a heightened sensitivity and embarrassment about changes that are occurring in the body.

**Lack of Coordination** due to an inability to adjust quickly to rapidly changing body size and proportions.

**Heightened Insecurity** which appears in the form of decreased self-confidence and may be also due to change in the behavioral standards set by parents as well as a decreased level of strength and energy.

Other effect includes a peace at early stage of puberty in self-about avoidance of responsibility, resentment of parents and anxiety about social relation.

**Emotional Tensions Peak by Mid-Puberty.** Throughout the period, individual shift their interaction increasingly away from parents and towards peers.

The psychological changes associated with puberty and early adolescence may result from an interaction between rapid physical changes and heightened social pressure from peers, parents and other adult. This time of an individual's life has been characteristically labeled as "storm and stress".

Physical development has a considerably impact on our psychological adjustment. Some individual's mature, physically, at an earlier age. Others mature at a later age. What is the impact of early and late maturation on our psychological and social development and adjustment?

Research studies have shown that early maturing boys were socially well adjusted and more popular as compared to late matures.

In a more extensive investigation Mussen and Jones and Jones (1957) have found that late Matures have a tendency to generate poorer self concept and are overly concerned with matters related to social acceptance. Further more they are less popular with their peers, immature, rebellious towards their parents and frequently resort to attention seeking behaviour. Early matures, on the other hand, present a more favorable picture during adolescence. We would now discuss the effect of early and late maturation among boys and girls.

#### Early and Late Maturation in Boy

- Boys who mature early have a high self-esteem are socially well adjusted and more skilful.
- Some research studies have found early maturing boys to be more poised, relaxed, good natured popular with peers, and likely to be leaders and less affected and impulsive than late matures. Other studies have found them to be more worried about being liked, more cautions, and more bound by rules and routines.
- Late matures have been variously found to feel, more inadequate, rejected and dependant, aggressive, and insecure; to rebel more against their parents; and to think less to themselves.
- Early matures are more muscular, they are stronger and better in sports and they have a more favorable body image.
- Early matures is likely to be given more responsibility by adults than a late mature.

#### Early and Late Maturation in Girl

- Advantages and disadvantages of early and late maturation are less clear-cut for girls then for boys. Girls tend not to like to mature early; they are generally happier when they mature neither earlier not late than their peers. Early maturing girls tend to be less sociable expressive, and poised; more introverted and shy; and more negative about menarche. They are pat to have a poor body image and lower self-esteem than later maturing girls (Simmons, Blyth, Van Cleave, & Bush, 1979).
- Early Maturing girls are considered to be sexually attractive interested in sex and parents treat early maturing girls more strictly and more disapprovingly than they treat less developed girls.

- A girl who is bigger than many of the boys she know and more bosomy than other girls will often feel uncomfortably conspicuous; but working through these problems may give her valuable experience in dealing with problems late in life. Some researchers have, in fact, found that early maturing girls make better adjustment in adulthood.

### Developmental Tasks of Adolescence

Following are the important developmental tasks of adolescence.

1. Achieving new and more mature relations with age-mates of both sexes.
2. Achieving a masculine or feminine social role
3. Accepting one's physique and using the body effectively
4. Achieving emotional independence from parents and other adults.
5. Preparing for marriage and family life.
6. Preparing for an economic career
7. Acquiring a set of values and an ethical system as a guide to behavior developing an ideology
8. Desiring and achieving socially responsible behavior

The developmental tasks that the individuals encounter in adolescence centre on acquiring and refining more advanced skills, abilities, and attitudes that led to preparation for adulthood.

Achieving these accomplishments is not an easy task and bring many conflicts to the individual and troublesome interactions with the family and others at this time in life.

The diagnostic and the Statistical Manual of Mental Disorder (DSM-III, 1980), has classified eating disorder into two types.

- (a) **Anorexia Nervosa and**
- (b) **Bulimia**

These two types of eating disorders are commonly found among college going adolescent female who is westernized and beauty conscious.

(a) **Anorexia Nervosa** is a life threatening disorder in which the individual usually a female adolescent, has an intense fear of becoming obese, either eats very little or binges and then induces vomiting, loses at least a quarter of her body weight and feels fat even though emaciated.

There is the intense fear of becoming fat. a distorted body image and a sense of being obese despite conspicuous thinness, and a loss of 25 percent of body weight without any known physical illness. Anorexics exercise vigorously to lose still more weight. They are often perfectionist and are easily depressed when they cannot meet the standards that they have set for themselves.

Many different psychobiological and socio-cultural factors have been suggested as the cause of anorexia nervosa. It seems likely that both kinds contribute. On the biological side, there is a strong evidence that our physiques are determined to very substantial degree of inheritance, on the socio-cultural side, there has been a recent growing emphasis on slimness as a requisite of beauty in women. Middle class educated women, in particular, seem to have accepted this concept. Evidence for the existence of this emphasis are the models selected for advertisements and the highly published contestant in the so-called "beauty pageants".

Anorexia is a life threatening disorder, because of the strict diet control that may create medical emergencies. The ratio of girls and women to boys and men with this disorder may be as high as twenty to one. Anorexia usually has its onset during adolescence, often shortly after the beginning of menstruation. The cessation of menstruation or amenorrhea may even precede noticeable weight loss. Like other people, anorexics are preoccupied with food, but for them it takes the form of collecting recipes and planning and preparing meals for others.

The DSM-III gives these six criteria for identifying anorexia nervosa:

1. Refusal to maintain body weight over a minimal normal weight for age and height.
2. Weight loss of at least 25 percent of original body weight; or if less than 18 years of age, weight loss from original body weight plus projected weight gain expected on pediatric growth charts may be combined to comprise, the 25 percent.

3. Disturbance of body image with inability to accurately perceive body size.
4. Intense fear of becoming obese. This fear does not diminish as weight loss progresses.
5. No known medical illness that would account for weight loss Amenorrhea (in females)

**Cause:** The causes suggested for anorexia nervosa fall in two categories; psychogenic theories and bioorganic theories. Psychogenic theories regarding anorexia are mainly consisting of psychoanalytic view. For example, one psychoanalytic theory contends that anorexia nervosa is a symptom of the child's refusal to grow up a type of protest. In this connection, some psychoanalytic theorists believe that having anorexia nervosa have much common with borderline syndrome adolescents. In the borderline syndrome, pathological dependency and odd behaviors exist that resemble both neuroses and schizophrenia (thus the term "borderline"). Anorectic adolescent, then, have been viewed as highly dependent person who fear separation and independence from parents. Further, both anorectic individuals and borderlines individuals tend to be shy tense, and hypochondriac it is often believed that they both have been overprotected. Other analytic theories have approached this problem in different ways. Because weight loss is profound, and is generally associated with under nourishment and death, a few clinicians have suggested that anorexia nervosa is really a form of sub intentional suicide. That is they believe this behavior disorder to be motivated by self-destructive urges. Thus for however, no empirical evidence exists to support any of these theories.

By contrast, the biogenic position contends that anorexia nervosa has a biological or genetic origin. Some evidence does show an association between certain abnormal chromosome patterns and anorexia in females (Korn et al., 1977). In particular, chromosome configuration XO ( a condition called Turner's syndrome) is associated significantly with anorexia. Another investigation revealed that individuals with this disorder exhibit symptoms of passivity, dependency, and, in the researcher's words, "psychionfantilism" (Kihlbom 1969). Further research is required before a more complete understanding of the etiology of anorexia nervosa is achieved.

**Treatment:** Almost every technique imaginative has been tried with anorectic patients (Bliss 1975). Lobotomies, hormone, injections, drugs, isolations, electro convulsive therapy, behaviour modification, hospitalization, insulin, coma, and even "gentle-reasoning" have been tried with these patients. In addition to physical methods, family therapy and individual psychotherapy (insight therapy) have been employed with varying degree of success and failure. Perhaps no single therapy has emerged because there are different causes of these disorder and different levels of severity.

**(b) Bulimia** has been described as the "binge purge" syndrome. It is a serious and spreading disorder marked by uncontrollable overeating, often followed by self induced vomiting or overdoses of laxatives to eliminate the caloric intake. Persons of normal weight, as well as those who are anorexics or obese, may be bulimics.

The bulimia person attempts to counteract stress and depression by overeating but does so in an uncontrollable manner. The typical bulimic is a female in her twenties while middle-class, with some college education, and has been a binge eater for a numbers of years. It is difficult to know exactly how many people are affected with this disorder, but some expert estimate that one in five college educated women have been or will be effected at some time in their lives. The incidence may be greatest in able, you working women who are expected to have career, to take care of a house, to raise children, and to retain slim form that is highly admitted in a male-dominated, sex conscious, achievement oriented society.

Reports indicate that some bulimics have consumed as many as 55,000 calories at a single sitting. This would approximate sixteen pounds of foods. More commonly, they consume up to 2000 to 5000 calories of pastries, breads, ice creams, cookies, etc; high calories of food are favored by them. The recurrent of heavy binging and vomiting brings in its wake a number of devastating physical consequences; ulcers, gastric and dental problems, an acute disturbance in the chemical balance of the blood can cause heart-attack. Other problems include sore throat, aching joints, feeling of weakness and dizziness and apathy.

Bulimic college students sometimes report that immediately before one of their caloric binges they feel depressed, angry and that vomiting may give them a pleasurable "high". Their problems, with eating and

weight effect their suicidal lives, their works, and their family relationship. Binge-purge eaters may also have a history of over using such substance as alcohol, marijuana amphetamines, diet pills and the barbiturates. The DSM-III lists several operational criteria for bulimia:

1. An episodic patterns of binge eating-accompanied by
  - (a) An awareness of disordered eating pattern, with a fear of not being able voluntarily to stop eating; and
  - (b) Depressive moods and negative after thought following the gorging.
2. The bulimic person must have at least three of the following symptoms.
  - Rapid consumption of food during the eating episode.
  - Ingestion of high caloric food during the episode.
  - Clandestine eating binges
  - Following the binges, abdominal pain, sleep, social interruption, or self-induced vomiting.
  - Repeated efforts to lose weight-diet, cathartic vomiting
  - Cyclical patterns of fasting.

**Causes:** Unfortunately no information is available concerning the possible causes of bulimia. Although a parent or sibling may be obese, no predisposing factors or stressors have found, according to DSM-III. It is not even known at this time whether, like obesity, bulimia is more common in lower social classes than in middle and upper classes; or whether, like anorexia nervosa, it is a mainly a middle class disorder. To be sure, much more research is needed in connection with this problem.

**Treatment:** Since bulimia does not follow a pattern like that of simple over eating (obesity), it seems doubtful whether appetite reducing method would be effective; many bulimic persons are not, in fact, overweight. There seems to be published evidence demonstrating a systematic and effective means for dealing with this problem. While private clinicians may have dealt with some bulimics with success, these data have yet to be made public.

**ADOLESCENCE CONDUCT DISORDER AND ITS TYPES**

While 5 to 15 percent of all adolescents show "occasional" acts of antisocial behavior (Meeks, 1980), the conduct disorders are characterized by more displays of antisocial activity. Four types of conduct disorders share the DSM-III core description of repetitive and persistent patterns of antisocial behaviour that violate the rights of others, beyond the ordinary mischief and pranks of children and adolescent.

In the conduct disorders of the aggressive and under socialized type, youths show a consistent disregard for the feelings of others, bully smaller children have few, if any same-age friends, and present serious school problem. They generally are hostile, verbally abusive, defiant, and negativistic.

Though anger is also present to some degree, it is not as obvious in youngsters classified as ingressive conduct disorder, under socialized type. While these youths are adapting at manipulating people for favors, they also share, with their, aggressive counterparts, a lack of concern for the feelings of others.

Generally, they show one of two patterns of behavior. In the first pattern they are timid. Unassertive, and shy, and often report feeling rejected and mistreated. At times they will be victimized, most often sexually. In a second pattern, the adolescent are less timid and more likely to exploit and manipulate others. Unlike their aggressive counterpart's response of uncontrolled anger when frustrated, these youth are more likely to react to pressure with deviousness and guile.

Adolescents designated as having socialized conduct disorders show an ability to make friends with some of their peers. In aggressive conduct disorders, socialized type, youth commit violation of basic right of others usually by some combination of physical violence or robbery. While they do not seem to have feeling of guilt or remorse for their illegal activities, socialized aggressive adolescents have an ability to develop friendship and maintain them for 6 months or more. They will extend themselves to help those of all friends, even to the point of taking punishment rather than informing of them. The final group of conduct disorders, ingressive and socialized type, is also marked by rebellion against authority, but lacks of the physically aggressive quality of the aggressive and socialized pattern. Some of the behavior of the non-aggressive adolescents seems prank like, but in actually they often end up in serious trouble with school and community authority.

**Adolescent Suicide**

The concept of suicide is not limited to self-murder. Some contend that suicide is simply the ultimate form of self-destructive behavior. This approach suggests, then, that there are degrees or levels of self-destructive behavior. The term 'life-threatening behavior (LTB)' is used to represent a class of actions that are less lethal than suicide. Weisman (1976) lists these types of life threatening behaviors:

- Self-injury and intoxication includes non suicidal overdoses of drugs, frequent alcohol abuse, and other repetitive acts that results in trauma.
- Rash, regretted, incautious, or bizarre acts, unskilled use of dangerous tools, instruments, and the automobile; where there is real danger, these people may show very poor judgment, which creates a death-related setting.
- Significant omissions - omits medication, disregards medical or other professional advice.
- Significant excesses - gross overeating, starvation diets, chronic alcoholism.
- Rebellious against rules, roles, and requirements during hospitalization.

In the late 1930s in his book "Man against Himself" Karl Menninger described how completed suicide, the ultimate act, is on continuum with lesser self-destructive behavior such as alcohol addiction, drug abuse, self-mutilation, poly surgery (exhibited by those who seek doctors who will operate on them), and other self-destructive acts. This profound point of Menninger that there exist degrees or symbolic acts of suicide certainly seems to apply to particular adolescent actions.

Although suicide is not a leading cause of death for those between 10 and 14 year of age, between age 15 and 19 it becomes the fourth leading cause of death, exceeded only by accidents, homicides, and malignant tumors (U.S. public health service, National Centre for Health Statistics, 1974). Of the approximately 25,000

suicide reported in one year, 4,000 were committed by the adolescent group. Another interesting statistic shows that in the last 20 years adolescent suicide has increased nearly 250 percent.

Although most of the clues to suicide have emerged from investigations of adult male suicides, a few of these conditions may also warn of adolescent suicide potential: 1. cropping physical disability; 2. Early rejection by the father, 3. Heavy drug use; 4. Verbal statements about one's own work lessens or the absence of hope or purpose in life; and 5. Major setbacks or failures in social, academic, or family affairs.

Greuling and De Blassic (1980) have found that suicide is the second leading cause of death among adolescent.

According to Miller (1975) the cause of adolescent suicide includes:

1. Feeling of social isolation
2. Stress due to strained relations with parents or in reaction to parental abuse
3. Drug abuse
4. Alienation from society
5. Depressions
6. High suggestibility to suicidal ideas
7. Internalized, self directed aggression
8. A need to communicate a desire for help

School counselors may be effective in helping to prevent suicide since these mental health professionals have more direct contact with suicidal teenager than others. Psychological counseling is very effective in helping adolescents to recover feelings of self worth and purpose during stressful periods when suicide often is contemplated. Though many children and adolescent remain at home and act out their frustrations and anxieties aggressively, nearly 1 million youngsters per year runaway. Involving nearly equal number of boys and girls, runways have been characterized as insecure, unhappy and impulsive (Jenkins and Stable, 1972), having low self-esteem (Beyer, 1974) and feeling out of control (Bartolla, 1975). The bulk of research on runaways shows them to be more disturbed than normal teenagers.

Why do over 1 million youngsters "run" each year? Disturbed parent child relationships seems to be one of the most important reasons (e.g. Brandon 1974, Gottlieb and Chafetax 1977), but runaways also have problems in school, (Walkar) seems to need to search for adventure and meaning (Watternberg, 1956), and suffer from boredom (Tobias, 1970). Hoshinl (1973) point out that the entire family of a runaway may be under stress and each family member would, if given the opportunity, choose to run to runaways more often appear to be organized around punishment and negativism and seem unable to support one another in crisis.

Occasionally, runways end up in serious trouble, but the great majority of them return home safety. Unfortunately, they often return of home situation that have not changed, and they may have to face the very same problems that pushed them to runaway in the first place. Hopefully, time away from the stressful family situation may give runaways an opportunity to reassess who they are and find new ways to deal with their difficulties. The laws in some states make running away a crime. When runaways return, they may face incarceration or they may be forced to see a counselor.

### Identity Disorders of Adolescence

DSM-III states that the major feature of this problematic reaction is the adolescent's uncertainty about his or her identity, and most important for diagnosis, severe subjective distresses regarding this uncertainty. Identity is viewed in general terms in this disorder; identity deals with issues such as "Who am I?" "What am I going to do with my life?" "By what value and standards should I live?" and many similar issues.

William Glasser, a psychiatrist and founder of reality therapy, wrote a book celled. The Identity Society (1972) that deals with many identity issues. Glasser maintains that most young people in our society no longer strive for goals as younger people in past generations had. Rather, today "roles" are sought before "goals". To illustrate this contrast, Glasser says that "almost everyone is personally engaged in search for acceptance as a person rather than as a performer of a task". Unfortunately for many of us, it is not "who we are" that counts, but rather what we do for a living. Our identity is inordinately tied to our occupational role, that, to say the least, is vulnerable. Thus, adolescents often experience profound distress over the task of making a career choice because there is uncertainty about what these careers are really like, and over whether or not they have access into educational prerequisites for these careers. The DSM-III says a

problem may become "chronic" if the person is unable to establish a career commitment or if, on another dimension, he or she fails to form lasting emotional attachments because of shifts in jobs and inter personal relationships.

One of the major achievements of early adolescence is the attainment of what Piaget calls formal operational thought (Inhelder & Piaget, 1958). Before adolescence children are largely concerned with the here and now, with what is apparent to their senses and with problems that can be solved by trial and error. During adolescence most people grow much better able to deal with problems on an abstract level to form hypothesis, and to reason from propositions that are contrary to fact. Thus according to Piaget, the stage of formal operation is seen during adolescence. This is the stage of highest level of cognitive development that people are capable of.

The attainment of formal operations gives adolescents a new way to manipulate or operate on information. They are no longer limited to thinking about the concrete here and now, as they were in the previous cognitive stage, concrete operations. Now, they can deal with abstractions, test hypothesis, and see infinite possibilities.

This advance opens many new doors. It enables teenagers to analyze political and philosophical doctrines, and sometimes to construct their own elaborate theories with an eye to reforming society. It even enables them to recognize the fact that in some situation ramifications, too. Not every one reaches the stage of formal operations by adolescence. It is not until the period of formal operations - the stage of cognitive development that is reached between the ages of eleven and fifteen that a person can think flexibly enough about the world to consider abstract universals such as freedom and justice, and to grasp their intrinsic qualities. Children develop the ability to generalize before the age of eleven, but they are not very ready to understand abstract characteristics such as congruence and mass. By the age of fifteen most individuals' can deal with these abstract concepts. They can also begin to think and operate on then level of theory, rather than being constrained by the observable facts or the apparent reality of a situation. These abilities are manifested in a number of areas, such as the realm of problem solving and scientific reasoning.

By the age of fifteen most of us can operate with these abstract concepts. We can also arrive at several possible conclusions when given a hypothesis, whereas a child would see only the various conclusions. A child shows a picture of a care wrecked in an accident, for example, may simply conclude that the car went into a skid and hit a tree. An adolescent can propose several possible causes for the accident - faulty brakes, wet road or drowsy driver.

The thinking of adolescents, unlike that of younger individuals, involves the ability to conceive of terms outside the realm of their own experience and information given. The adolescent can also deal with propositions that are contrary to act, can deal with the possible as well as the real. No longer bound by actual occurrences and data from the sensor)' world, adolescents can jump from proposition to proposition and from hypothesis to hypothesis and gain greater insight into many ideas and theories. Their interest in theoretical problems not related to everyday life and their ability to hypothesize new solutions increase throughout adolescence. The adolescents also develop a more mature notion of time the ability to conceive the distant future concretely and to set realistic long term goals. With this conception comes a new, sometimes poignant awareness that one and others are caught up the ongoing process of growth, ageing, and death. Piaget also found that the individual's ability to deal with symbols develops significantly during the stage of formal operations. One becomes able to understand political cartoons and metaphors for the first time, and can use symbols for other symbols, as in algebra. The adolescent's increased freedom in forming hypothesis often creates problems in making decisions. He or she sees not one but many alternative and this leads to doubt about his or her judgment. It often leads to external conflict, too, especially with parents and other authority figures (Weiner, 1977). Adolescents challenge adult decisions demanding to know the reasoning behind the decisions but also wanting to present the virtues of their opinions and the opinions of their peers. They are not likely to accept a decision without questions of theirs peers. They are not likely to accept a decision without questions and some debate. And are also likely to challenge religious and social values.

Interest in theoretical ideas also leads adolescent to construct ideal families, societies, and religions, as a formal operational thinker, the adolescent is freed from the bonds of personal experience and present time to explore ideas, ideals, role, beliefs, theories, commitments, and all sorts of possibilities at the level of thought (Neimark. 1975). They see that there are alternatives to the way thinking are presently done, and they want to find ways to end human suffering and poverty, social inequity, and false belief. Utopian

solutions to the world's problems planned communities, eastern religious, and new forms of conciseness - find many adherents in the adolescent group.

Adolescents caught up in idealism often place their ideals before family values. They may be outraged to find adults indulging in a few "harmless" or "pragmatic" voices while recommending virtue to young people, or practicing discrimination while invoking justice. The adolescents who take offence at such hypocrisy may rebel against the social structure in an intellectual sense but usually sense but usually they have no means to carry out the remedies they conceive.

### Adolescent Egocentrism

Egocentrism is generally characteristic of preoperational thought. It is also characteristic of adolescent thought process. In infancy egocentrism is expressed by the child's incapacity to distinguish reality from his or her own point of view and immediate experiences. Infants do not even know they have a point of view: But by adolescence, during the stage of formal operations, individuals, become able to think and reason not only about their own thought but about those of others as well. However, it is at this point according to David Elkind (1967a) that a new form egocentrism emerges. Adolescents searching to know who they are, become very self-absorbed. They indeed take into account the thoughts of others, but they assume these thoughts are all directed toward themselves. Specifically, egocentrism at this age means that adolescents believe that other people are as preoccupied with their behavior and appearance as they themselves are.

### Value System of Adolescents

The value system that an adolescent develops is closely related to his level of moral development. Generally adolescents develop their own values relative to values what others hold. However, parents, peers and teachers considerably influence the development of values among adolescents. Friendenberg (1959) believes that pressure of conformity with peers overwhelms the adolescent and prevents the full development of his or her values. Another study (Sorensen 1973) found that 86 percent of all adolescents felt that they did have their own personal values, though not so many were satisfied with the way they were putting them to use. An important part of the task, as Robert Kastenbaum suggests, is to integrate one's preferred values into an overall system. Adolescents' values generally centre on:

- Selection of friends
- Social judgment
- Leaders
- Sex

**Selection of Friends:** Adolescents select friends not only on the basis of availability in school or neighborhood but on the basis of their intellectual and emotional compatibility. Adolescents also become interested in the developing friendship with members of opposite sex. Adolescents generally want independence in selection of their friends.

**Social Judgment:** Just as adolescents have new values concerning their friends, so they have new values concerning acceptable or unacceptable members of different groups, such as cliques, or gangs, for example. These values are based largely on peer-group values which are used to judge members of the group.

Adolescents soon discover that they are judged by the same standards by which they judge others.

**Leaders:** Adolescents value leaders. They have their own ideas about the desirable qualities of leaders. According to adolescents, leaders are influential people who would lend credit and fame to them if they become familiar or associate with them.

**Sex:** Adolescents also develop certain sexual values which considerably determine their behavior towards members of opposite sex. Most of the sexual values of adolescents are greatly influenced by the peers or by the mass media. Especially films and television.

### Career Choice and Development among Adolescents

One of the most difficult and potential frustrating tasks of adolescents is the choice of the career. In the distant past vocational "choice" was really a matter of social dictate.

A farmer's son had no choice but to become a farmer and a barber's son could only become a barber. No scope existed for woman in the world of work of higher education, especially in India. Parents, even today, generally tend to see their unfulfilled dreams taking the shape of reality in their own children.

Today both women and men have, potentially speaking, a wider margin of freedom in the choice of careers. Yet various factors may make the job choice difficult and may impose limitations on one's actual range of choices. Those without sufficient education or training-dropout for example-will probably be limited to certain jobs, no matter what career they may fish for. Adolescence from economically or socially disadvantaged hems may not even be aware of the full range of career possibilities for they lack or have limited interactions with role models outside their primary relationships, and so they may unwittingly limit their aspirations (Laska & Miclin, 1979). Women and other minorities are still affected by discrimination in the job market, though the situation has improved somewhat in the past decade. Women also continue to be influenced by negative self-perceptions and as one study indicates, by internalized sex role stereotypes that evidence a clear male bias" (Hanes, Prawat. & Grissom, 1979). These influences can affect the choice of occupation.

According to Ginzberg et. al. (1951) there is three clear cut stages that adolescents to through in planning their career. These three stages are:

- **Fantasy period**
- **The tentative period**
- **The realistic period**

During the fantasy period, in the elementary school years, children's career choices are active and exciting rather than realistic, and their decisions are emotional rather than practical. The tentative period, which comes at puberty, ushers in a somewhat more realistic effort by youngsters to match interests with abilities and values. By the end of high school, students enter the realistic period and can plan for the appropriate education to meet their career requirements.

Many young people, however, are still not realistic in late adolescence. One of the most important questions to be asked with respect to the career choice of an adolescent is "how do adolescents make career choices?" Many factors enter in, including individual's ability and personality, education, socio-economic, racial or ethical background; societal values; and the accidents of particular life experiences. Parental attitudes and behaviour and one's gender considerably influence one's choice of career. The choice of a career is closely tied in with a central personality issue during adolescence; the continuing effort to define the self, to discover and mould an identity. The question "Who shall I be?" is very close to "What shall I do?" If we choose a career that we feel is worth doing and one we can do well, we feel good about ourselves. On the contrary, if we feel that it wouldn't matter to anyone whether we did our work or not, or if we feel that we are not very good at it, the core of our emotional well-being can be threatened. Thus, one's self-concept also considerably influences one's career choice.

### **Delinquency among Adolescence**

Delinquency is another common adolescent problem. Adolescents are more likely to face legal problems and indulge in criminal acts as compared to children.

Violence and vandalism (destruction of public and private property) are among the more common types of delinquent behavior observed among adolescents.

Once it was strongly believed that delinquency among adolescent is a product of lower socio-economic status. However, researchers today contend that delinquency today be more of a middle class phenomenon. Today, sex differences are also diminishing as girls are almost as likely as boys to be involved in delinquent acts. Most delinquent activities are performed by groups of adolescents rather than done singly. Participation in delinquent activities, however, is not the result of any single factor but rather the combination of influences that motivate a teenager to be destructive and perform illegal acts. Several themes recur consistently in research an adolescent delinquency. More delinquency comes from homes that are broken by divorce, separation or desertion than are non delinquents. More delinquents appear to have experienced deficient socialization and inadequate parenting than non delinquents. Peer relation has been found to have greater influence than parental influence on participation in delinquent activities.

Many delinquent acts involve malicious destruction of property (Vandalism) although delinquencies also cover truancy from school as well as petty theft and larceny. The acts of vandalism may be performed to

relieve boredom, and to provide excitement and as a way of passing recreational time with friends. Shop lifting is another type of delinquent behavior. It is associated primarily with females coming from homes where money is restricted or not readily available, where there is little family financial planning and little participation by the person in family chores.

In some cases, delinquency has been related to a history of physical and sexual abuse and to neurological and psychiatric problems. Stealing, lying, truancy and poor achievement in school are all important predictors of delinquency (Loeber and Dishion 1983). The strongest predictor of delinquency is the family's supervision and discipline of the children. Anti-social behaviour in adolescent is closely relate to parent's inability to keep the track of what their children do and with whom they do it.

Three are tow types of juvenile delinquents. One is the status -offender. This is young person who has been a truant, has run away from home, has been sexually active, has not abided by parent's rules, or has done some thing else that is ordinarily not considered criminal - except when done by a minor.

The second of juvenile delinquent is one who has done something that is considered a crime no matter who commits it - like robbery, rape or murder. People under age 16 or 18 (depending on the state) are usually treated differently from adult criminals. Court proceedings are likely to be secret, the offender is more likely to be tried and sentenced by a judge rather than a jury, and punishment is a usually more lenient. However, for some particularly violent crimes, minors may be tried as adult. Discuss sexual behaviour and practices among adolescents

### **Adolescents Relationship with Parents**

There is a general myth that adolescents are always in conflict with parents and that they do not get well along with their parents. However, research studies do not confirm with this myth. Most adolescents feel close to and positive about their parents, have similar values on major issues and seek their parents approval.

Young people feel a constant tension between needing to break away from their parents and realizing how dependent they really are on them. Adolescents ambivalent feelings are often matched by their parents own ambivalence torn between wanting their children to be independent and wanting to keep them dependent, parents often find it hard to let go. As a result, parents may give teenagers "double messages", that is the parents will say one thing but will actually communicate just the opposite by their actions.

Contact is more likely to surface between adolescents and their mothers than adolescents and their fathers (Steinberg. 1981-1987). This maybe partly because mother have been more closely involved with their children and may find it harder to give up their involvement. It may also be because fathers sometimes tend to withdraw from their teenager children -from their developing daughters, out of discomfort with the sexual sittings they may feel towards them; and from their sons, who may now be bigger than both parents and more aggressive.

By and large, parents and teenagers do not clash over economic religious, social or political values. Most arguments are about mundane matters like schoolwork, chores, friends dating, curfews and personal appearance. Later in adolescence, conflict is more likely to revolve around dating and alcohol.

Most disagreements are resolved with less trouble than popular mythology suggests. Quarrels may reflect some deep quest for independence (as is often speculated), or they may just be continuation of parents efforts to tech children to conform to social rules.

Discord generally increases during early adolescence, stabilizes during middle adolescence, and then decreases after the young person is about 18 years of age. The increased conflict in early adolescence may be related more to puberty than to chronological age, and some intriguing new research suggests that it may even be bi-directional (Steinberg, 1988).

### **Adolescents Relationship with Peers**

Adolescents spend a great deal of time with peer groups. The influence of peer group is considerable on the adolescent. Adolescents' values reflect the value of peer group. They are more likely to reject the parental values and accept the value system of peer group. Members of the adolescent peer group are constantly influencing and being influenced by each other.

**EARLY ADULthood**

Adulthood is a general term which covers a period of 21 years onwards. Early adulthood is a period that generally ranges from 18 to 45 years onwards. We would discuss the various characteristics of early adulthood period.

Following this we would discuss the physical development that takes place during early adulthood years. Both, biologically and physiologically an individual is at the peak of development. We would discuss the wide variety of physical changes that occurs in early adulthood. These changes are in weight and height, eyes, muscular strength, teeth, heart, lungs, skin etc. Health is at its optimal during early adulthood years. There are direct and indirect influences on health. The direct influences are diet exercise, smoking, alcohol and stress. Indirect influences are diet exercises, smoking, alcohol and stress. Indirect influences on health are socio-economic factors, education, gender, material status etc.

Following this we would discuss cognitive development during early adulthood. We would also examine the theoretical approaches concerning adult thought. Among the theoretical approaches we would examine the views of Piaget, Warner Shaie and Robert Sternberg.

Many personality changes occur during early adulthood. We would discuss the growth trends in adjustment. According to White (1975) there are Five growth trends these are stabilization of Ego identity, Freeing of personal relationship, Deepening of interests, Humanising of values, and Expansion of caring.

Adulthood is a period of physical as well as emotional and intellectual maturity. We would define the concept of adult maturity and discuss the Allport's seven dimension of maturity.

Adulthood is also a period of dating, falling in love and getting married. We would discuss these processes as they take place in early adulthood. We would also examine sex differences in marriage in early adulthood and the personality factors in marital adjustment.

Following this we would define love and discuss the theories of Passionate love and the Triangular model of love. Today many alternatives to traditional marriages are available. Some of the alternative to traditional marriage that we would discuss is single adulthood, contract marriage, community living, group marriage and swinging.

Related to marriage are two important topics which we would discuss in the form of short notes. These are divorce and cohabitation. We would end his chapter by discussing many important topics in the form of short notes some of these topics are developmental task of young adulthood, sexually in young adulthood, parenthood and new methods of becoming a parent. We would also discuss the issue of remaining childless, reconstituted family and the issue of single parent family. Working mothers, face many hardship and difficulties and they have to make man adjustments. We would discuss about working mothers and their life style. Towards the end, we would discuss occupational development during young adulthood.

Adulthood is a general term which covers a period of 21 years owners. It is a period which has been sub-divided into early, middle and late adulthood.

The term adult comes from the Latin verb "adulhs" which means "to grow to maturity" or "to grow to full size and strength". Adulthood is generally divided into three different stages.

**Early Adulthood (Eighteen to 45 years)**

**Middle Adulthood (45 to 65 years)**

**Late Adulthood (65 years to death)**

**EARLY (YOUNG) ADULTHOOD****Characteristics of Early Adulthood**

Following are important characteristics of early adulthood.

Early adulthood is a period of adjustment to new patterns of life and new social expectations. People enter into marital relationship, job etc.

Early adulthood is a "Setting down age". During this period of "Carefree days" are over. Settle down involves finding economic stability and a stable marriage partner.

Early adulthood is a "Reproductive age"

Most individual, are for women in 23-29 years. Majority of them marry and conceive during this period.

Early adulthood a period of change of values. Many values acquired during childhood and adolescence change during adulthood. Many core values become consolidated.

Early adulthood is a stage of individualism and creativity. Adults like to maintain their individuality; they are not innocent conformist as children or blind followers of certain practices as they did during adolescence. Adults spend a considerable amount of their time in the pursuit of intellectual and creative tasks.

Early adulthood involves certain developmental tasks some of which are as follows:

- Selecting a mate
- Leaving to live with a marriage partner
- Starting a family
- Managing a home
- Getting started in an occupation
- Taking on civic responsibility

During the early adulthood (the twenties and early thirties) the individual is at the peak of life biologically and physiologically. Individuals reach their peak of physical well-being during the years of early adulthood. This period is notable for being time of one's greatest strength and good health. Physical maturity in height and weight has been completed, and growth has ceased several years before early adulthood beings. Yet many changes occur in physical appearance and in the body's psychology over this period of time.

### **Weight and Height**

The effects of the ageing process can be seen beginning in early adulthood. One of the most prominent and visible signs of this process is the height and weight changes that occur between eighteen and forty-five. Weight changes are more noticeable and significant than height changes during this period.

The average weight of Men between eighteen and seventy-four years is 172 pounds (Abraham, 1979). Between eighteen and twenty four years, and average weight of men increases from 165 pounds to 178 pound between thirty-five and forty four years. The average weight of women between eighteen and seventy four years is 143 pounds (Abraham, 1979). This figure is twenty-nine pounds less than that for men at the same age. Average weights for adult females show gains from 132 pounds between eighteen and twenty-four to 149 pounds between thirty-five and forty four years.

Differences in this general pattern depend on the race of individuals. Black men as compared to white men weigh more on the average and show greater gains in weight during early adulthood. Likewise, black women weight considerably more than white women and show greater average weight gains during this period (Abraham, 1979). These differences may be attributed to differences in eating patterns and nutritional status of the two groups of people.

One of the principle causes of weight gains in an early adulthood is the age-related increase in fat deposition in the body (Timiars, 1972). The body increasingly stores excess fat under the skin as age increases in early adulthood. This process continues throughout the period. Another contributing factor to weight gains in adulthood may be associated with decreasing levels of physical activity. As individuals tend to become more sedentary in their daily lives through this period, excess fat and calories in the diet are not burned off but become stored in the body. Some slight changes occur in height measurements for adults between eighteen and forty-five. The mean height for men between eighteen and seventy-four years is 69 inches (five feet, nine inches). Men can be expected to decrease an average of one-half inch for each decade of age (Abraham, 1979). Between eighteen and twenty-four years of age, male have an average height of 69.7 inches decreasing to about 69 inches by age thirty-five to forty-four years.

Women have an average height of 63.6 inches or 5.4 inches less than men during eighteen to seventy-four years (Abraham, 1979). Height changes for women follow a similar pattern for men. At eighteen to twenty four years, the average height can be expected to be 65.3 inches, decreasing to 64.1 inches at twenty five to thirty-four years and remaining constant at thirty-five to forty-four years. Racial differences are noted also in height changes. While women are only slightly taller than black women by an average of about one-quarter inch. Timiras (1972) explains these slight decreases in height during early adulthood to be due to a setting of the spinal column and to the steady decrease in the density of the long bones of the body (arms and legs) and in the vertebrae of the spinal column.

### **The Eyes**

The ability of the eyes to accommodate begins to change in early adulthood. Accommodation refers to the adjustments made by the eyes for seeing things at different distances, or the ability of the eyes to focus properly and quickly. There is a gradual loss in the quickness of eyes to adapt for focusing objects until about the age of forty-five when accommodation is at its lowest (Allen, 1956).

Acuity or the sharpness of vision is at its peak for most individuals in the years of early adulthood. Males have better average eyesight than females (Roberts and Rowland, 1978).

### **Muscular Strength**

The peak of an individual's muscular strength occurs between twenty and thirty years. After this time, there is a continuous decline in strength that increases rapidly in the years of old age. Muscle strength of men is contrasted with work rates. Work rate is measured by the power or effort taken to crank a drive sprocket. While muscle strength does not change considerably between twenty and sixty-five, power output decreases after the age of forty and decreases considerably during the middle adulthood years (Shock and Norris, 1970).

### **The Teeth**

Most individuals still retain all of their permanent teeth during early adulthood. A small percentage has lost all of their upper or lower teeth (Kelly and Harvery, 1979). The loss of teeth is more likely to occur to women than to men during this period. There is a steady increase in decayed, missing or filled teeth during the years of early adulthood. The probability of experiencing periodontal disease also increases during early adulthood, especially for men. Periodontal disease is an inflammation of the gums and bone tissue surrounding teeth which results from poor or improper dental hygiene. Young adults become careless in caring for teeth with this condition often being the consequence. Periodontal disease can lead to loss of teeth if left uncorrected.

### **The Heart**

As an individual grows older, the heart becomes more sluggish in its ability to pump blood efficiently. This partially explains why people are unable to sustain large work load for long periods of time as they grow older. The efficiency of the heart to pump large amounts of blood remains fairly stable during early adulthood.

### **The Lungs**

The lungs change in their ability to function efficiently over the period of adulthood. Young adults are able to absorb about four and a half times as much oxygen as an older adult. The lungs play an important part in the body's ability to exercise and work muscles properly. The decline in oxygen absorption partially can be traced to lowering of the amount of blood pumped through the lungs as well as to losses in the mechanical efficiency of the lungs as people grow older (Timiras, 1972). Through the adulthood years there is a decline in the amount of air breathed into the lungs. This difference is noticeable among many individuals at the age of forty.

### **The Skin**

The skin begins to show some signs of aging in early adulthood. These signs are the first wrinkles that appear in the facial area, particularly around the eyes (crow's feet) and on the hands. The skin begins to lose its fine texture toward the end of the period. While acne and other related skin disorders are most frequent in adolescence, most individuals experience a recovery from these more frequently among young adults are fungus infections (dermatophytosis), especially of the feet and malignant or benign tumors. These conditions affect adults, males more than females, reflecting perhaps the effects of different working conditions and standards of hygiene as causes of these disorders (Johnson and Roberts, 1977).

### **Theoretical Approaches Concerning Adult Thought**

Adult thought is very much different from childhood thought. There are three approaches to adult thought, these are as follows:

- **Piaget's view and post-formal thought.**
- **Warner schiae on adult thought**
- **Robert Sternberg**

We will discuss each of these in brief.

**1. Piaget's views on post-formal thought results** - Piaget held that cognitive progress from infancy through adolescence from a combination of maturation and experience. What happens, then, in an **adult**? **Experience** plays an especially important role in intellectual functioning. But the experiences of an adult are different from and usually are broader than those of a child, whose world is defined largely by home and school. Because adults have such diverse experiences, it is very hard to generalize about the effects of experience on cognition in adults. Mature thinking is even more complex than the use of formal logic in Piaget's stage of formal operations, which Piaget considered the highest level of thought. Thought in adulthood is flexible, open and adaptive in new ways that go beyond logic. It is sometimes referred to as post-formal thought.

A shift occurs in mature thought: as a result, thinkers combine both the objective (rational, or logical, elements) and the subjective (concrete elements, or elements based on personal experience). This shift helps people take their own feelings and experiences into account (Labourie Vief, 1985,1986; Labourie - Vief & Hak8n - Larson, 1989). Mature thinkers personalize their reasoning using the fruits of their experience when they are called on to deal with ambiguous situations. Post-formal thought is also characterized by a shift from polarization (right versus wrong, logic versus emotion, mind versus body) to an integration of concepts.

**2. Warner Schaie.** Conceive of cognitive development (i.e. thought) in adults as progressing **through** stages of five steps.

K. Warner Schaie (1977-78) who believes that intellectual development proceeds in relation to people's recognition of what is meaningful and important in their own lives. The five stages of Schaie's theory chart a series of transitions from "what I need to know (acquisition of skills in childhood and adolescence), through "how I should use what I know" (acquisition of skills in childhood and adolescence), through "how I should use what I know" (integration of these skill in a practical framework), to "why I should know". (a search for meaning and purpose that culminates in the "wisdom of old age"). According to Schaie, real-life experiences are important influences on this progression. The sequence of stages in Schaie's model of cognitive development is as follows:

**(i) Acquisitive stage** (Childhood and adolescence). In the acquisitive stage, information and skills are learnt mainly for their own sake, without regard to the context, as a preparation for participation in society. Children and adolescents perform best of test that give them a chance to show what they can do, even if the specific tasks have no meaning in their own lives.

**(ii) Achieving stage** (late teens or early twenties to early thirties)

In the achieving stage, people no longer acquire knowledge merely for its own sake but use what they know to become competent and independent. Now, they do best on tasks that are relevant to the life goals they have set for themselves.

**(iii) Responsible stage** (late thirties to early sixties). In the responsible stage, people are concerned with long-range goals and practical real-life problems that are likely to be associated with their responsibilities to others (like family members or employees).

**(iv) Executive stage** (thirties or forties through middle age).

People in the executive stage are responsible for societal systems (like governmental or business concerns) rather than just family units; they need to integrate complex relationships on several levels.

**(v) Re-integrative stage** (late adulthood). Older adults - who have let go of some involvement and responsibility and whose cognitive functioning may be limited by biological changes are more selective about what tasks they will expend efforts on. In this re-integrative stage, they think about the purpose of what they do and bother less with tasks that have no meaning for them.

**3. Robert Sternberg** has proposed three aspects of intelligence: componential (critical), experiential (insightful), and contextual (practical). The experiential and contextual aspects develop and become particularly important during adulthood.

(a) Componential element how efficiency people process and analyze information. The componential element is the critical aspect of intelligence.

It tells people how to approach problems, how to go about solving them and how to monitor and evaluate the results.

b) Experimental element how people approach novel and familiar tasks. The experiential element is the insightful aspect of intelligence. It allows people to compare new information with that they already know and to come up in new ways of putting facts together - in other words to think in original ways (as Einstein did. for example, when he developed his theory of relatively). Automatic performance of familiar operation (like recognizing words) facilitates insight, because it leaves the mind free to tackle unfamiliar tasks (like decoding new words).

(c) Contextual element how people deal with their environment. The contextual element is the practical "real-world" aspect of intelligence. It becomes increasingly valuable in adult life - as in selecting a place to live or a field of work. It involves the ability to size up a situation and decide what to do: adapt to it, change it, or find a new, more comfortable setting.

Psychometric tests measure componential (critical) intelligence rather than experiential (insightful) or contextual (practical) intelligence. Since experiential and contextual intelligence are very important in adult life, psychometric tests are much less useful in gauging adult's intelligence than in gauging children's.

Early or young adulthood is the fullest, most individualistic, and at the same time, loneliest period of life (Havighurst, 1974). During this time tremendous pressures are brought to bear on individuals to "make a constructive place" for themselves in society. And yet the support system available for any person to accomplish this goal is few indeed. Havighurst has suggested that with the exception of the elderly, young adults receive less educative support to accomplish their developmental tasks (the prescriptions, obligations and responsibilities thought to be related to healthy adjustment) than any other age group.

### Definition of Adult Maturity

Maturity refers to a state that promotes physical and psychological well being. In most instances, the mature person possesses a well developed value system, an accurate self concept stable emotional behaviour, satisfying social relationships and intellectual insight. Coping with the demands of adulthood, mature individual is realistic in the assessment of future goals and ideals. According to White (1960) maturity implies like ability to cope more successfully with life's problems, increasing the effectiveness of our planning strategies, deepening our appreciation of the surroundings and expanding our resources for happiness.

### Allport's Seven Dimensions of Maturity

One of the more extensive published descriptions of maturity is that of Allport (1961), who postulates that maturity is an ongoing process best characterized by a series of attainments on the part of the individual. Each period of life has its share of obstacles that must be overcome road blocks that require the development of global formulation and decision making abilities. Methods for dealing with life's failures and frustrations as well as accepting its triumphs and victories have to be devised if maturity is to be nurtured. Age in itself is not a guarantee of maturity. Allport has identified seven specific dimensions or criteria of maturity that manifest themselves during adulthood. These seven dimensions include:

1. Extension of the self
2. Relating warmly to others
3. Emotional security
4. Realistic perception
5. Possession of skills and competencies.
6. Knowledge of the self
7. Establishing a unifying philosophy of life.

We would discuss each of them in brief.

**1.Extension of the self:** The first criterion of maturity, self extension, requires that individuals gradually extend their comprehension to encompass multiple facets of their environment. The sphere of the young child was primarily limited to the family, but over time the child becomes involved in various peer groups, in school activities and in clubs. Eventually, strong bonds develop with members of the opposite sex, and interest toward vocational, moral and civic responsibilities is generated. Each outlet provides the young adult with the opportunity to become involved in more meaningful person relationships and to fulfill the need of sharing new feelings and experiences with others (Henis and Tuner, 1976). While (1966) refers to the foregoing process as the deepening of interests. In a series of case studies, he discovered most mature young adults tend to become engaged in vocational, athletic, or academic pursuit. Each requires the extension of the self and the ability to experience involvement of some sort. Yet like Allport, White maintains that merely being involved in something does not necessarily imply satisfaction or happiness. Maturity is measured by one's active participation in an activity. Maturity implies movement away from a state in which interests are casual quickly dropped and pursued only from motive that do not become identified with the advancement of the interest or activity. True self-extension is a state in which a sense of reward comes from doing something for its own sake. In other words, maturity is promoted when the activity undertaken has true significance to the self.

**2. Relating warmly to others.** Allport's second criterion of maturity is the ability to relate the self warmly to others. By this, Allport means the capacity to be intimate with as well as compassionate towards others. How does one develop the capacity for intimacy? One of the more widely accepted interpretations of an adult psychological growth has been provided by Erickson (1963). During early adulthood mature psychosocial development is measured by the successful resolution of the stage known as intimacy versus isolation. Prior to this early adulthood, the individual was in the midst of an identity crisis, a struggle that reached its peak during adolescence. Erickson stresses the idea that as a young adult the individual is motivated to fuse this newly established identity with that of others. In short, the young adult is ready for intimacy, which means not only committing the self to personal relationships but also nurturing the motivation to maintain them. Most adults satisfy intimacy through marriage.

It is important to stress that intimate relationships other than sexual ones as possible. Individuals may develop strong bonds of intimacy in friendships that offer, among other features, mutuality, empathy and reciprocity.

**3.Emotional security:** Although numerous dimensions of maturity can be grouped under this third category, Allport maintains that four qualities in particular are important (a) Self acceptance (b) emotional acceptance, (c) frustration tolerance and confidence is self-expression.

Self acceptance is the ability to acknowledge one's self fully, particularly in terms one's imperfections. Mature people realize that they cannot be perfect in every respect, yet they nevertheless seek to fulfill their own potential. Total self acceptance requires exploring and accepting one's weaknesses.

By mature emotional acceptance, people accept emotions as being part of the normal self. People acquiring this dimension of maturity do not allow emotion to rule their lives, yet at the same time they do not reject emotions as being alien in nature.

Frustration tolerance is the capacity to continue functioning even during time of stress. To be able to handle life's frustrations and still manage to carry on is a formidable goal. For maturity to develop, one must learn how to best deal with life's frustrations and maintain a healthy life style.

The final dimension of emotional maturity is confidence in self expression. Maturity in this respect implies spontaneity; one is aware of one's own emotions, is not afraid of them, and has control over their expression. Immaturity conversely can manifest itself in a number of different ways, including timidity and shyness, emotional over reaction, or emotional under reaction.

**4.Realistic Perception** - Allport's fourth criterion of maturity is realistic perception. Quite simply, maturity in this sense means being able to keep in touch with reality, without distorting the environment to meet individual needs and purposes. Sometimes the complexities of events and situations combined with the ego defenses of the individual may produce an inaccurate interpretation of the environment. The mature mind is able to perceive the surroundings accurately. Allport is not implying that the mature person does not use any type of defense or coping mechanism. On the contrary, defense mechanisms become quite automatic for many of us and tend temporarily to alleviate anxiety and frustration. Allport's point is that the overuse or mis-use of such mechanisms usually distorts one's perception of the surroundings.

**5. Possession of skills and competencies** - Possessing some type of skill or competence represents Allport's fifth dimension of maturity. Unless one possesses some basic skill, it is virtually impossible to nature

the kind of security for maturity to develop. While the immature adolescent may argue, "I, no good at anything", mature adults strive to develop whatever skills they feel they possess.

Furthermore, skilled individuals are driven by a need to express their competence through some type of activity. They identify with their work and display pride in the skills needed to produce the finished product. In this sense, task absorption and ego-relevant activities are important to physical and psychological well-being.

**6. Knowledge of the self**- Knowledge of the self, or self objectification, is criterion number six. Most mature people possess a great deal of self-insight, of which many immature individuals have little. According to Allport, knowledge of the self involves three capacities; knowing what one can do, knowing what one cannot do, and knowing what one ought to do.

White (1966) believes that knowledge and stabilization of the self is one of the most important growth trends of young adulthood. In general, White proposes that the stabilization process owed much to those enduring roles that are characteristics of adult life. More specifically, he states that as individuals modify their behavior in order to fulfill their roles as workers marriage partners, and parents, for example, their experience beings to accumulate more and more selectively. In this sense the stored up source of stability and ego identity emerge increasingly out of behavior within roles.

**7. Establishing a unifying philosophy of life** - The final criterion or dimension of maturity outlined by Allport is the development of a unifying philosophy of life that embodies the concept of guiding purpose, ideals, needs, goals, and values. Since the mature human being is goal-seeking, such a synthesis enables him or her to develop in intelligent theory of life and to work toward implementing it. Mature people tend to view goals from a balanced perspective and are able to cope with failure if these goals are not met.

## Marriage

Most adults marry, usually for the first time in young adulthood. But people have been marrying at later and late ages. In 1988, the median ages of first-time bridegrooms was 25.9 and first-time brides, 23.6 years, compared with 24.7 and 22 years, respectively, in 1980 (US Bureau of the Census, 1988).

Studies done from the 1950s found that married people were happier than singles. Either marriage brought happiness, or happy people tended to marry.

In one study of 2000 adults around the country, for example, married men and women of all ages reported more satisfaction than people who were single, divorced or widowed. The happiest of all were married people in their twenties with no children especially women. Young wives reported feeling much less stress after marriage, while young husband, although happy, said that they felt more stress (A Campbell, Converse & Rodgers, 1975). Apparently, marriage was still seen as an accomplishment and a source of security for a woman but as reasonability for man. Sex Differences in Marriage - Women and men feel differently about marriage in other respects. Women see marriage as a place to express and talk about emotions; they consider the sharing of confidence as measure of intimacy. Men however, define intimacy differently; they tend to express love through sex, giving practical help (like helping her with the household chores), doing things together, or just being together (L. Thompson & Walker, 1989). As a result, men often get more of what is important to them; since women do the things that matter to men. Many men do not feel comfortable talking about feelings - or even listening to their wives talk about theirs - and this leaves the wives feeling dissatisfied. The ability of marriage to bring happiness seems to be changing (Glenn, 1987). Although more married people, than people who have never married call themselves "very happy", the gap has narrowed dramatically -among 25 to 39 year olds, from 31 percentage point in the early 1970s to 8 points in 1986. Apparently, never married people are happier today, while married people (especially women) are less happy. One possible reason is that some benefits of marriage are no longer confined to wedlock. Single people can get both sex and companionship outside of marriage and marriage is no longer the sole (or even the most reliable) source of security for women. Also, since most women now continue to work, marriage is likely to increase rather than decrease their stress. Marriage and Health - Marriage is a healthy state. Married people tend to be healthier than those who are separated, divorced or widowed (Anson, 1989). Married people have fewer disabilities or chronic conditions that limit their activities and when they go to the hospital, their stays are generally short. Married people live longer, too according to a

study going back to 1940 in 16 industrial countries (Hu & Goldmna, 1990). Those who have never married are next healthiest group, followed by widowed people and then by people who are divorced or separated. Personality Factors in Martial Adjustment - The newly married couple not only has to adjust to new roles those of husband and a wife-but also to one another. Every person has a unique personality that influences the way he or she will approach and adjust to new situations - including marriage. Some personality characteristics enable an individual to adapt well to marriage and to build a stronger relationship. Whereas, others do not. Characteristics such as emotional maturity. Self-control, willingness to engage in self-disclosure, ability to demonstrate affection and consideration for others and the ability to demonstrate affection and consideration for others and the ability to handle frustration and anger combined with high self-esteem and flexibility all have been linked to marital satisfaction ( Stinnet & Walters, 1977). So too, has the ability to communicate openly and honesty with one's spouse.

Interestingly, many studies indicate that the beginning of marriage at least traditional marriage it is the husband's personality traits rather than the wife's that are more strongly related to late martial happiness (Barry 1970). Among the important factors are the husband's stable male identity" which is, in turn, related to the happiness of his parent's marriage and to his close attachment to his father. High socio-economic status and educational level in the husband are also correlated with martial success. Perhaps most important in traditional marriages are the eyes of the beholder that is, wife's perception of the husband's maturity and role enactment. The higher the wife rates here husband on emotional maturity and the closer he seems to come, in her eyes to fulfilling the culturally prescribed role of husband, the happier the marriage. However, in light of the increasing equality between the sexes and the movement away from traditional marriages, one may question whether this pattern will continue to exist in the future.

Define love and discuss the theory of passionate love. Love can be defined as an emotional state involving attractions, sexual desire and concern about another person. Lover represents the most positive level of attraction. A close friendship turns into love when two people start viewing each other as potential sexual partners. Passionate love is one of the most important forms of love which has received a great deal of research attention. It is this type of love which is a topic of great concern, controversy and discussion among lay man as well as in most media like films, television and literature. It can be defined as an intense and often unrealistic emotional response to another person. Most of the recent theoretical and research interest has been centered on passionate love. This refer to an intense, sometimes overwhelming emotional state in which an individual thinks about his or her lover constantly want to spend as much time as possible with that person and is often unrealistic in judging the loved one (Murstein 1980).

## DIVORCE

Divorce is largely a phenomenon of young adulthood. Most people divorce during the seventh year of married life. About 2 out of 3 first marriages are estimated to result in divorce. It is not only common in USA. And European countries but the rate of divorce is increasing in India too. The typical divorced person is between thirty and thirty-three when divorce becomes final and lives in a city and has at least one child (Hunt & Hunt, 1977). The divorce rate is higher among the poor, the working class, and the poorly educated (Click 1975). Geographically, divorce is more common in the West. A California demographic study, for example, yielded the following data. Every married man in that state will marry an average of one and two-thirds time and every woman can expect to spend six and one half years as a divorcee (Schoen & Nelson, 1974). Nobody knows for certain what accounts for all these divorces, part from the increased social acceptance of divorce itself. However, it is clear that divorcing people tend to have married early (Click and Norton, 1971). For example, people who got married in their teen's shows quite a high probability of divorce. Those married in their late twenties shows consistently lower probabilities.

The process leading to divorce differs for each couple. Hunt and Hung (1977) identify three common scenarios (1) for relatively new couples, the marriage simply fades like an old photograph - that is, without conflict or awareness. Any chance happening, a job offer in another town passing flirtation is enough to bring the relationship to a quiet end. (2) In the second scenario, separation and divorce come as a shock to one of the partners: A woman who is happily in love with her husband finds a love note in his pocket. A man who is proud of his serene, smoothly running marriage comes home to find his wife strangely glum; she asks him to sit down, tells him she must leave to find some happiness in her life before it is too late (Hunt and Hunt, 1977).

In such relationship, one person is entirely ignorant of the other's feelings, much less, activities; in retrospect, no intimacy seems to have existed (3) Perhaps most common are divorces undertaken only after prolonged and organizing conflict. In this scenario, both partners realize that divorce is likely, but wait months or even years to take the final steps.

Naturally, reactions to divorce depend on the nature of process that preceded it. The happy wife who comes home to find a note taped to the refrigerator will react with shock; the wife who has been locked in self-destructive conflict with her husband for five years will react with relief. There is much evidence, though, that both will suffer a great deal of pains and will need time to "Mourn" the relationship. Some people, however, are never able to adjust to the divorce. They react to the breakup of their marriage with profound depression and anguish, in some cases, to the extent of contemplating suicide.

The divorced person will also encounter an identity crisis in the course of building a new life-style (Wiseman, 1975). For example, the woman who married young may have tied her identity to that of her husband to her married status. Now she has the new status of divorcee, and possibly single parent. She will need to establish a career identity, or at least find a job. And she will need to resolve issues of sexuality identity. At the same time that the individual is resolving her identity crisis and experimenting with sexual intimacy, she is confronting a host of practical problems. For most young couple divorce brings financial distress and a lowering of the standard of living, perhaps by as much as 25 percent. If there are children new parental relationships must be developed. There is especially true for the father, who does not usually have primary responsibility for child care.

Most divorced people eventually remarry. About one quarter do so within the year; within three years. One-half are remarried; within nine years, three quarters (Glick and Norton, 1971). Men tend to remarry sooner than women. Though divorce rates are higher for second marriages, the majority of those who marry again remain married. For many, divorce is seen as a growth experience by means of which they are able to find not only a new and more suitable partner, but a greater awareness of themselves - what they need and what they have to give to a relationship.

### Occupational Development and Young Adulthood

Early adulthood is the period in which people are expected to make decisions about what kind of work they want to do. They define their relationship to society by narrowing their occupational choice and finding their first job. In the mid-1960s in the middle class the consensus was that twenty to twenty-two was the

best age of finish one's education and go to work between the ages of twenty-four and twenty six such a person was expected to settle on a career (Neugarten Moore and Lowe. 1965). Today with increased acceptance of post college training or a few years of moratorium time for finding oneself a person may be twenty-five or thirty before being expected to do full-time work. Moreover, people from different "settle down" in an occupation. Nevertheless men and women are expected to make an occupational commitment, or the educational choices that lead to one, during their twenties. It is usually during the early adult years that the individual takes a first serious full-time job. In many cases, this job represents an implementation of the persons' self-concept for example the person who believes that she has the qualities of a nurse actually becomes a nurse. Research using trait description checklist, has in fact shown that there is a high correlation between people's self-concept and occupational stereotypes or images to which they aspire (Holland 1973). For most people career choice is not a one-shot decision made in early adulthood. The life cycle imposes different tasks at different periods of life; consequently, people develop and change in respect to their vocations. It has been suggested (super, 1957; 1963) that there are five stages of occupational development.

1. Crystallization of one's ideas about work (14-18 years)
2. Specification of a particular occupational preference and the beginning of job training (18-21 years ;)
3. Implementation of training and entry into first job (21-24 years)
4. Stabilization or becoming established in a particular field (25-35 years)
5. Consolidation and advancement within a field or on the job (35 years and on).

Super has also noted that as people go through these stages of vocational development there is continual updating and implementing of self-concept.

Other researchers (Levinson et al., 1976) whose sample consisted of forty men in four different occupations from blue-collar, worker to novelist, did not project such a straight line in occupational development as suggested by super, Levinson's stages include more tentative exploration and several crises periods of reassessment. He said that it is a cruel myth to believe that at the end of adolescence you choose your career, settle down and continue this way more or less indefinitely.

In early adulthood a stage Levinson calls Jetting into the Adult World, the young man tries to settle on an occupation or an occupational direction in line with his own interests and his sense of his own identity. He explores the possibilities in the work world and at the same time tries to match what he finds with his sense of his own potential. His task is to build a life structure, forging a link between the work world and his own identity. This is a period of provisional choices. From the ages of about twenty-eight to thirty two many men experience a transitional period, a crisis of reassessment. For every choice made, parts of the self are ignored. These aspects of the self come to the surface and must be dealt with. For example, a man who has gone into the manufacturing business with his father may not want to settle for just making money - he may want to have a try at writing a play. He stays home for eighteen months, completes the play and then finds a new career as an advertising copywriter and account executive. Some people find the right combination of career and identity and make a deeper commitment. Levinson's observations led him to hypothesize that if a significant start is not made by the age of thirty-four, chances are small that a man will find a satisfactory life structure with an occupation consistent with his identity and interests.

Sometime in his early thirties a man settles down. This is a period of deeper commitment to work and to family. The individual makes and pursues long-range plans and goals. Although he feels autonomous in his work he may, in fact, be subjected to many restrictions imposed by higher-ups or by the rules under which he works. This may push him into the next stage. Becoming your own man may occur from the late thirties to early forties. At this time a man wants desperately to have society affirm him in his work role - to be made manager or foreman or head of his department. This period comes to an end with another crisis, the Midlife Transition around forty to forty-five. Whether or not a man achieves the recognition he feels he deserves, he may go through a period of reassessment.

## MIDDLE ADULTHOOD

### Changes in Weight and Height

The trends of aging first appeared in early adulthood continue to occur in middle age. Sex differences are evident in both weight and height during this period; however, Men show an average decrease while women

show an average increase in body weight in middle age. The average weight of men in the group aged forty-five to fifty-four years is 175 pounds, decreasing to an average weight of 171 for the group aged fifty-five to sixty-four years. The average weight for women increases from 132 pounds at eighteen to twenty-four years to 149 pounds at forty-five to sixty-four years. These differences in weight changes may reflect differences in biological functioning of men and women as well as differences in lifestyle that result in earlier loss of body weight by men than by women a comparable time in life.

Accompanying the general decline in body weight are weight losses of most internal organs. The declining weight trend begins in the fifties for humans and primarily involves the skeletal muscles, liver, kidneys and adrenal glands. The heart is the exception here. This organ generally shows increases in weight with age as attempts to compensate for its declining efficiency by becoming enlarged (Timiras, 1972).

The trend the declining height continues during middle age, occurring more rapidly among women than men. The decline in height reflects the continued shortening of the spinal column caused by shrinkage of the elastic disk material. The declines in height are quite small and almost unnoticeable during these years.

### **Changes in Skin Tissue**

The organ most often associated with aging is the skin. Changes that occur to the skin are perhaps the most noticeable indication that aging is taking place. Changes in skin tissue that appears in middle age involve the texture, composition and appearance of these tissues. As age increases the skin becomes dryer and losses its ability to retain moisture. Because of these changes, the skin feels rougher to the touch. The loss of elasticity in the skin results in the increasing appearance of wrinkles, particularly in the neck, faces and hand areas. Loss of elasticity of the skin enhanced by the additional loss of subcutaneous fat resulting in an increasing tendency for the skin to fold and wrinkle. The degree to which our culture equates personal appearance with youthfulness is reflects in the refinement and frequency of cosmetic plastic surgery used to remove wrinkles and skin blemishes associated with aging. Changes in pigmentation of the skin lead to an increasingly pale appearance to the tissue, especially among while individuals.

The hair continues to thin and lose its natural pigmentation in middle age. Hairlines of both men and women recede even further during these years. Graying and thinning of the hair occurs over the entire body including the armpits and public areas of both men and women.

The nails are part of the same system as skin issue and also show signs of age. Growth of the nails begins to decline in rate during middle age. Nails become thicker and show color changes with increasing age. The prevalence of significant skin conditions and diseases increases through the years of middle age. While fungus disease continue to occur more frequently than others in middle age. The incidence of skin tumors increases dramatically during this period. These tumors may be benign, precancerous, or cancerous and are indicative of the aging process as well as environmental conditions such as continued exposure to sunlight and other factors.

### **Changes in Eyes and Ears**

Several changes in vision and in eye functioning can be expected in middle age. Common symptoms of against in the eyes include a gradual loss of accommodation (focusing ability). Depigmentation of the iris, loss of retinal reflexes, and changes in the lens of cornea (Thimiras, 1972). One of the more noticeable changed relates to loss of accommodation ability resulting in the necessity for regarding glasses or bifocal lenses for the first time by many middle age individuals. There is an increase in the appear-e of characters and glaucoma at this stage of the life cycle.

Cataracts are a change in the lens of the eye that renders it opaque to light. The condition is irreversible but can be corrected by surgery and special glasses. Glaucoma actually is a group of diseases characterized by increases in pressure from within the eye ball. This condition results in damage to the optic nerve, causing blindness. The disease can be controlled by medication in middle age, individuals should have regular eye examinations upon reaching this age status.

Hearing loss may begin to be highlighted in middle age with individuals noticing a decline in the ability to here high frequency sounds. This tendency may be more a product of living in a noisy environment for many years than a natural part of the aging process. The threshold at which sounds as well as speech are easily detected and understood also declines will age, appearing most prominently in middle age and increasing as individuals grow older (Fowland, 1980; Timars, 1972).

### Changes in Teeth

The primary change affecting the teeth in middle age is a condition called osteoporosis. This condition is a leading cause of tooth loss in middle and late adulthood. Osteoporosis is an inflammation of the bone tissue surrounding the teeth, causing the bone to soften and become more porous. As a result, teeth become loosened in the socket and eventually are lost. The condition occurs throughout the skeletal system and is not restricted to the mouth area. The cause of osteoporosis is not known but may be related, to low calcium in take in the diet (Kart, Metress, & Metress, 1978).

### Changes in Skeletal System

Individuals can expect to experience the first indications of arthritis and rheumatism in middle age. While both these conditions can occur at any age and from a variety of causes, they are usually associated with the aging process. Arthritis is an inflammation of a joint, accompanied by pain and changes in the structure of the joint. Rheumatism is the term given to a variety of pain in joints. Both these conditions become more chronic in nature as individuals experience middle and late adulthood. Osteoporosis (loss of bone mass), also characteristic of aging, is accompanied by degeneration of joint tissue limiting movement as aging occurs. More women than men experience this condition and show symptoms at an earlier age than men. The process begins at about age forty for women and at about age fifty five for men (Decker, 1980).

### The Circulatory System

The circulatory system shows signs of aging in the years of middle age. Three basic changes may be observed: 1. A decline in the elasticity of the arteries, 2. An increasing accumulation of fatty deposits in the arterial tissues, and 3., a general decline in the pumping ability of the heart muscle. Two conditions appear with increasing frequency in middle age that are major abnormalities of the heart and circulatory systems. These diseases of the heart are the basic of the major cause of death among individuals in middle and late adulthood. Arteriosclerosis is commonly called "hardening of the arteries," referring to the gradual loss of the elasticity of artery walls throughout the body. The condition is considered to be a normal part of aging. The result of arteriosclerosis is a lowering of the amount of blood that reaches body parts and organs. Arteriosclerosis refers to the process in which artery walls become congested and narrowed owing to the depositing of fats, cholesterol, and calcium salts. Scientists continue to debate conditions such as diets that are high in foods containing these substances (Timars, 1972).

Both of these conditions lead to a reduced blood flow throughout the body, raising the blood pressure. Blood pressure increases throughout life, reflecting these changes in arterial functioning as age increases. Blood pressure levels are greater among men than women until about age fifty five and then decrease slowly thereafter. The levels of black individuals exceed that of white individuals (Roberts and Maurer, 1976). The result of these lowered levels of blood reaching vital body organs are twofold, 1. blood pressure is increased throughout the body, causing a condition known as hypertension (high blood pressure); and 2. The aging process is quickened in the heart muscle since it must work harder to pump blood through smaller spaces in the blood vessels.

Hypertension and coronary heart disease account for the two leading causes of visits to physicians among middle age individuals. Females are affected more by hypertension than males while more males are affected by coronary heart disease than females (Cypress, 1979).

### The Respiratory System

Most changes in the respiratory system functioning are seen in late adulthood. However, the lungs begin to show signs of loss of their elasticity in middle age. A related effect is the increase in respiratory diseases in middle age.

### The Digestive System

The digestive system may show fewer signs of aging than other organ systems in middle age. Disturbances that may begin to appear in middle age include: 1. Intestinal obstructions, 2. Adsorption problems, 3. Gallstones, and 4. Ulcers in the stomach and duodenum areas. The appearance of gallstones and ulcers may be attributed in part to environmental conditions. Gallstones may be a product of diets that are rich in

cholesterol, and ulcers are closely related to prolonged environmental stress and personality traits that result from stress (Kinnel. 1980).

Ulcers of the stomach and duodenum increase with age and are the leading digestive disorder of individuals in middle age. Males are more likely than females to experience this condition.

The incidence to diabetes also increases dramatically during the middle age years. The tendency to develop this condition is genetic in nature but is closely related to being overweight as age increases. The condition involves the inability of the body to inetabolize sugars and uses them in the cells. Essentially, it involves the failure of the pancreas cells to secrete an adequate amount of an enzyme (insulin) responsible for digesting sugars.

### **The Reproductive System**

The reproductive system and sexual functioning of both men and women experience changes at midlife. While these changes are largely hormonal in nature, the effects influence organ functioning and sexual behaviour. The changes are much more evident in women than in men. At about age forty, many women begin to experiences irregular menstrual cycles. Menopause, or the ending of ovulation and menstruation, occurs between forty-five and fifty-five for most women. The process is irregular and does not have a disruptive effect for about 75 percent of women studied (Neugarten, Wood, Kraine,s and Loomis, 1968), Menopause occurs in a sequences of events that include reduced fertility, irregular and absent menstruation and atrophy or sexual organs. Changes in sexual organs accompany the decreased hormone levels during the menopause. The uterus decreases in weight by as much as 53 percent between thirty and fifty. The walls of the vaginal tract become thinner and lubrication diminishes during sexual excitement (masters and Johnson. 1966). Other symptoms that accompany the menopause include. 1. Disturbances of the autonomic nervous system such as hot flashes, chills, sweating, hypertension and rapid heart rate: 2. Neurological and psychological disturbances such as dizziness, headaches, nervousness, and periods of depression; and 3. Somatic disturbance such as osteoporosis (softening of bones), according to (Thimas and Meisami. 1972).

The male climacteric does not resemble that of the female. There is no equivalent process of menopause in men as in women. Sperm production begins in puberty and continues until death occurs. Likewise, there is not abrupt decrease in production of male sex hormones in men that is similar to the decline in estrogen production in women. Nevertheless, there are gradual changes in the reproductive systems of men during middle age. Most of the changes in the male reproductive systems are degenerative in nature. The prostate gland experiences notable changes in middle age are becoming enlarged and increasingly course. Because the prostate assists in forcing semen from the penis at ejaculation, middle-age charges of this gland reduce the strength of the ejaculatory response as age increases. Cancer of the prostate is at the highest level during this age period (Koch. 1980). Many men being to notice that erection as well as ejaculation occurs more slowly (Mates and Johnson, 1966). Some men report physical symptoms associated with these changes that include impotence, frequent urination, irritability, periods of depression, and so on. Undoubtedly, many of these symptoms and others are related to psychological stress and pressures associated with the middle crisis.

### **Sensory and Motor Capacity in Middle Age**

Although changes in sensory and motor capabilities during mid-life are real and affect people's concept of themselves and their interaction with others, these changes are usually fairly small and most middle aged people compensate well for them.

In the area of vision, people experience presbyopia i.e. for sightedness associated with ageing. Middle age people also experience a slight loss in the sharpness of vision.

There is also a gradual hearing loss during middle age especially with regard to more high-pitched sounds; this condition is known as presbycusis. About age 55, hearing loss is grate for men than for women (Torll, 1985). However, most hearing loss during these is not even noticed, since it is limited to levels of sound that are unimportant to behavior. Taste sensitivity begin go decline at about age 50, particularly the ability to discriminate "finer nuance of taste" (Trol, 1985, p. 32). Since the taste buds become less sensitive, foods that may be quite flavorful to a younger person may seem bland to a middle-aged person (Troll, 1985).

Sensitivity to smell holds up well; it is one of the last senses to decline (Troll, 1985).

### Health Problems in Middle Ages

The average individual during the middle age period is generally healthy. The most common chronic ailments of middle age are asthma, bronchitis, diabetes, nervous and mental disorders, arthritis and rheumatism impaired sight and hearing and malfunction of the circulatory, digestive and genito urinary system. These ills do not necessarily appear in middle age, however, and while three fifths of 45 to 64 years olds have one or more of them, so do two fifths of people between ages 15 to 44.

(i) One major health problem of mid life is hypertension (high blood pressure). This disorder, which often predisposes people to heart attack or stroke, affects 1 out of 5 adults. It is particularly prevalent among black people and poor people.

(ii) Another health problem in this age group is AIDS, which now occurs more often in people over age 50 than in children recorded age 13. People over 50 now account for 10 percent of recorded cases. Although most people occur in homosexual or bisexual men who contracted AIDS through sexual intercourse about 17 percent of patients in this age group routine screening began in 1985. The disease seems to be more severe and to progress more rapidly in older people (Brozan.1990).

Health problems in mid life are especially serve among Hispanic Americans because of poverty, low levels of education and cultural and languages barriers.

Psychologists till recently believed that with advancing age mental functions also decline. However recent research studies do not support this position.

Cross sectional studies are criticized on methodology ground for studying cognitive development or changes in elderly individual. Longitudinal studies are generally preferable to study intelligence in elderly people longitudinal studies have been conducted. In these studies the same individual are tested and retested a different points in their life span. Using this approach, little or no decline has been found in middle age. In fact two major studies showed that middle aged adult performed better than they had as young adults (Bayley & Oden, 1955; Nisbet 1957).

Terms and Oden's study of a group of men and women followed from preschool years to middle life has shown that mental decline does not set in during middle age among those with high intellectual abilities. A follow-up study made fifty years after the original study likewise showed little intellectual abilities decline among the members of the group. In specific mental abilities, such as problem solving and verbal ability, little or no decline was reported in middle age among those whose initial abilities were high.

A study reported by Kangas and Bratfway has indicated that intelligence may even increase slightly In middle age, especially among those of higher intellectual levels. While this study was made on only small group forty eight subjects they were tested over a span of years at the preschool level, during junior high school and young adulthood finally when they were between thirty nine and forty four years of age. No follow up into the latter part of middle age has been reported to date.

Like members of the Terman and Oden group, those with higher IQs men showed a slight gain in IQ scores as they grew older, while women showed a slight decline. Because men must be mentally more alert in order to complete vocationally than women must be in order to carry out their roles as homemakers, these findings suggest that use of mental abilities is an important factor in determining whether there will be mental decline in middle age.

Related to aging are two type of intelligence called as fluid intelligence and crystallized intelligence.

**Crystallized intelligence** is culturally derived that is, it is a result of knowledge and of problem solving techniques learned initially in school and more generally through socialization, and it involves knowledge of one's language and of the skills and technology of one's culture. Examples of crystallized intelligence include such abilities as vocabulary general information, reasoning ability related to formal logic and mechanical knowledge such as the use of tools and the understanding of mechanical knowledge principles. Crystallized intelligence is associated with the use of principles common to the culture in which one lives.

**Fluid intelligence** on the other hand is displayed by solving such problems as completing a series such as 3, 7, 11, and 13 and so on. Questions testing fluid intelligence aim to be culture free tapping abilities that are more directly related to neurophysiologic intactness. Fluid abilities are characterized by the use of personal strategies rather than those learned at school to solve problems. For example in estimating the amount of cement needed for building a sidewalk, a person relying on fluid intelligence would use a personally derived system for making the estimate the use of algebra might be used by a person using crystallized abilities.

Fluid intelligence is more affected than crystallized intelligence by hereditary factors as well as by injury to the central nervous system.

Research studies have shown that fluid intelligence peaks between the ages of twenty and thirty and thereafter declines. Crystallized intelligence on the other hand increases as one gets older (Horn & Donaldson 1980). Horn and Donaldson try to explain these differences by pointing to the learning process. If one is concentrating one's energies, the quality of learning is enhanced. These researchers point to the years from 20 to 30 as a period of great intensity in learning one's occupation as well as making sexual and marital adjustments. Thus one's fluid abilities are strained to the utmost in finding personal solutions to life's problems. At the same time one builds, one's crystallized intelligence on the retention of what was learned during these years.

### **Middle Age and Creativity**

Though intelligence declines in middle adulthood; these declines do not necessarily produce similar declines in creative abilities and contributions in middle age. In fact, the middle years may well be some of the most productive years of an individual's life. Dennis (1966) found that that decade of the forties was the most productive among creative. Individuals in the humanities area experienced increase in total creative productivity throughout the middle age. Lehnien (1962, 1966) however, reports that the number of superior works of creative individual's peak during their thirties and declines thereafter.

Some observers speculate that different kinds of creativity are associated with different age groups in adulthood. For example, Jacques (1964) speaks of hot-form-the-fire creativity of young adulthood and the "Sculpted creativity" of midlife. During the early years the creative work is intense and spontaneous the product seems to emerge as an effusion, readymade. Einstein is a prototype. The creativity of middle age on the other quality of the work effect an awareness of death and human destructiveness. Rather than effusion one finds a "working through". Shakespeare and Dickens are prototypes. It appears that spontaneous "hot-room-the-fire" creativity peaks in early adulthood but forms of creativity that require experience, revision and interpretation either remain unchanged or increase in middle age.

Mid-life marriage today is very different from what it used to be. When life experiences were shorter, with many women dying in childbirth, couples who remained together for 25, 30 or 40 years were rare. The most common pattern was for marriages to be broken by deaths and the survivors to remarry. Households were usually filled with children. People had children early and late, had many of them and expected them to live at home until they married. It was unusual for a middle-aged husband and wife to be alone together. Today, more marriages end in divorce, but couples who manage to stay together can often look forward to 20 or more years of married life after the last child has left home.

### **Marital Satisfaction in Mid Life**

What happens to the quality of a long-term marriage? Marital satisfaction seems to follow a U-shaped curve. From an early high point, it declines until late middle age and then rises again through the first part of late adulthood. The least happy time seems to be the period when most couples are heavily involved in child rearing and careers. Positive aspects of marriage (like co-operation, discussion and shared laughter) seem to follow the U-shaped pattern, while negative aspects (like sarcasm, anger and disagreement over important issues) decline from young adulthood through age 69 (Gilford, 1984; Gilford & Bengtson, 1979). This may be because many couples who are frequently in conflict divorce along the way. Marriages are often affected by stressful events in mid-life, but communication between partners can often mitigate such stress. The first part of the middle years, when many couples have teenage children making their way toward independence, tends to be stressful. The identity issues of mid-life appear to affect wives (but not husbands) feelings about their marriages; women become less satisfied with the marriage as child rearing makes fewer demands and their feeling of power and autonomy increase (Steinberg).

## SINGLE HOOD AND MIDDLE AGE

Approximately 5 percent of all middle aged adults in the United States are single people who have never been married. There is relatively little research on this group and almost none on the positive aspects of this adjustment. But ordinary observation produces many examples of single people who in the absence of immediate family obligations have achieved an especially intense career commitment. Obvious example includes the dedicated priest, school teachers or charity worker.

Naturally single middle aged people encounter special problems. They must accept the social identity of a single spinster or bachelor-which ever today carries some stigma. They must plan for a late adulthood that does not include support from or involvement with children-though of course they may be involved with other people's children. And they are less protected against illness or misfortune. For example, if they become mentally disturbed they are more likely to be hospitalized than are the married people. Who may be cared for by their families? The single middle-aged person, however, is not usually a loner. Most single people develop close relationships with friends' singles as well as married. And they often maintain close contact with their nuclear and extended family-particularly single women.

The motives for remaining single typically are some what different for males and females with a major exception noted for male and female homosexual most of whom choose never to marry. Never married women are better educated come from higher socio-economic groups and are more achievement oriented than single men. Moreover, it appears that the older single women have chosen not to marry rather than having never been asked (Haven 1973). In contrast, (Spreitzer and Rilley, 1974) report that successful achievement oriented men are the least likely to remain single.

### Widowhood and Middle Age

Just as children must adjust to the death of a parent, many individuals must adjust to the death of a spouse in middle age. Widowhood is applied to both men and women, although women more commonly survive men to become widows. This is a result of their longer life expectancy as well as a tendency to marry older men. Although more people become widowed at the age of sixty-five and over, the likelihood of remarriage decreases as people grow older (U.S. Bureau of the Census, 1980). Widowhood does not stymie the personal development of women. In our culture, there are many alternatives available to women at widowhood, according to (Lopata, 1973). Options include remarriage, retraining or reduction for jobs, reentry into the work force, participation in voluntary organizations, devoting time to mother and grandmother roles, and others. Lopata notes that the most pressing short-ranges needs of widows include:

1. Expressing grief and bereavement of family and friends
2. Meeting companionship needs especially if the women are alone for the first time in their lives
3. Being protected from the "good" intentions of people wanting to give advice that is often contradictory in nature
4. Providing as assistance in confidence building and competencies
5. Assistance in reengaging socially with others

Lopata found three patterns of adjustment to widowhood.

1. The self-initiating woman modified interaction pattern with family and friends, built a new life style that realistically remained flexible to future changes, and did not continually look back to the past.
2. In their traditional ethnic pattern, the women experienced little change in life style and continued interaction patterns in a pattern similar to that before the death of her spouse.
3. The socially isolated women found even greater difficulty in making new social contacts and in keeping old ones in widowhood because of her desire to remain alone after her spouse's death.

Men differ from women in adjusting to loss of a spouse. Relatively little is known about the details of men's problems for men because they are less likely than women to have a close, intimate confidence relationship with someone other than their spouse (Atchley, 1975). (Berardo, 1968) feels that men have a more difficult adjustment to such a loss than women, suggesting that men are poorly prepared to care for themselves (perform household tasks, for example). Courtship opportunities may be seen as inappropriate behaviour by many men, especially as they grow older. When widowhood coincides with retirement, the effects often

are found to be devastating for men. Because men to this age have a greater identity in work roles than women, the event of losing of spouse tends to destroy plans that have been made by men for their retirement years with the presence of a couple in mind instead of a single individual (Atchely, 1975). In middle adulthood, individuals have two types of occupational patterns.

### **Stable pattern**

#### **Changing career (or second careers)**

#### **Pattern I: Stable Careers**

People who follow the pattern of a stable career are reaping personal benefits and also letting society benefit from their years of experience in a chosen field. Most of them continue to enjoy the work they have settled into. And because of their accumulated experience and wisdom, many reach positions of power and responsibility. Most business, academic and political leaders and many other prominent people in our society tend to be in their middle years.

Middle-aged men with stable careers tend to fall into two major categories: workaholics and the mellowed (Tamil, 1989). "Workaholics" may work at a frenzied pace because of a last ditch effort to reach financial security before they retire or because they find it hard to relinquish any of their authority.

The "mellowed" have come to terms with their level of achievement, even if they did not go as far in their careers as they had hoped. The best adjusted among the "mellowed" have a sense of relaxation rather than failure. They are often happier, less cynical and steadier in temperament than their more successful counterparts (Bray & Howard, 1983). Although these middle-aged men want to do challenging work, they do not pin their emotional well-being on their jobs as much as they used to.

#### **Pattern 2: Changing Careers**

Stories of mid-life career changes abound these days, as people seek new careers for a variety of reasons. With longer life expectancies, many middle-aged people realize that they do not want to keep doing the same thing for the next 20 years and therefore strike out in totally new directions. Some are forced by unemployment to seek second careers. Some would rather change jobs that deal with competition from younger people moving up the career ladder. Some think "I'm in a rut" or "I've gone 1 can go with this company" and seek the challenges of a job that offers more. Some become acutely aware of the time remaining to accomplish their goals. For these individuals who are within reach of achieving their goals, this period of occupational life is experienced as exciting, challenging and rewarding. For others, however, the large gulf between aspirations and achievement may become a major factor in triggering a mid-life crisis.

In mid-life most people are seeking some assurance of job stability seniority for example. They are less likely than younger workers to leave their company. This may reflect the fact that older people generally take fewer risks than young ones. But it also represents positive satisfaction with the work itself. The highest percentage of satisfied workers can be found in the over forty categories.

The reported satisfaction of most middle-aged workers does not mean that they are complacent. Mid-life is time when many workers are likely to reevaluate their career goals. One writer finds that an important task of this period is to set one's own milestones in the absence of school or institutional guidelines (Mills, 1970). Another note that at forty a person may fix on some key promotion or accomplishment that will magically affirm his work to society, example he may set deadline for becoming a full professor or for writing a best seller (Levinson et al., 1977). Typically the middle aged worker begins to assert control over the path of advancement by seeking or turning down promotions or transfers and accepting the limitations of the self with respect to work (Mills, 1970). All of these are factors in the extent to which individuals are satisfied or dissatisfied with their work.

#### **Self Actualization and Middle Adulthood**

Self actualization is one of the growth needs and the highest need in the hierarchy of needs proposed by Abraham Maslow.

Self actualization refers to an individuals' need to develop his or her potentialities in other words to do what he or she is capable of doing. "Self actualizers" then are the people who make the fullest use of their capabilities.

Self actualization is not attained until the middle years of adulthood. In the years prior to middle age, energy is frequently dissipated in diverse directions, including sexual relationships, educational advancement, career, alignment, marriage and parenthood. The need to achieve financial stability during these young adult years consumes considerable psychic energy. By middle age, though, many people have managed to fulfill most of these needs and can spare the energy to strive toward ego maturity.

In order to study the self-actualizing personality, Maslow selected 48 individuals who appeared to be making full use of their talents and were at the height of humanness. His subjects were students and personal acquaintances, as well as historical figures. In his analysis, he described 12 "probable", 10 "partial" and 26 potential or possible self-actualizers. His analysis of these individuals' reveals 15 traits that they felt were characteristics of the self actualizing personality. These traits are as follows:-

**More efficient perception of reality:** Many self-actualizing persons are able to perceive people and events realistically. That is to say, their own wishes, feelings, or desires do not distort reality. They are objective in their analysis of the environment and are able to detect that which is dishonest or false.

**Acceptance of self and others:** People with self-actualizing personalities lack such negative characteristics as guilt, shame, doubt and anxiety characteristics that sometimes interfere with the perception of reality. Individuals with healthy personalities are capable of accepting themselves for what they are and know their strength and weaknesses without being guilty or defensive.

**Spontaneity:** Self-actualizing people are relatively spontaneous in their overt behaviour as well as in their inner thoughts and impulses. Although many conform to societal standards there are those who are unconcerned about the role society expects them to play. Maslow discovered that some self-actualizing people develop their own values and do not accept everything just because others do. While others may accept the status quo, self-actualizers perceive each person, event, or object as it really is and weight it accordingly.

**Problem centering:** Unlike the ego centered personality, who spends much time in such activities as introspection or self evaluation, problem-centered individuals direct their energies towards tasks or problems. Problem centered persons are also likely to consider their goals important.

**Detachment:** Maslow discovered that his subjects need more solitude than the average person. The average person frequently needs to be with others and soon seeks the presence of other people when left alone (this reflects the need for belongingness and esteem derived from others). Self-actualizers, on the other hand enjoy privacy and do not mind being alone.

**Autonomy:** As can be inferred from nearly all the other characteristics of the self actualized personality, such people have a certain independence of spirit. Individuals are propelled by growth motivation more than by deficiency motivation and are self contained personalities.

**Continued freshness of appreciation:** Self actualizing people have the capacity to continually appreciate all of nature and life. There is a pleasure, even an ecstasy about experience that have become states to others. From some of the subjects' studies, these feelings are inspired by nature for others the stimulus may be music for still others, it may be children. But regardless of the sources, these occasional ecstatic feelings are very much a part of the self actualizing personality.

**The mystic experience:** Self-actualizers are not religious in the sense of attendance at formal worship, but they do have periodic peaks of experience that Maslow describes as "limitless horizons opening up to the vision, the feeling of being simultaneously more powerful and also more helpless than one ever was before, the feeling of great ecstasy and wonder and awe, the loss of placing in time and space with, finally, the conviction that something extremely important and valuable had happened, so that the subject is to some extent transformed and strengthened even in this daily life by such experiences" (Maslow, 1970, p. 164).

**Gemeinschaftsgefühl:** This German word, first coined by Alfred Adler, is used by Maslow to describe the feelings towards mankind that self actualizing persons experience. This emotion, which might be loosely described as "the love of an older brother", is an expression of affection, sympathy and identification.

**Unique interpersonal relations:** Self-actualizers have fewer "friends" than others but they have profound relationships with those friends they do have. Outside of these friendships, they tend to be kind to and patient with all whom they meet. An exception is the harsh way they sometimes speak to hypocritical,

pretentious, or pompous people. However, for the most part, what little hostility they exhibit is based not on character but on situation.

**Democratic character structure:** Maslow found that without exception, the self-actualizing people he studied were democratic, being tolerant of others with suitable character regardless of their social class, race, education, religion or political belief.

**Discrimination between means and end:** Unlike the average person, who may make decisions on expedient grounds; self actualizing people have a highly developed ethical sense. Even though they cannot always verbalize their moral positions, their actions frequently take the "higher road". Self actualizes distinguish means from end and will not pursue even a highly desirable end by means that are not morally correct.

**Philosophical unhostile sense of humor:** The humor of self actualized person is of a higher order. They do not consider funny what the average man considers to be funny. Thus they do not laugh at hostile humor (making people laugh by hurting someone) or authority rebellion humor (the unfunny, oedipal of smutty joke). Characteristically what they consider humor is more closely allied to philosophy than to anything else. It may also be called the humor of the real because it consists in large part in poking fun at human beings in general when they are foolish, or forget their place in the universe to try to be big when they are actually small. This can take the form of poking fun at them but this is not done in any masochistic or Lincoln never made a joke that hurt anybody else. It is also likely that many or even most of his jokes has something to say, had a function beyond just producing a laugh. They often seemed to be education in a more palatable form akin to parables or fables.

**Creativeness:** Without exception, every self-actualizing person that Maslow studied was creative in some way. This creativity is not to be equaled with the genius of a Mozart or an Einstein. Since the dynamics of that type of creativity are still not understood. Rather it is what Maslow says "the naïve and universal creativeness of unspoiled children." He believes that creativity in this sense is possibly a fundamental characteristics that we all are born with but lose as we become uncultured characteristic that we all are born with but lose as we become uncultured. It is linked to being spontaneous and less inhibited than others and it expenses itself in everyday activities. Described quite simply, it is a freshness of thought, ideas and actions.

**Resistance to enculturation:** Self actualizes accept their culture in most ways, but they still, in a profound sense, resist becoming uncultured. Many desire social change but are not rebellious in the adolescent sense. Rather, they are generally independent of their culture and manage to exhibit tolerant acceptance of the behaviour expected to their society. This, however, must not be constructed as a lack of interest in making changes they believe in. If they feel an important change is possible.

## LATE ADULTHOOD

### Physical Changes in Late Adulthood

Fatty concentrations (cholesterol) in the heart and arteries also reduced blood flow throughout the body; degeneration of the blood vessels leads to increased blood pressure (Kohn, 1977). Yet these changes, while most evident in later life, do not originate during this period. In fact, losses in cardiac output (the amount of blood the heart can pump through the body) are evident from early adulthood Brandfonbrenner, Landowne and Shock (1955), for example, found a linear drop-off in cardiac output at the rate of one percent per year in males aged nineteen to eighty-six. Other organ systems also show reduced efficiency in late adulthood. Vital lung capacity decreases with age. Older people frequently report shortness of breath, particularly after mild exercise such as climbing stairs raking the yard (Klocke, 1977). Changes in the gastrointestinal system. Such as deterioration of the mucosa lining in the intestinal tract, and reduction of gastric juices, contribute to the frequent intestinal complaints of the elderly. Normal immune functions also decline with age and in the elderly may be related to an increase incidence of cancer (Mainodan, 1977; Teller, 1972).

Nearly all the sensory system shows loss of efficiency in old age. Probably the most usual sensory loss associated with aging is hearing. According to Corso (1977), 17 percent of people who are sixty-five or over show signs of advanced presbycusis hearing loss due to degenerative changes in the auditory system.

Besides the above changes some other physical changes that occur in late adulthood are as follows:-

**Visual Problems:** Some visual problems encountered in late adulthood are as follows:

(a) After age 65, serious visual problems that affect daily life are all too common. Many older adults have trouble perceiving depth or color.

(b) About 17 percent develop cataracts, cloudy or opaque areas in the lens of the eye that prevent light from passing through and thus cause blurred vision.

(c) Many elderly people also suffer from retinal disorders that result in blindness as a result of building up fluid pressure in the eye, which often damages the eye internally. Some other defects in vision are concerning movement perception, visual search (locating signs near vision, light sensitivity etc.)

**Hearing Problems:** Hearing loss is very common in late life; about 3 out of 10 people between ages 65 and 74 and about half of those between 75 and 79 have it to some degree.

**Problems Concerning Taste and Smell:** Elderly people have problems with regard to taste and smell. This can lead to nutritional problems in elderly people. Doty (1984) found that more than 4 out of 5 people over 80 years old have major impairments in smell and more than half of them have almost no sense of smell at all.

**Decline in Coordination and Reaction Time:** Information processing is slowed down. Their motor coordination declines increasing the risk of accidents in work. This leads to a decline in their physical activities.

**Decline in General Functioning:** There is a decline in general physiological functioning in late adulthood. Some of these changes, a few of which we have discussed earlier, are as follows:

The chemical composition of the bones changes, causing a greater chance of fractures. People may shrink in size as the disks between their spinal vertebrae atrophy, and they may look even shorter because of stooped posture. Osteoporosis, a thinning of the bone that affects some women after menopause, may cause a "widow's hump" at the back of the neck.

All the body systems and organs are more susceptible to disease but the most serious change affects the heart. After age 55, its rhythm becomes slower and more irregular; deposits of fat accumulate around it and interfere with its functioning and blood pressure rises.

Insomnia, i.e. lack of sleep is also a common problem of elderly people.

In both men and women, the hair becomes thinner, what is left turns white, and it sprouts in new places, on a woman's chin and out of a man's ears.

The skin becomes paler and splotchier; taking on a parchment-like texture skin tends to hang in folds and wrinkles.

The human body has the equivalent of money in the bank for a rainy day. Normally, people do not use their organs and body systems to the limit; but extra capacity is available for extraordinary circumstances. This backup capacity, which lets body systems function in times of stress, is called reserve capacity (or organ reserve) it allows each organ to put forth 4 to 10 times as much effort as usual. Reserve capacity helps to preserve homeostasis, the maintenance of vital functions within the optimum range (Fries & Crapo, 1981).

With age, however, reserve levels drop. Although the decline is not usually noticeable in everyday life, older people cannot respond to the physical demands of stressful situations as quickly or efficiently as they used to.

## LATE ADULTHOOD

### Health Problems in Late Adulthood

Late adulthood is a period where health generally declines along with deteriorating physical conditions. However, in recent decades due to advances in medical sciences the general health of elderly individuals has increased to a considerable extent.

Health in late adulthood is influenced by many factors some of these include use of antibiotic, educational level, socio-cultural factors, dietary restrictions and physical exercises.

Following are the important points to be noted with regard to the health problems in late adulthood.

Overall, older people need more medical care than younger ones. They go to the doctor more often, are hospitalized more frequently stay in the hospital longer, and spend more than 4 times as much money on health care.

Although most elderly people are in good health, chronic medical conditions do become more frequent with age and may cause disability. Older people have at least one chronic condition: the most common are arthritis (48 percent); hypertension (37 percent); heart disease (30 percent); cataract (16 percent); hearing impairments (30 percent); and impairments of the leg, hips, back, or spine (17 percent). But people over 65 have fewer colds, flu infections, and acute digestive problems than younger adults. The danger with older people is that a minor illness-along with chronic conditions and loss of reserve capacity - may have serious repercussions.

Susceptibility to illness is one of the most serious problems confronting older persons. This is generally the case in chronic conditions. Most people over 65 years suffer from one or more chronic conditions.

The most common chronic conditions restricting activity in late adulthood especially in individuals over 65 years are heart diseases, arthritis, hypertension, visual impairment and orthopedic problems.

Another common problem of late adulthood is the dental problems, especially tooth decay. This affects digestion and leads to many other problems.

Many elderly people suffer from a wide variety of functional disorders. The most common functional disorders in the elderly include depression, paranoid reactions, hypochondriacs, and chronic anxiety (Butler & Lewis, 1977; Pfeffer, 1977). Of these, depressive reactions are the most frequent and are characterized by extreme sadness, social withdrawal, inhibition, lowered self-esteem, pessimism, indecision, and occasionally, a slowing down of mental processes as well as physical movement. The suicide rate for the elderly, which is linked to depression, is higher than for any other age group especially for white males.

It is estimated that some 1 to 2 percent of elderly people living in the community suffer from major depression, another 2 percent have less severe depression, and more than 10 percent have some important symptoms (Blazer, 1989).

Organic brain syndrome is another major problem faced by individuals in late adulthood. The two most common organic brain syndromes associated with aging are cerebral arteriosclerosis and senile dementia. Cerebral arteriosclerosis is related to increased arterial cholesterol levels - as the arteries "harden" blood flow to the brain is reduced and localized brain death occurs. Initially, mood or affect changes are noted, as well as increased irritability, fatigue, and headaches. As the condition progresses, cognitive processes are affected especially memory, abstraction ability, and assimilation of new information. The onset of this disease, which more often affects men than women may occur as early as the mid-fifties.

Senile dementia, on the other hand, begins much later in life, usually in the mid-seventies, and is more often found in females than males -probably because women live longer than men. This condition is associated with diffuse or general brain loss of unknown origin. Over the course of the disease brain weight can reduce as much as 15 to 30 percent. Typical symptoms include errors in intellectual and social judgment, mood changes, memory impairment spatial and temporal disorientation, general confusion, loosening of inhibitions, and deterioration of personal habits.

Both cerebral arteriosclerosis and senile dementia are chronic conditions; full recovery from them is not possible, although improvement can be obtained at times with proper medical treatment.

Many individuals in late adulthood also suffer from irreversible mental problems. The most common irreversible mental problem is Alzheimer's disease. It is a degenerative brain disorder that gradually robs people of intelligence, awareness, and even the ability to control their bodily functions and finally kills them.

This malady, the most prevalent and most feared irreversible dementia, occasionally strikes in middle age, but most of its victims are over 65. Estimates of its prevalence vary from 6 to 10 percent of all people and over 65 and from 20 to 50 percent of all people over 85.

In late adulthood there is a slowing down of the control nervous system. This leads to memory losses and difficulties in memory retrieval and learning. Scores on intelligence test also declines.

A wide variety of mental changes occur in late adulthood which is generally detrimental. Some of these mental changes are as follows:

**Learning is slow in them:** Inductive and deductive reasoning slows down. Older people tend to lack the capacity for, or interest in, creative thinking. Thus significant creative achievement is less common among older people than among younger ones. Old people tend to have poor recent memories but better remote memories. Mental rigidity is another common problem in late adulthood.

Psychomotor ability is also affected in late adulthood. Some of the psychomotor abilities that are affected are as follows:

- (i) While all motor abilities decline to some extent, some decline earlier and more rapidly than others.
- (ii) Decline in strength is most pronounced in the flexor muscles of the forearms and in the muscles which raise the body. Elderly people tire quickly and require a longer time to recover from fatigue than younger people.
- (iii) Decrease in speed with aging is shown in tests of reaction time and skilled movements, such as handwriting. It is especially marked after age sixty.
- (iv) In late adulthood, people tend to become awkward and clumsy, which causes them to spill and drop things, to trip and fall, and to do things in a careless, untidy manner. The breakdown in motor skills proceeds in inverse order to that in which the skills were learned, with the earlier learned skills being related longest.

### Personality Development in Late Adulthood

Personality development continues even in late adulthood. Late adulthood is the development stage during which people clarify and find use for what they have learned over the years. People can continue to grow and adapt if they are flexible and realistic - if they learn how to conserve their strength, adjust to change and loss, and use these years productively. People now have a new awareness of time; and they want to use the time they have left to leave a legacy to their children or to the world, pass on the fruits of their experiences, and validate their lives as having been meaningful.

Erik Erickson, Robert Peck and George Vaillant have carried out research and have presented their theories and personality development in late adulthood. We would discuss each of them in brief.

**Erik Erickson:** According to Erickson, the state of integrity versus despair emerges during the adulthood.

Erickson sees older people as confronting a need to accept their lives-how they have lived - in order to accept their approaching death: They struggle to achieve a sense of integrity; of the coherence and wholeness of life, rather than give way to despair over inability to relieve their lives differently (Erickson & Kivnick, 1986). People who succeed in this final, integrative task- building on the outcomes of the seven previous crises - gain a sense of the order and meaning of their lives within the larger social order, past, present, and future. The "virtue" that develops during this stage is "wisdom" as "informed and detached concern with life itself in the face of death itself (Erikson. 1985. p. 61).

Wisdom, Erickson says, include accepting the life one has lived, without major regrets over what could have been done or what one should have done differently. It involves accepting one's parents as people who did the best they could and thus deserve love, even though they were not perfect. It implies accepting one's death as the inevitable end of a life lived as well as one knew how to live it. In sum, it means accepting imperfection in the self, in parents, and life. People who do not achieve acceptance are overwhelmed by despair, realizing that time is too short to seek other roads to integrity. While integrity must outweigh despair if this crisis is to be resolved successfully, Erickson believes that some despair is inevitable. People need to mourn not only for their own misfortune and lost chance but also for the vulnerability and transience of the human condition.

Yet, Erickson also believes that late life is a time to play, to recapture a childlike quality essential for creativity.

**Robert Peck:** Three adjustments of Late Adulthood. Peck (1955) expanded on Erickson's discussion on psychological development in late life, emphasizing three major adjustments that people must make. These adjustments allow them to move beyond concerns with work, physical well-being and mere existence to a broad understanding of the self and of life's purpose. Peck's three adjustments are:

(a) The issue in this adjustment is the degree to which people define themselves by their work. Everyone has to ask "Am I a worthwhile person only in so far as I can do a full time job; or can I be worthwhile in other, different ways - as a performer of several other roles, and also because of the kind of person I am?" (Peck, 1995, in Nugarten, 1968, p. 90). Retirees especially need to redefine their worth as human beings. People need to explore themselves and find other interest to take the place of the work (whether centered in the marketplace or at home) that had given direction and structure to life. People are more likely to remain vital if they can be proud of personal attributes beyond their work. They need to recognize that their ego is richer and more diverse than the sum of their tasks at work.

(b) Transcendence of the body versus pre-occupation with the body - Physical decline creates the need for a second adjustment: overcoming concerns with bodily conditions and finding other sources of satisfaction. People who have emphasized physical well-being as the basis of a happy life may be plunged into despair by diminishing faculties or aches and pains. Those who focus on relationships and on activities that do not demand perfect health adjust better. An orientation away from pre-occupation with the body should be developed by early adulthood, but it is in late life that this orientation is critically tested. Throughout life people need to cultivate mental and social powers that can grow with age, along with attributes like strength and muscular co-ordination that are likely to diminish over the years.

(c) Transcendence of the ego versus pre-occupation with the ego - Probably the hardest, and possibly the most crucial adjustment for older people is to go beyond concern with themselves and their present lives and to accept the certainty of death.

How can people feel positive about their own death? They can recognize that they will achieve lasting significance through what they have done so far - through the children they have raised, the contributions they have made to the society and the personal relationships they have forged. They transcend the ego by contributing to the well-being of others and this, Peck says, human beings apart from animals.

**George Vaillant:** Factors in emotional health - The Grant study, a longitudinal study that began with college sophomores, examined the physical and mental health of 173 of these men at age 65 (Vaillant & Vaillant, 1990). Emotional health at this age was defined as the "clear ability to play and to work and to love" (p. 310) and as having been happy over the previous decade.

It is surprising to see the very limited role that various factors play in emotional health. A happy marriage, a successful career, and a childhood free of such major problems as poverty or the death or divorce of parents were all unimportant in predicting good adjustment later in life. More influential was closeness to siblings at college age, suggesting a close family. Factors associated with poor adjustment at age 65 included major emotional problems in childhood and, before age 50, poor physical health, severe depression, alcoholism, and heavy use of tranquilizer.

Probably the most significant personality trait was the ability to handle life problems without blame, bitterness or passivity - or, in the researcher's terms, to use "mature defense mechanisms". The subjects, who over the years, had not collected injustices, complained, pretended nothing was wrong, or become bitter or prejudiced - and could thus respond appropriately to crises - were the best adjusted at age 65. The best adjusted 65 year olds had also been rated in college as well-organized, steady, stable and dependable; and they continued to show these traits (which were more important than being scholarly, analytic or creative) throughout life.

But some characteristics linked with good adjustment in young adulthood - like spontaneity and making friends easily - no longer mattered. Possibly the men who were egocentric and isolated early in life improved their social skills over the years, while the extroverted men did not develop other abilities that may, in the long run, be more valuable (Vaillant & Vaillant, 1990).

### Research on Change of Personality

Although basic personality traits (like extroversion, neuroticism, and openness to new experiences) are generally stable throughout life, values and outlook do seem to change in ways like those Erikson proposed. In studies by Carlo Ryff and her associates (1982: Ryff & Bakes, 1976; Ryff & Heincke, 1983), men and women of various ages reported that they were most concerned with intimacy in young adulthood, with generativity in middle adulthood, and with integrity in late adulthood. They felt that other aspects of their personalities - like impulsiveness, humility and orderliness had not changed.

Between the middle and late years, many women's focus shifted from "doing" to "being", from instrumental values (like ambition, courage, and capability) to terminal values (such desirable end states of existence as a sense of accomplishment, freedom, and playfulness). Men did not show this kind of shift: middle-aged men were already focused on terminal values, possible because in this cohort, men may have changed their values earlier in life.

For some older people, the tendency toward introspection from middle age onward result in their becoming more preoccupied with meeting their own needs. This may be a reaction to lifetimes of caring for and about people; it may also reflect the fact that personal needs are greater in old age.

### Family Life of Late Adulthood

Old age is a period of life where drastic changes occur in the family life and it is a period which requires a great deal of adjustment in the sphere of family life. The family life of late adulthood covers the following topics which we will discuss in brief.

- Married life in late adulthood
- Psychosexual adjustment
- Relationship with spouse, offspring, Grandchildren and siblings

Characteristics of marital life in late adulthood, as listed by Brubaker (1983).

- (i) Many people in late adulthood continue to remain married.
- (ii) Family is still the primary and most important source of emotional support.
- (iii) Family life in late adulthood is multigenerational. Most older people's families include at least three generations, may span four or five. The presence of so many people in family is a source of entertainment for many people.
- (iv) Late family life has a long history. Couples continue to remain married for more than 20 to 25 years. This gives their family life cohesion and attachment in spite of conflicts & problems.
- (v) Late adult family life is generally satisfying and many say that their marriage had got better over the years.

### Married Life in Late Adulthood

There are many stereotypes about the married life of late adulthood period. Many people wrongly believe that older husband and wife are unhappy, isolated, and lonely & rejected by loved ones. No doubt this is true in rare cases. In most cases, married life is satisfying and happy. In one important study the majority of the older couples described this time as the happiest period in their marriage.

For older, marriage is more successful. One reason, why older people report more satisfaction with marriage is that people of this age are more satisfied with life in general. Besides this by late adulthood each changed. An age-related decrease in ego energy, one's mental and emotional resources may also contribute to a more easygoing relationship in late marriage. So too does the tendency to "blow out" conflict and other stressful situations.

According to Adams (1975) the two factors that give the older marriage its unique character are the gradual shift in focus away from the children, and the retirement of the husband (and more recently the wife) from occupational life. Both events provide the couple with increased freedom from outside responsibilities and obligations. Consequently, husbands and wives often find that they have more time for each other during this period than at any other time in their marriage, a factor that seems to facilitate marital happiness. Finally, it should also be noted that marriage is not only quite satisfying during this age period, but it is also psychologically and biologically beneficial. Research indicates that older individuals who are married are less

likely to experience loneliness and depression that is the unmarried elderly (Tibbits, 1977). They also show less evidence of mental illness (Hobe, 1973) and they are likely to live longer (Civia, 1967).

Satisfaction with marriage among older people is interested if their children are successful and happily married and if they have good relationships with their grandchildren, even if their contacts with them are infrequent.

Studies of marital happiness in old age have revealed that older people feel that their marriage have been very satisfactory, that their lives are calmer now that their parental responsibilities are over, and that they have a new freedom to do as they please. Psychosexual adjustment. Late adulthood is a period which required sexual adjustment. Medical and physical conditions influence sexual behaviour and adjustment among spouses. Men & women generally show a decline in sexual behaviour with advancing age. However recent study by Brody (1978) has pointed out that there is a much longer interest in sex and a greater desire for sexual activity in old age than is popularly believed to exist.

In older age, as at other ages, sexual activity has a marked influence on marital adjustment, which in turn affects sexual activity. Diminishing sexual power can have a serious effect on marital adjustment during old age.

**Sexual activity continues in old age:** Sexual activity becomes more matured and mutually satisfying rather than a desire to seek personal gratification.

**Relationship with (a) spouse (b) offspring (c) Grandchildren and (d) siblings:** (Elderly individual's interpersonal relationship with these near and dear ones are greatly influenced by their family environment and personality make up.

Relationship with one's spouse is greatly influenced by the common interest's health and intellectual development of spouse. Many couples spend time together have cordial relationship with each other and take interest in each other's activities. It is this period of time where companionship for them is more important

Middle and upper-class adults, on the whole spend more of their leisure time with spouses and have more recreational interests in common with them than those of the lower class groups.

Relationship in the old age with offspring especially in a nuclear urban family is more formalized. Sussman (1965) have found that older people rely primarily upon their children in times of illness and they receive almost instant help. Older people are of great use of their children. They offer wide variety of help to the offspring's. Among the types of aid that older parents give their children are money, services such as babysitting or legal advice, and household services such as needlework and woodwork. One group of researchers concluded that "altogether, the proportion of old people who give help to their children tends to exceed the proportion who receives help from their children". (Riley, Riely & Johnson, 1968). The amount of mutual aid does not seem to depend on how close the generations live, or how often they visit. Some sex-linked patterns do emerge: sons tend to receive money from their elderly parents, whereas daughters receive services (Sussman, 1960: 1965).

Research studies have indicated that relationship of elderly people with their children is not free from conflict. Children of elderly people, who are themselves grown up and parents, resent the interference of elderly parents in their life. Researches studies have also found that for the most part, elderly women are absorbed in their relationship with their children than elderly men are. Because women have a closer relationship with their grown children than men have, there is usually more friction between women and their children than between men and their children.

Relationship with grandchildren is a source of enjoyment, involvement as well as conflict and tension. In old age, one's grandchildren are no longer toddlers or schoolchildren. They are adolescents, or older. A number of studies suggest that grand parenting in old age is not always as rewarding as it was earlier. As the grandchild becomes older, and perhaps less attentive, the grandparent experience a kind of "reality shock" that leads to disenchantment (Troll, 1971). One study of older grandparents found that subjects did not feel particularly close to grandchildren, but were 'glad to see them come and glad to see them go'. When grandparent and grandchildren live under the same roof, there is likely to be friction between them.

When conflicts develop about the grandchildren's behaviour, they more often involve the mother and the grandmother than the father and the grandfather. Grandfather, on the whole, have fewer and more remote contacts with grandchildren than grandmother do, and they are far less likely to be called on for help in an

emergency. As a result, grandmothers generally are more interested and absorbed in the lives of their grandchildren than grandfathers are. While the grandfather may be proud of the achievements of his grandchildren and feel that they reflect favorably on family, the reactions of grandmothers are usually more personal and more emotionally toned.

Elderly individuals also have siblings who are generally more or less of their age. Relationships with siblings play a very important role in the life of the aging adult particularly for those individuals who have lost a spouse, are divorced, or were never married (Shanas, 1979). Siblings often provide the support and help that normally would come from a spouse. They act as "confidants", share family occasions, holidays and recreational activities; aid in decision-making, home-making and home repairs; boost morale; lend money in time of financial need; and provide nursing care and emotional support in times of illness.

Research suggests that the influences of siblings on older adults differ depending upon the sex of the sibling and the sex of the individual (Cicirelli, 1977, 1979). Generally female siblings exert a greater influence on both aged men and women. They are more effective in preserving family relationships and providing emotional support. Furthermore, the presence of sisters tends to reduce the threat of aging for the older man; that is, older men seem happier and less affected by economic and social insecurities when they have living sisters. For aged women, the presence of sisters results in greater concern about social skills, social relationship outside of the family, and community activity. In other words, sisters stimulate each other and tend to facilitate a more stimulating and challenging environment for the older woman.

### Ageing

Ageing is a continuous process which continues throughout life. However, it becomes more pronounced during the adulthood. One of the most dramatic changes that occur due to ageing is the loss of physiological function and changes in physical appearance. Besides, physical changes, many psychological changes also, occur due to ageing. Ageing is a process that is not well understood. As a result, there is a variety of theories that explain the process of ageing in adulthood. Ageing is a series of complex and interrelated changes that occur over time. Timiras (1972) defines ageing as "a decline in physiologic competence that inevitably increases the incidence and intensifies the effects of accidents, disease, and other form of environmental stress, "Most theories of ageing focus on factors that are either hereditary or environmental in nature as the cause of ageing. Most likely, there is an interaction between genetic and environmental factors that result in the ageing process.

Kimmel (1980) lists a variety of hereditary and environmental factors that influence ageing. Hereditary functions to influence the life expectancy of all species. Humans have the longest life span of all the mammals while birds, reptiles, insects, and plants each have drastically different life span Kallman.

### Patterns of Ageing

Ageing is greatly influenced by one's personality factors. Successful ageing does not follow any single pattern. How people adapt in old age depends on their personalities and how they have adapted to situations throughout life. (Neugarten et al, 1968) have identified four major patterns of ageing on the basis of their study in which they interviewed 159 men and women aged 50 to 90. The patterns of ageing are as follows:

**Integrated** - Integrated people were functioning well, with a complex inner life, a competent ego, intact cognitive abilities, and high level of satisfaction. They ranged from being very active and involved with a wide variety of interests, to deriving satisfaction. They ranged from being very active and involved with a wide variety of interests, to deriving satisfaction from one or two roles to being self-contained and content.

**Armour-defended** - People were achievement oriented striving, and tightly controlled. Both those who stayed fairly and those who limited their expenditures of energy, socializing, and experience showed moderate to high levels of satisfaction.

**Passive dependent** - Passive dependent people either sought comfort from others or were apathetic. Some, who depended on others, were moderately or very active and moderately or very satisfied. Others who have been passive all their lives, did little and showed medium or low.

## LATE ADULTHOOD

Retirement is one of the transitional period of late adulthood and is a period in one's life, occurring generally during the late adulthood which required adaptation and changes in behaviour role and perception. Retirement always involve role changes. Changes in interest and values and changes in Whole pattern of individual's life.

The age at which one retires is variable. In certain countries or occupation it is 58 years and in others it is 60, 62 or 65. Many individuals even at 70 years may be healthy and efficient in their work output but are forced to retire due to social security reasons.

Psychologists have investigated in depth the question why people retire? Reasons for retiring vary from individual. Those who choose to retire do so because of adequate financial resources good pension plan desire to spend more time with family, or dislike for the job. Involuntary retirement however, usually results from the mandatory retirement policy of the company or from poor health.

Very few retire out of compulsion due to health problems.

Retirement for some is joyous occasion, especially people who have money. For other who need money, retirement can create more problems and lead to unhappiness, lack of self-esteem and economic hardships.

Not all workers retire, or retire completely. Self-employed people artists, professional scholars, independent craftsmen, and contractors - are not affected by mandatory retirement. But of those who do retire, an important life adjustment is required.

Retirement has little effect on physical health, but it sometimes affects mental health (Boose, Aldwin, Levenson & Ekred, 1987). Among 1513 older men surveyed in the Boston Veterans Administration Normative Ageing Study (a cross-sectional study) retirees were more likely than workers to report depression, obsessive-compulsive behaviour, and physical symptoms that had not organic cause. Those who had retired early (before age 62) or late (after age 65) reported the most symptoms.

### Factors Influencing Retirement

Amount of choice one can exercise considerably influence one's decision to retire. This varies with the occupation. People who are self-employed, such as novelist, carpenter, or physician, have the option of working well beyond normal retirement age. A school teacher can choose to retire after twenty or twenty-five years with different benefit schedules. Members of trade unions may be able to take advantages of a "flexible retirement" clause that allows them to work past normal retirement; or provides an early retirement option that allows full benefits after only thirty years. Other workers have no choice but to accept retirement at the specified moment. Satisfaction with respect to one's occupation is another factor that influences one's decision to retire. Unskilled workers, who generally show less job satisfaction, are likely to opt for early retirement within a corporation that provides this option. More satisfied workers chose retirement at sixty-five or, if possible, later.

### Economic Status

People who are financially well off prefer early retirement as compared to individuals who are in need of financial security. Sex also considerably influenced one's decision to retire. Women generally prefer retirement as compared to men. Women, on the whole adjust better to retirement than men. There are three reasons for this.

First, the role change is not as radical because, for the most part, women always played the domestic role, whether they were married roles.

Second, because work provides fewer psychological benefits and social supports for women, retirement is less traumatic for them than for men.

Third, because few women have held executive positions, they do not feel that they have suddenly lost all their power and prestige. On the basis of attitudes to retirement elderly people, according to Havighurst, can be divided into two general categories.

The first category he labeled the transformers - those who are able and willing to change their lifestyles by reducing their activities by choice and by creating for themselves new and enjoyable lifestyles. This they do

by dropping old roles and undertaking new ones. They seldom relax and do nothing but, instead, they develop hobbies, travel, and become active in community affairs.

The second category, the maintainers, Having Hurst has explained, holds onto work by pursuing part-time assignments after retirement and other activities to fill their time. They, like the transformers, seldom relax and do nothing, but what they do is a continuation of what they have done for years some form of work for which they are paid as they were throughout their working years.

### Stages of Retirement

To understand the adjustment required of elderly persons, it is helpful to develop an awareness of the various phase of retirement. Atchely (1976) views retirement not only as a process but also as a social role that unfolds through a series of six phases. Furthermore, he believes that various adjustments must be made by the retiree as these stages are encountered. Because the retirement period is an individual phenomenon that varies in duration, relating these six phases to chronological ages is impossible. Also, individuals may not experience all the phases or encounter them in the order proposed.

**Pre-retirement:** This period can be further divided into two sub-stages, remote and near. In the remote phase, retirement is perceived as an event that is a reasonable distance in the future. This phase can begin before a person takes his or her first job; it ends when retirement nears.

Anticipatory socialization and adjustment for retirement at this point are usually informal and unsystematic. The individual may also become exposed to negative stereotypes concerning retirement, (of course, an individual's positive or negative reaction depends in part on the prevalent view of retirement on the part of relatives, friends, and coworkers). The near phase emerges when workers become aware that they will take up the retirement role very soon and that adjustments are necessary for a successful transition. This phase may be initiated by a company's preretirement program or by the retirement of slightly older friends. Some workers may develop negative attitudes at this time because the realities of retirement are much clearer and financial prerequisites for the retirement role may not have been met. Many workers also fantasize about retirement and attempt to imagine what their lifestyles might entail after work stops. Preretirement programs appear to be successful in reducing anxious feeling about subject.

**The Honeymoon Phase:** This period, immediately following the actual retirement event, is frequently characterized by a sense of euphoria that is partly the result of one's newfound freedom. It is a busy period for many people, filled with such activities as fishing, sewing, visiting family members, and traveling, although these activities will be influenced by numerous factors including finances, life-style, health, and family situation. The honeymoon period may be short or long, depending on the resources available to people and their imaginativeness in using them and life begins to slow down, some retirees become disenchanted and feel let-down or even depressed. The depth of this emotional let-down is related to a variety of factors such as declining health, limited finances, or being unaccustomed to such an independent lifestyle. In some cases, eagerly anticipated post-retirement activities (i.e. extensive travelling) may have lost their original appeal. Unrealistic preretirement fantasies as well as inadequate anticipatory socialization for retirement may also promote disenchantment.

**The Reorientation Phase:** For those whose retirements either never got off the ground or landed with a loud crash, a reorientation phase of adjustment is necessary. At this time, one's experience as a retired person is used to develop a more realistic view of life alternatives. Reorientation may also involve exploring new avenues of involvement, sometimes with the assistance of groups in the community. Many seek to become actively involved in jobs especially designed for the retiree, either on a volunteer basis or for pay.

**The Stability Phase:** Stability, as defined by Atchely, does not refer to the absence of change but rather to the reutilization of criteria for dealing with change. People who reach this stage have established a well-developed set of criteria for making choices, which allow them to deal with life in a fairly comfortable and orderly fashion. They know what is expected of them and know what they have to work with, strengths as well as weakness. In the stability phase, the individual has mastered the retirement role.

**The Termination Phase:** Although death may end retirement in any phase, the role itself is most often cancelled out by the illness and disability that sometimes accompany old age. When people are no longer

capable of housework or self-care, they are transferred from the retirement role to the sick and disabled role. This role transfer is based on the loss of able-bodies status and autonomy, both of which are instrumental for carrying out the retirement's role. Retired status is also lost, of course, if a full-time job is taken.

### **Cognitive Changes in Late Adulthood**

There are two principle changes in cognitive functions in old age.

1. A decline in general intellectual functioning
2. Changes in memory as age increases in late adulthood

The decline in mental functioning with age increase is well documented by many researchers. These changes are among the major stereotypes characteristics of the elderly.

The aged individual is pictured often as being forgetful, intellectually slow, indecisive and so on. IQ scores made by individuals in old age do show a constant decrease along with ageing. Scores on verbal portions of these tests do not show greater declines, indicating that "stored information" is relatively unaffected by advancing age. However, problem solving skills are affected more significantly by increasing age. Research findings generally indicate that the decline in mental functioning may be due more to a diminished performance speed and changes in solving problems that are new and unfamiliar.

Among the most striking mental characteristics of the elderly are the changes affecting their memory? Undoubtedly, these changes can be frustrating for older person as well as for those with whom they interact frequently.

As individuals progress through late adulthood, there is increasing difficulty in processing long term memory. Problems with memory may account also for older person's communication. Problem of being repetitious in relating facts during a conversation. Often they may forget what was said only a few minutes earlier.

### **Developmental Tasks of Old Age**

The developmental tasks of late adulthood differ from those of earlier stages in two fundamental ways.

There is a focus on maintenance of life rather than discovering more about it. The task centre on happenings in the person's own life rather than on the lives of other (Hurlock, 1980).

Following are important development tasks of late adulthood.

- Adjusting to decreasing physical strength and health.
- Adjusting to retirement and reduced income.
- Adjusting to death of spouse
- Establishing an explicit affiliation with one's age group
- Meeting social and civic obligation
- Establishing satisfactory physical living arrangement

The above development tasks require person to adjust to several unique challenges in order to remain healthy during this time in life. The challenges include:

Accepting on increasing degree of dependency on others, adjusting to decreased economic conditions, leading to changes in life type and living conditions and developing new interests

### **Sensory Changes in Old Age**

Many deteriorating changes in the sensory system occur, in the old age. Age related vision changes include:

An increase in threshold or an increased amount of light needed to stimulate retinal calls.

A decreases in actually or sharpness of vision due to changes in the lens, pupil size, and accommodation (focusing) ability and

A decrease in adaptation to dark and light environments

Elderly people may expect to experience a number of eye disorders.

These include:

Muscular degeneration or a decreased blood supply to the retina causing loss of vision sharpness in looking directly but not in the peripheral areas and eyelids disorders such as drooping eyelids. Blindness increases

considerably after the age of sixty. Perhaps the most significant sensory changes occur with hearing and can cause serious adjustment problems that sometimes lead to a complete withdrawal from others. Hearing handicaps increase considerably as age increases in late adulthood.

About 50 percent of individuals over the age of 65 experience some degree of hearing loss. These losses occur earlier in men than women perhaps due to men's exposure to more hazardous environmental working conditions. There is a continued loss of hearing for high frequency sounds that first became noticed in middle age. Loss of mid to low range frequencies occur more frequently as age increases. Deafness can occur in many elderly people due to hardening of the bones and nerve damage to the inner ear.

Changes can be expected in both taste and smell perception in old age. Many elderly people remark that food tastes bland and will heavily season it with salt, pepper and other condiments to improve flavor. This change in taste often is attributed to a decrease in the number of taste buds as well as the need for stronger stimulation to activate the taste receptors in the mouth.

### **Divorce in Late Adulthood**

Divorce in late life is rare; couples who take this step usually do it much earlier. Despite the increase in divorce over the past 20 years, very few people over age 65 are divorced and not remarried.

People who divorce after age 50 have more trouble adjusting to the change than younger people, and older divorced people tend to have less hope for the future (Chiriboga, 1982). Divorced and separated people express much less satisfaction with family life than married people do. The men are less satisfied with friendships and activities not related to work, and the women's standard of living drops. For both sexes, mental illness and death rates are higher, perhaps because social support networks for older divorced people are inadequate (Uhlenberg & Myers, 1981).

### **OLD AGE AND DEATH**

#### **Characteristics of the Old Age**

According to Hurlock the following are the characteristics of old age:

**Old age is a period of decline.** During the old age physical and psychological functioning declines. There is a decline in intellectual activity, social activity, and physical health. Psychological make up plays an important role in decline. People who have low motivation and who have poor self concept decline faster as compared to individuals who are higher in these abilities.

**Individual differences.** There are individual differences in ageing. Some decline early as compared to others. People age differently. Some may look old at 60 years other at 70 years also can be active and healthy. Individual's age is poor criteria for determining when one reaches old age. Hereditary endowments, different socio-economic and educational backgrounds and different patterns of living considerably influence the process of ageing.

**There is also intra individual in ageing.** As a general rule, Physical ageing precedes mental ageing. Physical functions are the first to decline.

**Old age is filled with many stereotypes.** There are many stereotypes concerning old age. Old age perceived unfavorably. Many wrongly believe that old people are sick and physically weak. Old people are a subject of Joke and different forms of rumour. Another stereotype is that old people are perceived to be physically and mentally unproductive, accident prone and had to live with. It is generally believed in our society that "Young is beautiful and old is ugly".

**Unfavorable social attitudes toward old people.** Social attitudes toward old people are unfavorable. Unfavorable social attitudes towards the elderly are fairly universal in American Culture today. There are cross cultural differences in unfavorable social attitudes towards elderly. People who come from countries where respect for the elderly is customary (e.g. India) usually treat elderly people with more consideration and respect than do those who are materialistic and less tradition bound.

**Old age as a privileged group.** Older people, especially in India are considered "Senior citizens" and are a privileged group. They receive concessions in almost all areas. For e.g. they have concession in Train fares, seating arrangement in Train & buses, Income Tax concessions etc.

**Prone to be victims of crimes.** Elderly people are victims of crimes ranging from purse snatching to rape and murder. This is especially true of elderly women who create the impression that they are not strong or agile enough to defend themselves.

**Changes in role and status.** Old age brings about changes in Role and Status. Old people play a less active role in social life. Their social status due to retirement and decline of Income is generally reduced. In many cases, due to retirement, there is role reversal, retired men would stay at home and would be expected to help in the household activities including looking after grandchildren and attending to their needs.

**Period of enjoyment, relaxation and family involvement.** Old age is a period when one starts taking more interest in one's household activities and one's family. It is a period where one generally spends time in relaxation and in pursuing his/her leisure activities.

### Development Tasks

In every stage of development the individual has to achieve certain developmental tasks. The major developmental task in the old age as follows:-

**To attain Ego-integrity and avoid Ego despair** - According to Erikson ego-integrity means that even without daily responsibilities of job and family, psychologically healthy older people can maintain their sense of 'wholeness' and adequacy and are satisfied that they have done good job of living. Ego-desire is the result of general dissatisfaction with one's life, wishing it were possible to do it over again and knowing that it will never be possible. Ego-integrity is to be developed, at this period, while the feeling of ego-despair is to be avoided.

**Facing Death** - Old people who have developed ego-integrity can face death peacefully, but those who have developed ego despair cannot die calmly because they feel that they spent their lives uselessly.

**Adjustment to Retirement** - This is one of the most critical tasks of old age. After retirement the old person's self-image may change and he may feel helpless. Retirement being greater leisure but excessive free time leads to boredom and financial problem, lack of social contact and other related problems. Some people continue job-related activities and hobbies expand their roles as volunteers and develop new interests; e.g. a retired teacher may engage in adult education programme, guide needy students etc. or a lawyer may do some social work related to his old profession.

**Disengagement** - Old people reduce the number of contacts with the world and tend to be less active and withdraw from their social surroundings. This is called disengagement. This is very difficult task, and only a few people can adjust with it smoothly. Those who were more active in social activities in their earlier life, find it most difficult. In the societies where old people are respected and consulted by the young generation, old people find it easier to adjust, but in the society where old people are treated as a useless burden on the family, they find it very difficult to adjust.

**To overcome depression and frustration and to adjust to decreasing physical strength and health** - In many cases physical and financial difficulties of old men make them dependent on others. In some cases even if there are no financial difficulties, the loss of life-partner makes it very difficult to live. Depression is very common in old people. Suicide rate is higher among old people than among young people. To overcome depression and attain peace of mind is very difficult task for old people. Thus the developmental tasks to be attained in the old age are numerous and also difficult.

### Various Problems of Old Age

Old age brings various types of changes in physical structure, cognitive abilities, emotions and personality pattern. These changes give rise to various special problems for the old people. The most important problems for old age are the following:

**Physical problems** - In the old age muscular strength decreases activities become slower. Sensitivity to physical stimuli is reduced, therefore, they cannot quickly respond to physical stimuli. The ability to resist diseases and environmental changes also reduces; therefore, they cannot recover from illness easily and quickly. Thus old age produces physical helplessness and dependence on others.

**Economic problems** - In the old age the person retires from this job or occupation and there is no regular income. If he has no sufficient savings, he has to depend on others for economic support.

**Social Problems** - Since the activities of old people are reduced and they cannot contact other people frequently, the social relationships are gradually reducing. They cannot make new friendship or establish new social relationships.

**Adjustment problems** - Due to physical and economic, dependence on others, old people have a feeling of inadequacy. They suffer from depression; they may be tired of living. Due to generation gap there may be

various conflicts between old people and their children. Old people develop a complaining and irritating behavior.

**Leisure time problems** - Old people are retired from job and they have no specific activities to pass time. They have a lot of free time which brings tiredness, boredom and fatigue. Because the activities are reduced, they cannot pass the time easily. Thus there is the problem about leisure time. Some old people can resolve these problems successfully and live happily till their death. But in general the adjustment in the old age is very difficult.

Highlight by Buzybee

## DEATH

Death has, today, largely become a phenomenon of late adulthood. Advances in social sciences and increases in human understanding have helped us to understand the process of death more objectively and academically. A new field of study called as Thanatology has developed as a result of interest in death.

The terminal stage of life is a significant and valuable portion of the life course. If people live long enough, they are bound to have to deal with the death of people close to them. Furthermore, the awareness that they themselves will die one day can impart a special appreciation of life's pleasures and can make them think about the values they live by.

People can prepare themselves for the last stage of life by seeing others who are close to death, by nothing how death. By considering death, people can enhance life. They can live in the knowledge that each day may be their last, giving them a final chance to express their best qualities and is savor the sweetness of life.

All deaths are different, just as all lives are different. The experience of dying is not the same for an accident victim, a patient with terminal cancer, a person who commits suicide and someone who dies instantaneously of a heart attack. Not is the experience of bereavement the same for their survivors. Yet all people are human; and just as there are commonalities in our death.

### Types of Death

Today Death is considered to be a process, rather than a distinct process. Schultz has differentiated three different types of death these are as follows:

**Clinical Death:** refers to the cessation of spontaneous respiration and heartbeat in some cases artificial respiration will revive the patient.

**Brain Death:** occurs when the individual is deprived of oxygen for four or six minutes or more. The cortex, that portion of the brain controlling voluntary actions, thought and memory, dies first, followed by the midbrain, and finally the most primitive portion of the brain, the brainstem.

**Cellular Death:** Finally cellular death involves the physical death of organ systems the irreversible loss of organ functioning. This form of death - which occurs at different rates for different organs - is the last to occur.

The distinction between these forms of death is hardly trivial. Today people lapse into comas, showing no evidence in life in higher order brain areas, and yet their vital organs continue to live - they may even be able to breath on their own. Are these people alive? The answer appears debatable.

At present, physicians use brain death as evidence for a final determination of death. Veatch (1976) notes that the various criteria includes unreceptively and unresponsively, no movements or breathing no reflexes, and a flat electroencephalogram (EEG) reading that remains flat for twenty-four hours.

### Aspect of Death

There are at least three aspects of dying:

(a) **Biological**

(b) **Social**

(c) **Psychological**

All of which have become increasingly controversial.

The legal definition of biological death varies from state to state but in general, biological death is considered to be the cessation of bodily processes. A person may be pronounced death is when the heart stops beating for a significant period of time or when electrical activity in the brain stops. The criteria for death have become more complex with the development of medical apparatus that can prolong the basis sings of life indefinitely. People in a deep coma can be kept alive for years, even though they may have suffered irreversible brain damage and may never regain consciousness.

The social aspects of death revolve round funeral and mourning rituals and legal arrangements for the inheritance of power and wealth. A major problem in present-day society is a lack of widely accepted conventions of behavior for people who know that they are dying, for those around them, and for the survivors after the death of loved one. Several convention which do exit are rarely helpful either for dying

people or for those close to them: isolating the dying in hospitals or nursing homes, refusing to discuss their conditions with them, separating from them before death by visiting less often, and thus leaving them to cop with death alone.

The psychological aspects of death involves the way people feel both about their own death as it draws near and about the death of those close to them. Most people today have a great deal of trouble coming to terms with the meaning of death. We need a most positive acceptance of the reality of death as a natural and expected phase of life.

Discuss the Psychological changes that occur when one encounters death.

### Psychological Changes

Developmental psychologists and Behavioral scientists have found many psychological changes that occur as death approaches. Some of these psychological changes are detailed below. Classic studies in this area has been done by Liberman (1965), Butler (1971) and Riegel and Riegel (1972).

Patients who were near death had increasing difficulty in organizing and integrating the stimuli in the environment. Their energy levels appeared to decrease.

According to Liberman aged people approaching death probably experience a kind of psychological upheaval not because of the fear of death, but because of a disorganization of the mental processes as death approaches.

Liberman further states that people who withdraw from life in the later years may be building a kind protective shell around themselves as they "attempt to hold them together- to reduce the experience of chaos" (Liberman, 1965).

Riegel and Riegel (1972) conducted a ten-year study of old people in Germany. The subjects were 190 males and 190 females who ranged in age from fifty-five to seventy-five years. The investigators administered a battery of tests to the subjects, including an intelligence test, a word-association test, tests to assess verbal achievement, and a number of attitude and interest tests. Finding showed that subjects whose performance of intelligence tests was below average were closer to death than their more successful peers - indicating a terminal drop intellectual functioning. The Riegels analyzed the subjects' scores by going backward in age starting with the time of death. They concluded that the decline in performance on intelligence test was due to a sudden drop in ability that occurred with five years before death. These findings appear to dovetail with Liberman's conclusion that old people near death experience a systematic disintegration. However, these findings remain somewhat controversial.

Liberman and Coplan (1970) in a study of 80 people aged 65 to 91 found that:-The subjects who had died within the year had lower scores on cognitive tests. They were also less introspective and more docile. Those who were dealing with some sort of crisis and were close to death were more afraid of an more preoccupied with death than people who were best by similar crises but were not close to death. (Person who were close to death but whose lives were relatively stable at the time showed neither special fear of death nor preoccupation with it).

People who are nearing death indulge in a process that is called as life review. They organize their memories and reinterpret the actions and decisions that have shaped the course of their life. Ideally, the life review is a positive experience resulting in further integration of the personality in the face of death. For some, the life review leads to less ego involvement with one's own situation and to more concern with the world in general. For other, it produces nostalgia and perhaps a touch of regret. In still others, it leads to anxiety, guilt, and depression: instead of reflecting on a full life, the person feels cheated and emerged. In a small number of cases, the person taking stock of his life may be thrown into a state of panic that may result in suicide (Butler: 1971).

Near death experiences is another psychological changes that many people experience, either before dying or during some or other period of their life. These experiences often include a feeling of well-being a new clarity of thinking, a sense of being out of one's body, and visions of bright lights. Three ways in which such experiences have been explained are a prediction of a state of bliss after death (the transcendental theory); as a result of biological states accompany the process of dying (the physiological theory); and as a responses to the perceived threat of death (the psychological theory). A recent study of such experiences, both in people who actually did come close of death and in others who only thought that they were close to death, found some support for all three theories. (J.E. Ownes, Cook, & Stevenson, 1990). Researchers studies the medical records and personal accounts of 28 hospital patients who would have died if doctors had not saved them,

and of 30 who mistakenly thought they were in a danger of dying. The two groups of patients had very similar sensations, a finding that lends support to the psychological theory. But those who had actually been, nearly death reported near death experience more often - evidence for the physiological theory. And the researchers have support for the transcendental theory in the fact that dying patients reported clearer thinking, despite the likelihood that their brain functioning was in a fact diminished.

Discuss the attitude towards death and dying across the life span. People of different ages think and feel differently about death. We will briefly review how people conceive about death and react to it during different stages of their life span.

**Childhood:** Most young children seem to think of death as a temporary state. It is usually not until sometimes between the ages of 5 and 7 those children evidently understand that death is irreversible - that a dead person animal or flower cannot come to life again. At about the same age; children realize two other important concepts about death; first that it is universal (all living things die); and second that a dead person is nonfunctional (all life functions end at death). Before then, children may believe that certain groups of people (like teachers, parents, and children) do not die, that a person who is smart enough or lucky can avoid death, and that they themselves will be able to live forever. They may also believe that a dead person can still think and feel.

These observations about children's views of death emerge from a review of 40 studies that have been done since the 1930s, most of them based on interviews with children (Speece & Brent, 1984). Cultural experience, too, influence attitudes towards death. Children from poor families are more likely to associate it with disease and old age (Bluebond-Langner, 1977).

**Adolescence:** Adolescents tend to have highly romantic ideas about death; "adolescents make have soldiers because they do not fear annihilation" so much as they are concerned about being "brave and glorious" (Pattison, 1977, p. 23). In their attempt to discover and express their identity, they are concerned with how they will live, not with how long they will live. This may partially explain in appeal of suicide to adolescents. Adolescents mourning the death of a family member sometimes feel embarrassed talking outsiders and may feel more comfortable grieving with their peers than with adults.

**Young adulthood:** For young adulthood death is a most frustrating experience. Most young adults - having finished their education, training, and courtship and having recently embarked on careers, marriage, or parenthood - are eager to live the lives they have been preparing for. When they are suddenly taken ill or badly injured, young adults are likely to feel more intensely emotional about imminent death than people in any other period of life (Pattison, 1977). They feel extremely frustrated at the inability to fulfill their dream. Their frustration turns to rage, and that rage often makes young adults trouble some hospital patients.

**Middle adulthood:** It is middle age that most people really know keep inside themselves that they indeed going to die. With the death of their parents, they are now the oldest generation. As they read the obituary pages which they are likely to do more regularly at this age than they used to they find more and more familiar names.

### Steps of Dying

Elizbeth Kubler Ross, a psychiatrist who works with dying people, is widely credited for having inspired the current interest in the psychology of death and dying. She found that most patients welcome an opportunity old Age and Death to speak openly about their conditions, and the most are aware of being close to death even when they have not been told how sick they are. After speaking with some 500 terminally ill patients, Kubler-Ross (1969, 1970) outlined and gave examples to illustrate, each of five stages in coming to terms with death:

#### Denial (Refusal to accept the reality of what is happening)

##### Anger

##### Bargaining for extra time

##### Depression

##### Ultimate acceptance

We will discuss each of these in brief.

**Denial:** Most people respond with shock to the knowledge that they are about to die. Their first thought is "Oh, no this can't be happening to me." When people around the patient also deny reality, he or she has no

one to talk to and, as a result, feels deserted and isolated. When allowed some hope along with the first announcement and given the assurance that they will not be deserted no matter what happens, people can drop the initial shock and denial rather quickly.

**Anger:** After realizing that they are dying, people become angry. They ask "Why me?" They become envious of those around them who are young and healthy. They are really angry not at these people but at the young and the health that they themselves do not have. They need to express their rage to get rid of it.

**Bargaining for extra time:** During this stage, the terminally ill patients start bargaining with their fate. For example, they may ask God for a certain amount of time in return for good behavior. They may promise to do good deeds, devote time and money for religious activity.

**Depression:** When the terminally ill patient can no longer deny his illness, when he is forced to undergo more surgery or hospitalization, when he begins to have more symptoms of becoming weaker and thinner, he cannot smile any more. His numbness or stoicism, his anger and rage will soon be replaced with a sense of great loss. At this stage the person enters a deep depression. He is depressed because of the losses he is incurring, for example, loss of body tissue, loss of job, and loss of life savings. And he is depressed about the loss which is to come. The patient is in the process of losing everything and everybody he loves. It is important that he be allowed to express his sorrow.

**Ultimate acceptance:** Finally, the dying person accepts death. The struggle is over and the person experiences a final rest before the long journey. At this point, the person is tired and weak. He sleeps often. In some cases, the approach of death feels appropriate or peaceful. The person may limit the number of people he will see and withdraw his interest from matters of the world. Silence and constancy are appreciated. The person seems to detach himself so as to make death easier. Not all terminal patients' progress through the stages Kubler-Ross describes. For example, a person may die in the denial stage because he is psychologically unable to proceed beyond it or because the course of his illness does not grant him the necessary time to do so. As important as Kubler-Ross's work is, it is not without its critics. Schulz (1978) notes that many researchers have found it difficult to use her system. The stages are highly subjective and therefore difficult to identify in patients. Shneidman (1980) also reports while he has observed evidence of isolation, envy, bargaining, depression and acceptance in a dying person, he sees no reason to think of these behaviors and affect states as 'stages'. Moreover, he does not believe that everyone goes through these stages in the same order. Instead, Shneidman sees the dying person as expressing a constantly alternating display of affect and thought. Feelings of anguish, depression, hope, envy, bewilderment, anger, acceptance, denial, pain and even yearning are all evident in the dying person - but their appearance according to Shneidman, would seem to be less predictable than Kubler-Ross' theory suggests.

### Grief Therapy

Grief therapy is a professional program to help the bereaved cope with their losses. Most bereaved people are able, with the help of family and friends, to work through their grief and to resume normal lives. For some, however, grief therapy; a program to help the bereaved cope with their losses is indicated (Schulz, 1978).

Professional grief therapists focus on helping bereaved people express their sorrow and their feelings of loss, hostility, and anger. They encourage their clients to review their relationships with the deceased and to integrate the fact of the death into their lives so that they can be freed to develop new relationships and new ways of behaving toward surviving friends and relatives.

There are also organizations - Such as Widow to Widow, Catholic Widow and Widower's Club; and Compassionate Friends (for parents of children who have died) which provide non-professional grief therapy, emphasizing the practical and emotional help that one person who has lost someone close can give to another.

### Controversial Issues of Death and Dying

Technological advances and an increased understanding of the study of human development have helped us to understand certain controversial issues of death and dying. Two such important issues are as follows:-

(a) Euthanasia and the Right to Die

(b) Suicide

Euthanasia in simple language means "mercy killing". It can be active or passive. Active euthanasia refers to action deliberately taken with the purpose of shortening a life in order to end suffering or to carry out the wishes of a terminally - ill patient. Passive euthanasia in mercy killing which takes the form of withholding treatment that might extend life such as medication, life-support, or feeding tubes. Although active euthanasia is highly controversial, most people are not in favor of preserving life in all cases. In a New York Times - CBS Poll taken in 1990. 53 percent of the respondents said that doctors should be allowed to assist in ill person in taking his or her own life (Malcolm, 1990).

One physician predicts that active euthanasia will become increasingly common, perhaps following the pattern in the Netherlands (Sprung 1990). In 1984, the Dutch Medical Association issued guidelines for doctors to participate in euthanasia: the request must be made freely and consistently by patient, the patient's condition must be unbearable and without hope of recovery, and another physician must agree on the advisability of euthanasia and on the method. Active euthanasia remains a criminal offence in the Netherlands if these medical guidelines are not followed, but it is estimated that there between 2000 and 10,000 cases a year.

In India, too, many seminars and conferences on euthanasia have taken place but it has not received any legal sanction yet.

Suicide is another controversial issue related to dying and death. Suicide was once considered to be a criminal offence in our country under section 309 of the Indian Penal Code. However, recently, the Supreme Court of India has struck down this section to be unconstitutional. This decision of the Supreme Court has raised many legal issues and controversies. Though legally suicide may be tolerate or may not be punitive in nature, but it has definitely not achieved social sanction in our country. Individuals who commit suicide are considered to be cowards, lack courage and poor will.

### **Old People's Attitude towards Death**

Many individuals wrongly assume that old persons are generally afraid of death. However research studies by Crandall has shown-that old people generally do not fear death. In fact, a classic study by Munichs (1966) found that the most common orientations in individuals seventy and older were acceptance or acquiescence. A more recent study suggests that elderly subjects were less preoccupied with death fears than middle-aged subjects (Bengston et. at. 1977). Similarly, Hinton (1967) reported that young terminal patients were more anxious about their illness than were older patients. Munichs research suggests that excessive fear or denial of death in older people represents a general failure to come to terms with their own limitations, which, in turn may be an indication of psychological immaturity. Fear of death is also associated with lack of ego integrity (Erikson, 1963). On the other hand, a relative lack of fear of death is associated with people's having experienced "purpose in life" (Durlark, 1973), and having achieved integrity.

Crandall (1980) lists seven reasons why people fear death and dying. They are:

1. The fear of what happens after death.
2. The fear of a painful death.
3. The fear that dying will drain the family's finances.
4. The fear of what will happen to one's family after death.
5. The fear of the indignities associated with dying.
6. The fear that no one will really care or will remember you after death.
7. The fear that during will strip away one's defenses and facades, leaving one's real self exposed.

Majority of older people regard death as a natural phenomenon for which they have already prepared themselves.